ESSAY · SUBMITTED TO MULTICOIN CAPITAL

The Growth Trap in Proof of Physical Work

Adding operators makes most DePIN networks worse at the thing they sell. I measured it on the largest deployed mesh in the world.

MAX MOHAMMADI · 7 AUGUST 2026 · FULL ANALYSIS FOLLOWS AS AN APPENDIX

Multicoin named this category in 2022: protocols that pay people to do verifiable work that builds real infrastructure. Four years of deployment have shown that verifiability is a spectrum. The physical act sets where a network lands on it, and that position predicts revenue better than market size, hardware cost or token design.

A second failure runs alongside the first, and nobody has priced it. In most of these networks, recruiting more operators makes the network worse at delivering what it sells. I measured this on the largest deployed mesh network in the world. The parameter that governs it turns out to be 47 percent worse than everyone assumed.

Three tiers, not one adjective

Sort DePIN networks by what an outside referee can check. Three tiers appear.

Coverage is a claim that a radio sits at a place. Nothing checks it. Proof of location has no general solution: GPS is spoofable at negligible cost, and hardware attestation falls to tampering and cloning. Helium's own denylist confirmed 25,000 fraudulent hotspots, and in the Hong Kong and Shenzhen region it flagged 1,291 of 3,490 listed hotspots.

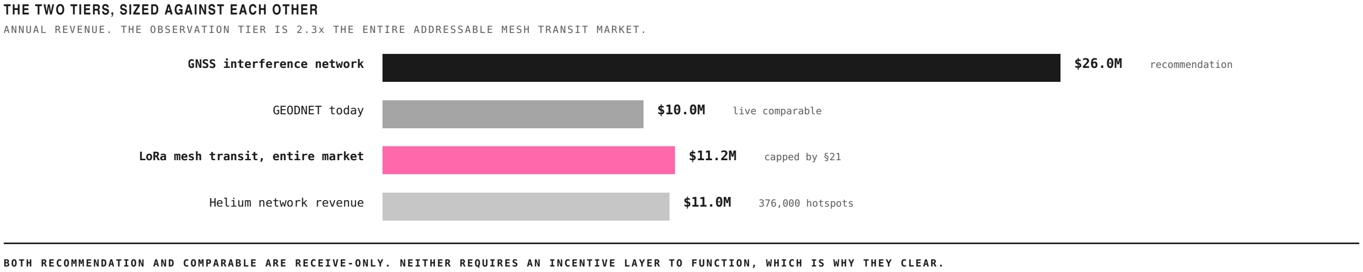
The verification failure is the less interesting half of the diagnosis. A single LoRaWAN gateway serves thousands of sensors transmitting a few bytes a day. The network built more than half a million gateways for a sensor population that a few thousand would have covered. A good in near-infinite oversupply clears at approximately zero. Helium IoT peaked around $6,500 a month against $365 million raised, which was the correct price.

Delivery is a packet relayed and confirmed received. The destination has its own interest in the transaction, so a relay cannot fabricate a delivery to a counterparty that received nothing. Verification collapses from proof of location, which is unsolved, to a signature over a path, which is ordinary cryptography. Relaying also consumes shared channel time, so the resource is scarce.

Observation is a signal heard and reported. The referee here is physics. Aircraft broadcast GPS position on 1090 MHz, so a receiver's claim is checkable against every other receiver that heard the same aircraft. Satellite ephemeris is published, so a GNSS station's claimed observations must agree with where the constellation was. Fabricating a consistent view becomes an orbital mechanics problem.

The three tiers are not equally hard to build. Coverage is the easiest to bootstrap and the hardest to price. Observation is the reverse: the hardware is commodity and the operators already exist, but nobody has paid them.

Revenue follows that order. GEODNET, whose operators never transmit, reaches $10 million annualised from 20,500 receive-only base stations. FlightAware runs more than 40,000 volunteer ADS-B receivers across 193 countries and sold to Collins Aerospace. ADS-B Exchange ran 14,000 feeds and sold to JETNET. Both compensated their operators with nothing.



Annual revenue by tier. The observation-tier opportunity is 2.3× the entire addressable mesh transit market, and both it and its live comparable are receive-only.

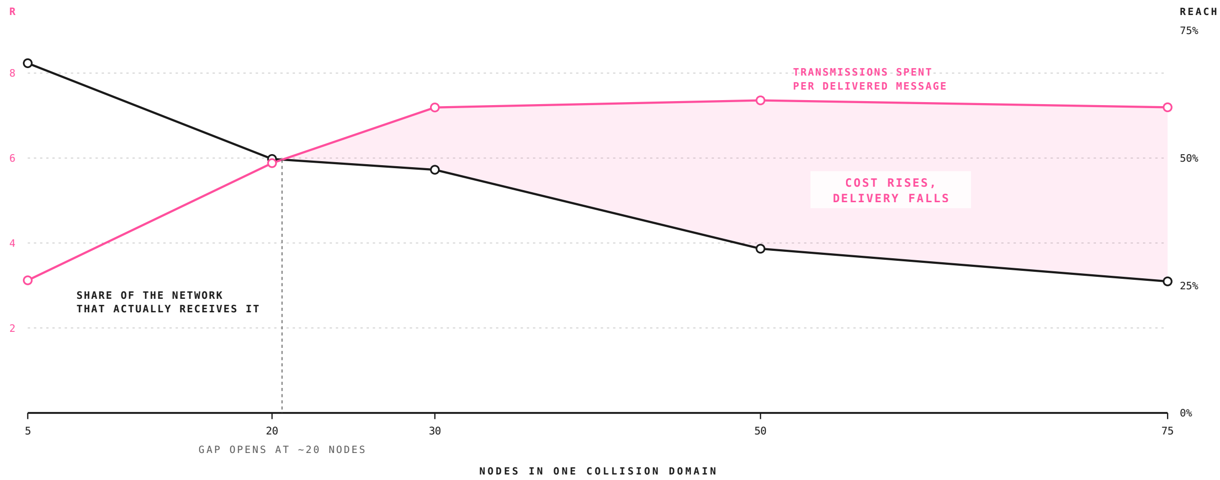
The parameter nobody measured

The delivery tier holds most of the category's optimism, and its ceiling can be computed. Its economics turn on one number: the rebroadcast factor, transmissions spent per delivered message. Published work assumes 3 to 5. My own earlier drafts assumed the same.

Meshtastic, the dominant LoRa mesh firmware, publishes a discrete-event simulator that models its production routing logic, airtime, channel-activity detection, contention windows and collisions. I wrote a headless driver and ran it across node densities from 5 to 75 and hop limits of 3, 5 and 7, three repetitions each.

At realistic density the rebroadcast factor is 7.36. That correction cuts the revenue ceiling of the tier by 32 percent. At the US firmware default, one transmission holds the channel for 600 milliseconds, so a collision domain delivers 163 messages an hour and grosses $14,300 a year at a cent a message. The hardware inside that domain costs more.

The simulation produced a second result I did not design for.

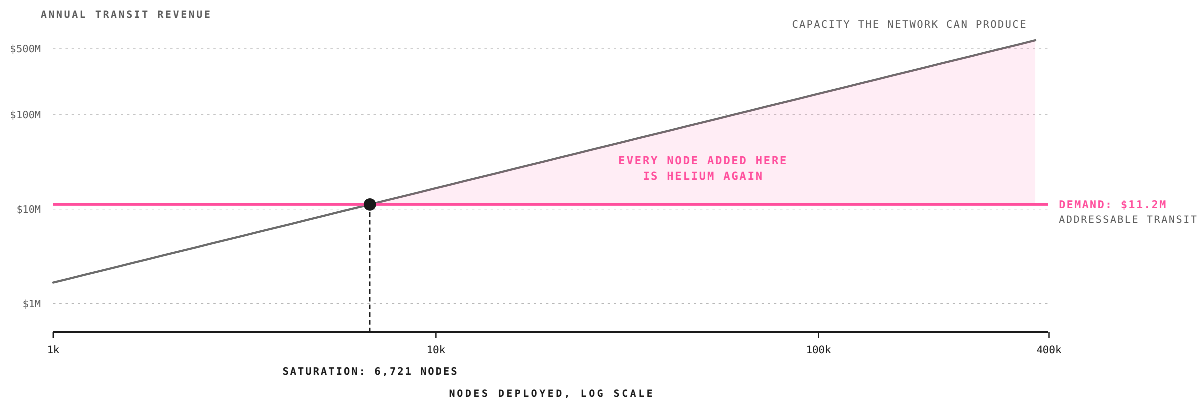


As nodes are added, transmissions per delivered message rise and the share of the network reached falls. The shaded region is the widening gap between what a message costs and what it delivers.

Reach, the share of the network that receives a given message, falls from 68.6 percent at five nodes to 25.8 percent at seventy-five. Adding nodes makes a flooded mesh worse at its core function. Production networks show the same effect: a regional Meshtastic mesh collapsed at Hamvention 2024 when one user's bridge overwhelmed it, and DEF CON now requires specialised firmware with rebroadcasting restricted to hold 2,000 nodes in a single venue.

Where the trap closes

Capacity scales linearly with deployed nodes. Demand does not scale at all. Building the addressable market from transaction counts gives $11 million a year for the entire off-grid settlement opportunity. Set that against what the physics permits and the two lines cross.



Capacity against demand, log scale. The shaded wedge is supply nobody buys. Past the crossing, every node added is capacity issued against fixed demand.

6,700 nodes serve the entire addressable transit market. Past that point, each additional operator adds capacity against fixed demand and dilutes the revenue available to everyone already there. Protocol revenue tops out near $1.1 million a year however large the network gets.

Hold that against what the category has been building. Helium deployed 376,000 hotspots for a sensor market that a few thousand gateways would have covered. A LoRa mesh needs 6,700 nodes to saturate its own. Both networks were sold on the same growth story, and both were built roughly fifty times past the point where growth stopped paying.

This restates Helium's non-scarcity failure one tier up, and it generalises beyond LoRa and beyond mesh. Any protocol that pays for presence hits it, in any market where presence past a threshold produces nothing.

Every DePIN network is sold on a growth story. In this one, growth is the failure mode.

The emission schedule is the product

So this is a token design problem, and every schedule in the category got it wrong the same way. Helium paid per hotspot and got 376,000 of them earning $29 a year each. Hivemapper paid per image, so drivers circled well-mapped streets until MIP-24 moved rewards to hexes. Crankk handed 80 percent of emission to miners with no buyer on the other side and produced a 1,877-day payback. None of these were execution failures. Each schedule worked exactly as written, and the writing was the mistake.

Two networks solved half of it each. **ORE** issues at a fixed rate per unit time, split competitively, so additional participants take a smaller share and never cause more tokens to exist. That answers saturation directly, and Helium lacked it. **Helium** built the best sink in the category: buyers burn for non-transferable credits at a fixed dollar rate, so demand comes from usage and the buyer never holds a volatile asset.

Nobody has shipped both, weighted by marginal contribution, withheld until a buyer exists.

Three protocols solving unrelated problems have converged on the marginal-contribution half independently. DoubleZero pays fibre contributors by Proof of Utility weighted with Shapley values, rewarding the marginal performance each link adds. Shelby compensates storage providers for serving data instead of holding it idle. GEODNET caps rewarded stations per geographic area. Three live derivations of one rule beat any argument I could make for it.

The obvious objection

Supply must exist before anyone buys it, and withholding emission until a contract is signed means launching with no network. Every protocol in this category made that trade deliberately.

Look at what happened to Bitchat. Its Bluetooth mesh reaches 30 to 300 metres, which works in a crowd and nowhere else. Nobody waited for the project to fix that. Four independent bridges now put Bitchat traffic onto LoRa, one of them adding LoRa to the protocol directly on solar ESP32 relays under an MIT licence. Range solved, by volunteers, on $15 hardware. And every bridged message now pays the 7.36 flood tax measured above, so the traffic moved inside the ceiling instead of escaping it. Demand for this is real enough that people build and give away the hardware. Nobody is paying the relays.

The observation tier answers the bootstrapping question the same way. Volunteer RF networks already exist at scale and already have paying enterprise customers: FlightAware's 40,000 receivers, ADS-B Exchange's 14,000 feeds, MarineTraffic, WSPRnet, PSKReporter, SatNOGS. **The supply side of RF observation has been built, monetised and sold twice, and the people who built the physical layer own none of it.** No bootstrapping problem remains. A compensation problem remains, which is far easier.

Where I would point this

At GNSS interference monitoring, for four reasons that are unusual in combination.

1. **The buyer has published the gap.** IATA measured a 65 percent year-on-year rise in loss-of-GNSS events per thousand flights. Per Eurocontrol's own summary, detecting affected areas from a distance remains impossible, so pilot reports are still the main source of information. Most opportunities require convincing someone they have a problem. This one requires building what they have already asked for.
2. **Nobody occupies the space.** Septentrio and Safran sell mitigation hardware. GPSPATRON and GMV sell centralised detection. No distributed measurement network exists, though the technique is published and the hardware is commodity.
3. **Receive-only escapes every constraint above.** No licence, duty cycle, power ceiling or airtime cost applies. The verification is the strongest in the category, because a fabricated observation must satisfy orbital geometry and agree with every neighbour watching the same satellites.
4. **Precedent solves the supply side.** A node is $260 of commodity parts and needs an operator with roof access, sky view and a power outlet. GEODNET has recruited exactly that person 20,500 times across 150 countries.

Built from account counts and contract values across airlines, air navigation providers, insurers, maritime, timing and defence, the addressable subscription revenue is $26 million a year. That is small. It is also 2.3 times the entire addressable mesh transit market, against a live comparable, GEODNET, trading near $85 million on $10 million of revenue.

What this means for the thesis

Proof of physical work was right about the axis and did not say where along it anything sat. Adding the tiers turns verifiability into a category a network can be sorted into, and the sorting predicts revenue.

The growth trap is the sharper correction. The category has spent six years optimising for deployed device count, because that metric looks like traction and because emission schedules were built to maximise it. For most of these networks the metric is wrong, and the schedule that maximises it destroys the thing it was meant to build.

Three rules follow. Pay for physical work that something external can check. Meter the reward by marginal contribution. Withhold the token until somebody is paying for the output. DoubleZero, Shelby and GEODNET have each implemented one of the three, and none has implemented all of them.

The next DePIN network worth owning will be the first one to announce a cap on its own supply of operators, and it will be attacked for it.

The full analysis follows as an appendix. 47 sections, 26 figures, 52 sources: the simulation method and driver, the capacity derivation, the regulatory ceilings that confine any paid mesh to unlicensed spectrum, chain selection read from source, the addressable-market build, and a full protocol and token specification. Every figure and table is generated from a single parameter set.

APPENDIX · THE ESSAY PRECEDES THIS

The Growth Trap in Proof of Physical Work

Verification tiers, mesh capacity, and a protocol specification

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ABSTRACT

Decentralized physical infrastructure networks are conventionally sorted by the resource they sell. We argue the more predictive axis is *what the network can check a contribution against*, and that three tiers exist: coverage, checkable against nothing; delivery, checkable against a counterparty; and observation, checkable against physics via published reference data. Realised revenue tracks that ordering closely. We test the delivery tier from first principles, measuring its governing parameter, the rebroadcast factor R, with the Meshtastic project's own discrete-event simulator and obtaining R = 7.36 against the 3 to 5 commonly assumed. The correction cuts the tier's revenue ceiling by 32 percent; reach falls from 68.6 to 25.8 percent as nodes are added, so a flooded mesh degrades before it monetises; and addressable transit demand saturates at 6,721 nodes, past which a growing network destroys its own unit economics. All three results moved against the idea we began with. **Recommendation: fund a receive-only GNSS interference monitoring network, $26M addressable against a $260 bill of materials and a live comparable at $85M; require demand-weighted emissions; decline any incentive layer built on managed-flood routing.**

KEYWORDS DePIN; LoRa; Meshtastic; MeshCore; time on air; verification; GNSS interference; decentralized wireless

PREPARED FOR	VERSION	DATE	PAGES
Multicoin Capital	1.0	2026-08-07	67
MEASURED PARAMETER	**METHOD**	**CODE**	**SOURCES**
R = 7.36	Discrete-event sim	github.com/ maxmoneycash	52

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1 What We Are Bullish On

THE CLAIM, THE CHAIN, AND THE PROTOCOL

We are bullish on verified RF observation as the DePIN category that produces revenue, and on Aptos as the chain a new protocol in it should be built on. Neither of those is the consensus position. Both follow from work in this report.

The category is conventionally sorted by resource sold: coverage, compute, storage, mapping. We argue the variable that predicts revenue is *what the network can check a contribution against*, and that three tiers exist. Coverage is self-asserted and checkable against nothing, which is why Helium's LoRaWAN network peaked at $6,500 a month against $365 million raised. Delivery is checkable against a counterparty, and we price it from first principles and find it capped. Observation is checkable against physics, and it is the only tier with networks producing revenue today.

THE THREE CLAIMS

- **Verification tier predicts revenue.** GEODNET reaches ~$10M receive-only; Helium reached $6,500 a month selling coverage. The tier explains the gap better than market size, hardware cost or token design.
- **The delivery tier is capped, and we measured the cap.** R = 7.36 against the 3 to 5 assumed, reach falling 68.6% to 25.8% as nodes are added, transit demand saturating at 6,721 nodes.
- **No existing token in this category has both halves of a sound design.** ORE has the right emission and no real demand; Helium has the right sink and an emission schedule that destroyed its unit economics. Nobody has combined them.

WHY APTOS, AND WHY THIS IS NOT FLATTERY

We reached Aptos by reading `aptos-core` and `MystenLabs/sui` against a criterion nobody uses to select a chain: how a signed transaction behaves when it sits in a relay for six hours. Aptos's `expiration_timestamp_secs` may be set so that a transaction never expires, stated in the source comment. Solana needs a durable nonce account to reach the same place. That property, plus Block-STM for parallel reward epochs, Move resources for non-copyable device identity, and the Move Prover for *provable* emission invariants, gives Aptos a weighted score of 14 against Solana's 8 (§16).

We note the obvious conflict: Multicoin led Aptos's seed. The defence is the derivation. Solana wins the two heaviest practical objections in that table, and we say so: it has every DePIN comparable and a device registry proven to 500,000 nodes. Aptos wins the criteria that determine whether the tokenomics are sound, which is what a new protocol is choosing on.

WHAT WE WOULD FUND

A **verified RF observation protocol on Aptos**, starting with GNSS interference measurement. $26M addressable against a $260 bill of materials, buyers who have published the gap, a live comparable at $85M, and a token design combining ORE's emission with Helium's sink. Full specification in §17, emission schedule in §18.

WHAT WE WOULD DECLINE

Emission per device, which is the root failure in four of the eight tokens in §15. **Any incentive layer on managed-flood routing** (§23). **Any consumer-facing device business**: three companies across three technologies abandoned the consumer layer and kept the wholesale layer (§21).

HOW TO READ THIS REPORT

Part I is the framework. **Part II** is the recommendation: the observation tier, the GNSS opportunity, the team, the chain and the token. **Part III** is the infrastructure it sits on. **Part IV** is the delivery-tier analysis that produced the framework, including the measurement. **Part V** is what would make us wrong. **A reader with ten minutes should read §1, §11 and §17.**

PRINCIPAL RESULTS

- Observation tier: $26M addressable, 2.3× the entire mesh transit market.
- R = 7.36 measured; transit saturates at 6,721 nodes and protocol revenue caps near $1.12M.
- Aptos 14 : Solana 8 on weighted criteria for a new protocol.
- Proposed supply 42,048,000, no premine, no genesis VC allocation, emission gated on the first paying contract.
- Mesh capacity has 212× architectural headroom on the same radios (§30).

2 Why Now

THREE CHANGES SINCE HELIUM IOT WAS WRITTEN OFF

The consensus objection to anything LoRa-shaped is short and good: *Helium already did this*. Three things changed between 2024 and mid-2026 that did not hold when that judgement was formed.

1 DEMAND STOPPED BEING HYPOTHETICAL

Bitchat, Bluetooth mesh, no servers, no accounts, no phone numbers, recorded roughly 48,800 downloads in Nepal in a single day during the September 2025 protests, spiked again during Iran's January 2026 blackouts and Uganda's January 2026 election, and reached **430,000 daily active users in India on 26 July 2026** during the Delhi student protests. India's Cyber Crime Coordination Centre ordered GitHub to remove three Bitchat repositories inside a three-hour window on 23 July. The app stayed up.

> **THE MISREAD**
>
> This is a demand signal with **no matching supply**, and almost nobody in the category is reading it that way. Bluetooth mesh spans 30–300 metres. It works at a protest because a protest is a crowd. It does not work across a city, a county, or a disaster zone. LoRa mesh spans 2–15 km per hop. The people who downloaded Bitchat 430,000 times in a day are asking for a network Bluetooth physically cannot provide and LoRa physically can. Nobody has connected the two curves.

2 SUPPLY CROSSED THE HOBBYIST THRESHOLD

Meshtastic passed 40,000 GitHub stars and an 80,000-member subreddit by early 2026, with 100+ supported hardware variants, entry devices under $50, and active meshes in most major US cities. The physical layer now exists at meaningful density *without anyone having paid for it*, which is simultaneously the opportunity and, per §15, the cultural risk.

3 ROUTING BECAME SETTLEMENT-CAPABLE

MeshCore, released in 2025, separates repeater roles from client roles so end devices do not relay, floods once to discover a path and then embeds it in subsequent packets, supports 64 hops against Meshtastic's 7, and returns deterministic delivery confirmation where Meshtastic's acknowledgement can be ambiguous. Read as a network engineer, that is source routing adapted to a low-bandwidth channel. Read as an economist, it is a settlement-layer specification: identified counterparties, an auditable route, and a deterministic receipt.

The three together are the "why now." Demand exists and is measurable. Supply exists and was free. And for the first time the routing layer can produce the artefact a payment system needs. Sections 3 through 12 test whether the economics follow, and largely find that they do not at consumer prices, which is the finding that makes the recommendation in §16 specific.

WHAT THE CONSENSUS GETS RIGHT

Two prior attempts failed and the failures are instructive. Helium's LoRaWAN arm is diagnosed in §4. Crankk, which integrated with Meshtastic firmware on Kadena in December 2024 and did attempt to reward participation, shows an average device cost near $150 against eight cents a day in earnings, a 1,877-day payback. Crankk built on LoRaWAN star topology and a Meshtastic fork, inheriting exactly the flood-routing tax quantified in §18 and §25. It is the control experiment for this thesis, and it is why architecture appears in the recommendation.

3 A Verification Ladder

THE ORGANISING IDEA

A DePIN protocol must convert an off-chain physical act into an on-chain reward. The conversion is only as sound as the check that sits between them. We propose sorting networks by that check.

WHAT THIS ADDS TO PROOF OF PHYSICAL WORK

The 2022 essay that named this category defined it as protocols that incentivize people to do *verifiable* work, and that adjective has been carrying more weight than it can bear. Four years of deployment have shown that verifiability is not a property a protocol either has or lacks. It is a spectrum, the position on it is set by the physical act sition on it more closely than it tracks any other variable including market size, hardware cost or token design.

This report makes three additions. It supplies the tiers, so that verifiability becomes a category a network can be sorted into of them. It quantifies the delivery tier from first principles and measures its governing parameter, which yields a hard ceiling nobody had computed. And it shows that the tier a network occupies predicts realised revenue across every comparable in the sector. The original framing was right about the axis and did not say where along it anything sat.

FIGURE 1: THE THREE TIERS AND THEIR OBSERVED OUTCOMES

TIER	WORK PAID FOR	CHECKED AGAINST	OBSERVED OUTCOME	
COVERAGE	an assertion that a radio is at a place	nothing. the claim is self-certifying	Helium IoT peak revenue	**$6,500 / month**
			capital raised against it	**$365,000,000**
			hotspots on the denylist, HK/SZ	**37%**
DELIVERY	a packet relayed and confirmed received	the receiving counterparty’s signed receipt	verification cost	**one signature**
			scarcity	**channel airtime, rivalrous**
			ceiling, one collision domain	**$14,291 / year**
OBSERVATION	a signal heard and reported	physics, against published reference data	GEODNET annualised revenue	**~$10,000,000**
			FlightAware volunteer receivers	**40,000**
			compensation to those volunteers	**none**

TIERS ARE DEFINED BY WHAT AN EXTERNAL PARTY CAN CHECK A CONTRIBUTION AGAINST, NOT BY THE RESOURCE SOLD. REVENUE FIGURES ARE THE MOST RECENT PUBLIC DISCLOSURES FOR EACH NETWORK.

4 The Coverage Tier

AUTOPSY

UNVERIFIABLE IN THE GENERAL CASE

Proof of location has no general solution. GPS is spoofable at negligible cost. Hardware attestation is defeated by tampering, cloning, and optical probing of the secure element. The best deployed approaches are fractional: they combine reported signal strength, mapping consistency, and latency into a *justification* for withholding rewards, not a proof of honesty. Helium’s Proof of Coverage, in which beacons are witnessed by neighbouring hotspots, is the most sophisticated attempt, and its own denylist confirmed 25,000 fraudulent hotspots out of 524,000 network-wide, rising to 1,291 of 3,490 in the Hong Kong and Shenzhen region.

NON-SCARCE, WHICH IS THE DEEPER PROBLEM

Verification failure is the diagnosis usually offered. It is incomplete. Even a perfectly honest coverage network would have failed, because coverage in a star topology has no marginal scarcity. Gateway capacity vastly exceeds sensor demand at every plausible deployment density, so additional gateways add no additional sellable resource.

This distinction matters for what follows. A network can be fixed if its problem is verification: better attestation hardware, better challenge protocols. A network cannot be fixed if its problem is that the thing it sells is not scarce. Sorting by the verification ladder surfaces which failure a network has.

WHERE THE COVERAGE TIER STILL WORKS

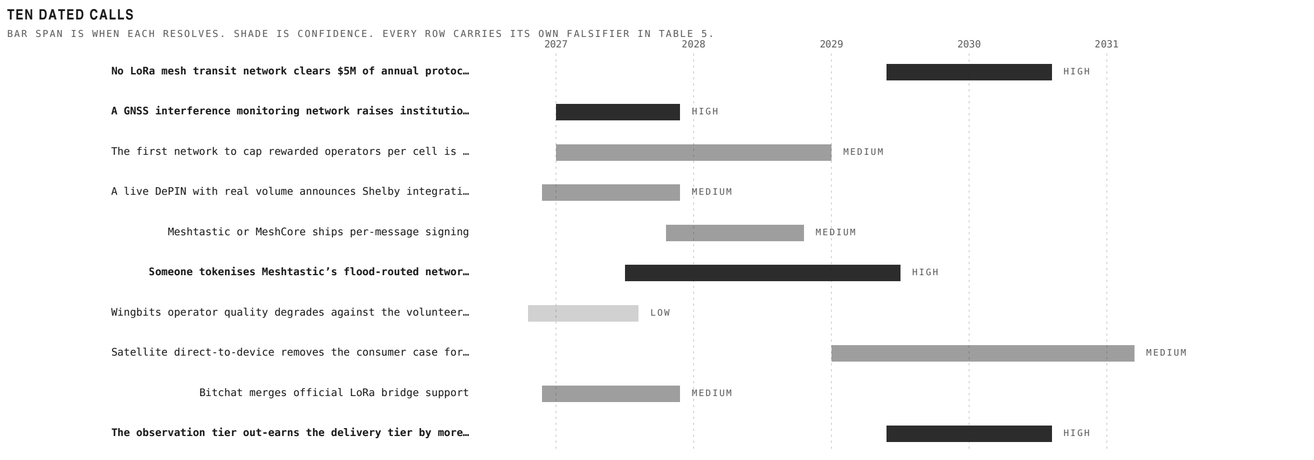
Helium’s cellular business survived precisely because it stopped selling coverage and started selling *offload*: bytes carried for a paying carrier, which is measured at the carrier’s own meter. That is a delivery-tier product wearing coverage-tier clothes, and it is the only part of Helium that retained value through the June 2026 divestiture of Helium Mobile. Section 12 returns to this.

5 The Positions

WHAT WE PREDICT, AND WHAT WOULD PROVE US WRONG

Everything after this section is the work that produced these calls. Each carries a date and a falsifier, so the thesis can be scored rather than admired.

FIGURE 2: TEN DATED CALLS



SOLID BARS ARE HIGH CONFIDENCE. THE FIRST AND LAST ROWS ARE THE THESIS: MESH TRANSIT STAYS SMALL AND RF OBSERVATION DOES NOT.

BAR SPAN IS WHEN EACH RESOLVES; SHADE IS CONFIDENCE. THE FIRST AND LAST ROWS ARE THE THESIS ITSELF.

TABLE 5: THE PREDICTIONS

PREDICTION	RESOLVES	CONFIDENCE	BASIS	WHAT WOULD FALSIFY IT
No LoRa mesh transit network clears $5M of annual protocol revenue	2030	**high**	capacity saturates at 6,721 nodes against $11.2M of total addressable demand	any paid mesh reporting audited protocol revenue above $5M
A GNSS interference monitoring network raises institutional capital	end 2027	**high**	IATA, FAA and EASA have published the gap and committed budget to closing it	no priced round in the category by December 2027
The first network to cap rewarded operators per cell is attacked for it, then copied	2027–29	medium	GEODNET, Hivemapper and Wingbits each arrived at geographic capping separately	no major network adopts hard per-cell caps
A live DePIN with real volume announces Shelby integration before mainnet	2027	medium	Shelby names DePIN data feeds as a target workload and is Aptos-native	Shelby reaches production with no DePIN network publicly integrated
Meshtastic or MeshCore ships per-message signing	2028	medium	no paid relay market can settle without it; the gap is now widely noticed	a paid mesh launches on unsigned channel traffic and is not gamed within a year
Someone tokenises Meshtastic’s flood-routed network directly, and it fails	within 24 months of launch	**high**	paying per rebroadcast in a network where every node rebroadcasts everything	a flood-routed paid mesh sustains positive operator margin past two years
Wingbits operator quality degrades against the volunteer ADS-B baseline	within two reward epochs	low	the crowding-out risk in re-intermediating unpaid contributors	measured feed quality holds or improves against FlightAware and ADS-B Exchange
Satellite direct-to-device removes the consumer case for off-grid messaging	2029–31	medium	65 MHz of exclusive spectrum, 15,000 filed satellites, 150 Mbps stated target	D2C stalls on handset silicon or regulatory access past 2031
Bitchat merges official LoRa bridge support	2027	medium	four independent bridges already ship; issue #973 is open	the project ships its own long-range transport instead, or declines both

6 What Is Being Built Right Now

AUGUST 2026

The category is usually described through its tokens. The more useful picture is what shipped this year, and who paid for it.

TABLE 6: LIVE AS OF AUGUST 2026

PROJECT	WHAT IT IS	STATUS	DETAIL	FUNDED BY
Bitle	LoRa added to the Bitchat protocol	shipping, MIT	solar ESP32-S3 relays, 2–10 km hops	**unpaid**
MeshCore-BitChat bridge	BLE-to-LoRa bridge firmware	shipping	$15–80 boards, #mesh hashtag channel	**unpaid**
MeshCore	source-routed mesh firmware	shipping since Jan 2025	3,324 commits, 214 contributors, 64 hops	**unpaid**
Meshtastic	managed-flood mesh firmware	shipping since Feb 2020	14,045 commits, busiest month ever Jul 2026	**unpaid**
Wingbits	ADS-B observation with hex rewards	TGE April 2026	$9.2M raised, cryptographically secured receivers	token
GEODNET	GNSS RTK corrections	live, ~$10M revenue	20,500 stations, area caps, halving schedule	token
DoubleZero Edge	shred subscription over private fibre	beta April 2026	$30–100/city/epoch, 10% of revenue burns 2Z	token + USDC
Shelby	hot storage with paid reads	Early Access testnet	RPC and storage-provider roles, Clay codes, Aptos-native	TBD
Xona Space Systems	LEO PNT constellation	$170M Series C Mar 2026	258 satellites, 42 mm demonstrated accuracy	equity
TrustPoint	C-band LEO PNT	2027 soft launch target	up to 300 satellites, $6.5M raised	equity
GPSPATRON	centralised interference detection	commercial	GP-Probe hardware plus GP-Cloud	enterprise
tri-net	four-arm DePIN incl. RF sensing	pre-launch	Zynq-7020 + AD9361, ETX routing, 0% VC	token, 0% allocation

STATUS IS WHAT SHIPPED, NOT WHAT WAS ANNOUNCED. THE FINAL COLUMN IS THE FINDING.

FOUR OF THE TWELVE ARE UNPAID

Meshtastic, MeshCore, Bitle and the Bitchat bridges are the entire physical and protocol layer for off-grid messaging, and every one of them is built by volunteers on $15 to $80 hardware. Meshtastic recorded its busiest month ever in July 2026, six years after first commit. **The supply side of this category is being built for free, at accelerating pace, by people with no token.** That is the opportunity and the warning at once: the infrastructure exists, and nothing about its existence proves anyone will pay for it.

THREE TRENDS WORTH NAMING

Architecture is moving from flood to structure. MeshCore shipped source routing in 2025 and the bridges built this year target it as often as Meshtastic. The direction of travel is toward exactly the property a settlement layer needs, driven by operators who want their networks to work rather than by anyone planning to charge for them.

Reward design is converging on marginal contribution. DoubleZero pays Shapley-weighted Proof of Utility. Shelby pays for serving data instead of holding it. GEODNET, Hivemapper and Wingbits all cap rewards per geographic cell. Four independent teams, four different problems, one rule. Nobody has yet combined it with a fixed issuance rate.

The money is going to resilience, not measurement. Xona raised $170M and TrustPoint is building a second constellation, both to replace a signal that can be jammed. Nobody is funding the network that would tell you where the jamming is, though the

7 The Observation Tier

WHERE THE REVENUE ALREADY IS

Coverage fails because nothing checks it. Delivery is checkable against a counterparty, and §15 through §22 price that tier in detail and find it capped. Observation is checkable against physics, and it is the only tier of the three with networks already producing revenue at scale.

THE EXTERNAL REFERENCE

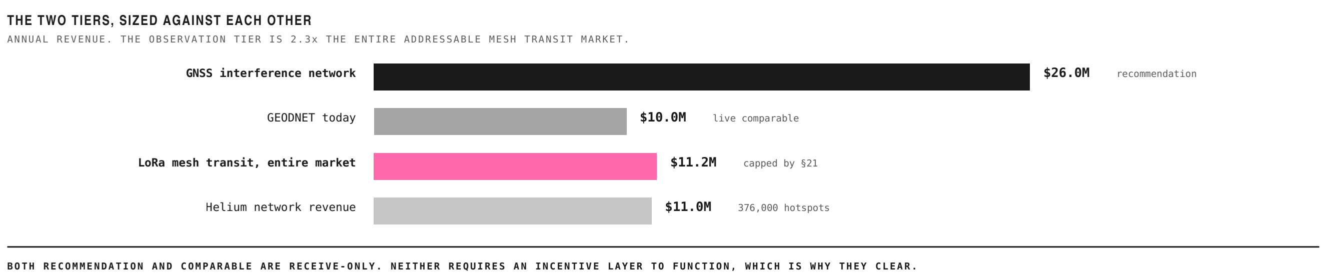
Aircraft broadcast GPS position on 1090 MHz, so a receiver's claim is cross-checkable against every other receiver that heard the same aircraft and against multilateration timing. Fabricating a track consistent with dozens of independent receivers' timing geometry is not a spoofing problem but an orbital mechanics problem. Satellite ephemeris is published, so a GNSS station's claimed observations must agree with where the constellation was. Beacon transmitters announce power, frequency and schedule in advance.

This explains why GEODNET works and Helium's IoT network did not, and the explanation is not the one usually offered. **GEODNET's operators never transmit.** They receive GNSS signals and upload corrections checkable against public ephemeris. It is a receive-only radio network, and it produces the highest-quality revenue in decentralized wireless: roughly $10 million annualised across 20,500 base stations.

RECEIVE-ONLY ESCAPES EVERY CONSTRAINT ON THE DELIVERY TIER

No licence is required to operate a receiver. There is no duty cycle on listening, no encryption restriction on data one did not transmit, and no power ceiling. 47 CFR §97.113 governs what amateur stations *transmit*; §20 shows that provision permanently confines any incentivised mesh to unlicensed spectrum. None of it applies here. Neither does the airtime ceiling in §18, because a receiver consumes no channel time at all.

FIGURE 2: THE TWO TIERS, SIZED AGAINST EACH OTHER



ANNUAL REVENUE. THE RECOMMENDATION IS 2.3 TIMES THE ENTIRE ADDRESSABLE MESH TRANSIT MARKET, AND HAS A LIVE COMPARABLE TRADING AT $85M.

DEMAND IS PROVEN BY ACQUISITION, NOT BY ARGUMENT

The observation tier does not require a demand thesis. It has had two liquidity events in which the supply side received nothing. FlightAware operates more than 40,000 volunteer ADS-B receivers across 193 countries whose contributors are compensated with a free premium account; the company is now part of Collins Aerospace. ADS-B Exchange ran roughly 14,000 volunteer feeds under a royalty-free, worldwide, perpetual and irrevocable data licence before being acquired by JETNET. The same pattern holds at MarineTraffic, WSPRnet, PSKReporter, the Reverse Beacon Network and SatNOGS.

> **WHAT THAT PATTERN IS WORTH**
>
> These are functioning global networks with tens of thousands of hardware operators, real enterprise customers and demonstrated exit value, in which the people who built the physical layer own none of it. That is the clearest demand evidence available anywhere in this category. It is also the clearest statement of the opportunity: the supply side of RF observation has never been compensated, and the question is which application is worth re-intermediating first. §6 answers it.

8 GNSS Interference Monitoring

THE RECOMMENDATION

ADS-B is taken: Wingbits raised $9.2 million and launched its token in April 2026 with cryptographically secured receivers and hex-based rewards. One application in the observation tier remains uncontested, and it sits on a capability gap that regulators have publicly stated and publicly committed to closing.

THE GAP, IN THE REGULATORS' OWN WORDS

The International Air Transport Association measured a **65 percent year-on-year increase** in loss-of-GNSS events per thousand flights during the first half of 2024. Affected airspace now spans the eastern Mediterranean, the Black Sea, the Russia-Baltic region, the India-Pakistan border, Iraq, Iran and the Korean peninsula. The FAA has issued a substantially revised interference resource guide. EASA and Eurocontrol have published a joint action plan explicitly aimed at pooling monitoring data into a shared operational picture.

And per SKYbrary's summary of the operational position: *it is not currently possible to detect affected areas from a distance, so pilot reports remain the main source of information.* Existing detection work is largely *inferred* from ADS-B anomalies, reconstructing interference from aircraft behaving strangely, .

> **WHY THIS IS THE RECOMMENDATION**
>
> A stated capability gap, in a safety-critical system, with documented exponential growth in incidents, where the regulators have already committed budget to building the exact data layer that does not exist. The supply side is a $260 bill of materials and the same operator profile GEODNET has already recruited 20,500 times. No incentive layer is required for the network to function, which removes the single hardest problem in the delivery tier.

FIGURE 3: WHERE THE INCUMBENTS ARE, AND WHERE THEY ARE NOT

VENDORS PLOTTED BY WHAT THEY SELL AND HOW THEY DEPLOY IT. THE SHADED QUADRANT IS DISTRIBUTED MEASUREMENT AT DENSITY, WHICH NOBODY OCCUPIES.

TABLE 1: WHO SELLS WHAT

VENDOR	PRODUCT	FORM	BUYER	POSITION
Septentrio	AIM+ interference mitigation	receiver firmware	OEM / integrator	mitigates, does not map
Safran	IDM detection and countermeasure suite	fixed-site systems	defence, government	site protection, not coverage
Hexagon / NovAtel	resilient PNT hardware	receivers and antennas	survey, defence	mitigation layer
BAE, Thales, Northrop	anti-jam antenna arrays	platform integration	defence primes	beamforming, not measurement
GPSPATRON	GP-Probe + GP-Cloud	centralised probe network	critical infrastructure	closest analogue; enterprise-owned nodes
GMV (STAGER / SILENT)	COTS receivers, C/N0, AGC, spectrum, carrier phase	institutional pilot	research, infrastructure	validates the technique
A distributed network	crowdsourced measurement at density	none	—	**does not exist**

9 How the Network Works

NODE DESIGN AND VERIFICATION

The technique is not speculative. GMV's STAGER system, published in 2026, builds interference detection from commercial off-the-shelf multi-constellation receivers using exactly the signal-processing methods below, and estimates angle of arrival when several units are deployed across a region. What does not exist is anyone running those nodes at civilian density.

TABLE 2: WHAT A NODE MEASURES

TECHNIQUE	WHAT IT OBSERVES	DETECTS	REQUIRES
Carrier-to-noise density (C/N0)	per-satellite signal quality drop	jamming	commodity receiver
Automatic gain control trend	front-end compensating for a strong emitter	jamming	commodity receiver
RF spectrum monitoring	the interfering waveform itself	jamming, classification	SDR front end
Carrier-phase double differences	dispersion inconsistent with geometry	**spoofing**	multi-constellation receiver
Ephemeris cross-check	claimed observation against published orbit	**spoofing, and node honesty**	network-side
Time difference of arrival	emitter position from multiple nodes	localisation	≥ 3 nodes, disciplined clock

THE FIRST FOUR ROWS RUN ON THE NODE. THE FIFTH RUNS NETWORK-SIDE AND IS ALSO THE HONESTY CHECK. THE SIXTH REQUIRES THREE OR MORE NODES AND A DISCIPLINED CLOCK.

VERIFICATION, WHICH IS THE POINT OF THE WHOLE FRAMEWORK

A node claims to have observed a set of satellites with particular carrier-to-noise ratios at a particular instant. That claim must agree with published ephemeris, which is external, and with every other node observing the same satellites at the same time, which is adversarial. A node fabricating observations must fabricate a self-consistent view of the constellation that also agrees with its neighbours. Contrast the delivery tier, where §21 shows collusion is bounded by staking economics bound is weakest exactly at the price point where the business works. Here the bound is orbital mechanics.

TABLE 3: BILL OF MATERIALS

COMPONENT	COST	NOTE
Multi-constellation GNSS receiver	**$38**	u-blox ZED-F9P class
Survey antenna	**$55**	multi-band, with ground plane
SDR front end for spectrum	**$30**	RTL-SDR class, L-band downconvert
Single-board computer	**$45**	Pi-class, or an ESP32 for the reduced node
GPS-disciplined oscillator	**$60**	required only for TDOA nodes
Enclosure, cabling, mount	**$32**	outdoor rated
Total, full node	**$260**	$200 without the disciplined oscillator, for detection-only sites

COMPONENT PRICING AT SMALL VOLUME. THE REDUCED NODE OMITS THE DISCIPLINED OSCILLATOR AND CONTRIBUTES DETECTION BUT NOT LOCALISATION.

At $260 the node costs a fraction of a GEODNET base station and shares its operator profile: roof access, sky view, power outlet. That is the population GEODNET has already recruited 20,500 times across 150 countries. **The supply-side acquisition problem is solved by precedent.**

WHAT THE NETWORK SELLS

Not devices and not raw observations. A regional interference picture: where, when, what waveform, how strong, and with

10 Sizing the Recommendation

BOTTOM-UP, AND A CORRECTION

TABLE 4: ADDRESSABLE REVENUE BY SEGMENT

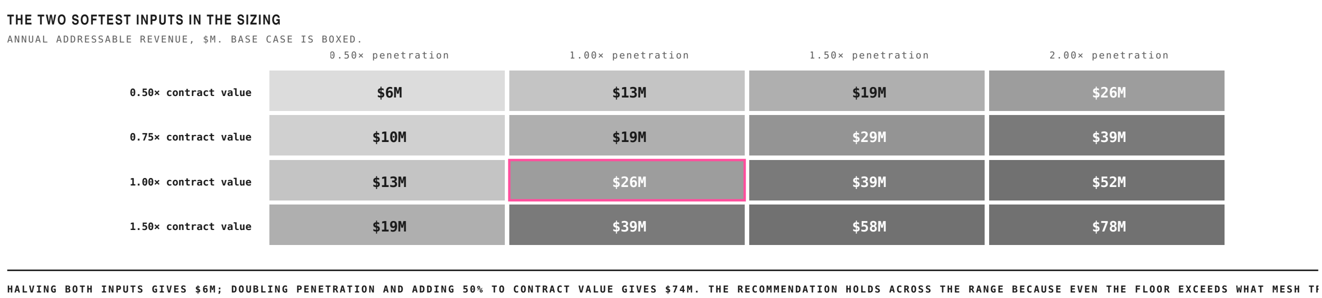
SEGMENT	ACCOUNTS	ACV	PENETRATION	REVENUE	BASIS
Airlines and air navigation providers	520	$85,000	15%	**$6.6M**	safety of life; IATA has called for urgent action
Aviation and marine insurance	60	$220,000	20%	**$2.6M**	hull, delay and liability underwriting
Maritime and port operations	300	$60,000	12%	**$2.2M**	AIS spoofing, sanctions evasion, port safety
Critical-infrastructure timing	400	$75,000	10%	**$3.0M**	grid, telecom and exchange timestamping
Defence and national PNT offices	45	$900,000	25%	**$10.1M**	situational awareness and attribution
Receiver OEM data licensing	12	$400,000	30%	**$1.4M**	feed sold into Septentrio, Safran, Hexagon product lines
Total addressable	–	–	–	**$26.0M**	2.3× the entire mesh transit market in §21

ACCOUNT COUNTS ARE ADDRESSABLE ORGANISATIONS. ANNUAL CONTRACT VALUES ARE SET FROM COMPARABLE ENTERPRISE DATA SUBSCRIPTIONS IN AVIATION AND DEFENCE. PENETRATION IS AT MATURITY, NOT AT LAUNCH.

The total is **$26.0 million a year**, roughly 2.3 times the entire addressable mesh transit market computed in §24 and about 2.6 times GEODNET's current annualised revenue on the same hardware class and operator profile.

> **A CORRECTION TO OUR OWN EARLIER ESTIMATE**
>
> An earlier draft of this thesis sized this line at $40 to $120 million on the basis of segment intuition. Constructing it from account counts and contract values gives $26 million, which is lower than the bottom of that range. We report the smaller figure. The recommendation does not depend on the larger one and the earlier estimate should not have been offered.

FIGURE 5: THE TWO SOFTEST INPUTS, VARIED TOGETHER



ANNUAL ADDRESSABLE REVENUE ACROSS CONTRACT VALUE AND PENETRATION MULTIPLES. THE RECOMMENDATION SURVIVES THE FLOOR OF THE RANGE.

SENSITIVITY, BECAUSE THESE ARE OUR ASSUMPTIONS

Contract value and penetration are the two numbers we set ourselves and they are the first a reader should push on. Halving both gives $6 million; doubling penetration and adding half again to contract value gives $74 million. **The recommendation does not turn on the base case: even the floor of that range exceeds what mesh transit can produce at any node count.** What would change the conclusion is not a smaller multiple but a demonstration that aviation and defence buyers will not purchase interference data at all, which is condition 4 in §33.

WHY THE COMPARABLE IS UNUSUALLY TIGHT

GEODNET is not an analogy here, it is the same business with a different data product. Same L-band receivers, same rooftop operator, same receive-only regulatory posture, same requirement for geographic density, same tolerance for consumer-grade

11 Who We Would Fund

TEAM PROFILE AND SOURCING

A recommendation without a team is a research note. This section states the profile, where those people currently are, and what we would do if none of them wants to build it.

TABLE 5: REQUIRED CAPABILITIES

CAPABILITY	WEIGHT	WHY IT MATTERS HERE
RF and GNSS signal processing	must	C/N0, AGC and carrier-phase analysis is the product, not the wrapper
Regulated-buyer sales	must	aviation and defence procurement is the go-to-market, not developer adoption
Distributed-systems and attestation	must	node honesty is an ephemeris cross-check problem, not a consensus problem
Token design under saturation	must	the emission rules in §10; most teams get this exactly backwards
Hardware manufacturing at small scale	helpful	a $260 BOM can be contract-manufactured; not a differentiator
Crypto-native distribution	helpful	matters less here than in any other DePIN category

THE FOUR "MUST" ROWS ARE UNUSUAL IN COMBINATION, WHICH IS WHY THE COMPANY DOES NOT EXIST YET.

The binding constraint is the combination. RF signal processing and regulated-buyer sales rarely appear in the same founding team, and crypto-native distribution, which is the strength most DePIN teams lead with, matters less here than in any other category in this report. A team that is excellent at community growth and weak at aviation procurement will build a network nobody buys.

THE SHAPE OF THE RIGHT FOUNDER

The archetype we would back has spent time inside the problem it: a GNSS or SDR engineer from a receiver vendor, a national PNT office or a defence prime, who has watched the interference problem worsen and has concluded that the measurement layer is missing. That person will not arrive fluent in tokenomics, and §18 is short enough to teach. The reverse pairing, a crypto founder who has read about GNSS jamming, is far more common and far worse, because the technical claims in §7 are the product and cannot be outsourced.

The second seat is commercial and it is unusual. Aviation and defence procurement runs on multi-year cycles, framework agreements and security clearances, and rewards patience over growth-hacking. A founder who has closed a contract with an air navigation service provider or a national PNT office is worth more to this company than one who has grown a Discord to a hundred thousand members, which inverts the usual DePIN ranking.

11 Who We Would Fund

CONTINUED: SOURCING

TABLE 6: WHERE THOSE PEOPLE ARE

POOL	TYPE	CONVICTION	NOTE
GEODNET operators and staff	adjacent	**highest**	already run receive-only GNSS hardware at 20,500 sites; nearest possible team
Wingbits team and alumni	adjacent	**high**	shipped a cryptographically-secured receiver network and a TGE
GPSPATRON, GMV interference groups	incumbent	**high**	own the detection technique; would need to be convinced to decentralise
Septentrio / u-blox firmware engineers	incumbent	medium	the receiver expertise; unlikely to leave for a token network
SDR and amateur radio builders	community	medium	the tri-net author is the archetype: capable, uninterested in capital
Defence-tech founders post-exit	adjacent	medium	goTenna and Forterra alumni understand the buyer

CONVICTION IS OURS. THE FIRST TWO ROWS ARE WHERE WE WOULD START AND ARE REACHABLE THROUGH EXISTING PORTFOLIO RELATIONSHIPS.

THE HONEST POSITION ON SOURCING

We have not found a company building this. The absence is the finding. The detection technique is published, the buyers have stated the need, the hardware is commodity, and the operator population has been recruited before by an adjacent network, and still nobody is doing it. The most likely explanation is that the people who understand GNSS signal processing work at Septentrio and Safran, and the people who understand decentralized supply-side recruitment work in crypto, and those two groups do not meet.

> **WHAT WE WOULD ACTUALLY DO**
>
> Incubate rather than wait. The GEODNET operator base is the single highest-value introduction available, the node design in §7 is a weekend of engineering away from a working prototype on commodity parts, and the first customer conversation is a call to a national PNT office that has already published the gap. If the right team exists we would rather find them than build one. If they do not, this is a company we would help start, and the willingness to do that is the difference between a recommendation and an intention.

WHAT WOULD MAKE US PASS ON A SPECIFIC TEAM

Leading with token design before a buyer conversation. Proposing to sell devices rather than data. Treating node count as the growth metric, which §10 shows is the precise error that produced every failure in this category. Or an inability to explain why ephemeris cross-checking makes node honesty tractable, which is the one technical question that separates people who have thought about this from people who have read about it.

12 Token Design Under Saturation

THE MECHANISM THIS CATEGORY KEEPS GETTING WRONG

Every incentive failure documented in this report has the same shape: emissions weighted by deployment, in a market where deployment past a threshold produces no additional revenue. The saturation result derived in §25 is the general form of it, and it applies to the observation tier as directly as to the delivery tier, because coverage of an already-covered area is worth nothing in both.

TABLE 7: HOW EACH NETWORK GOT THIS WRONG

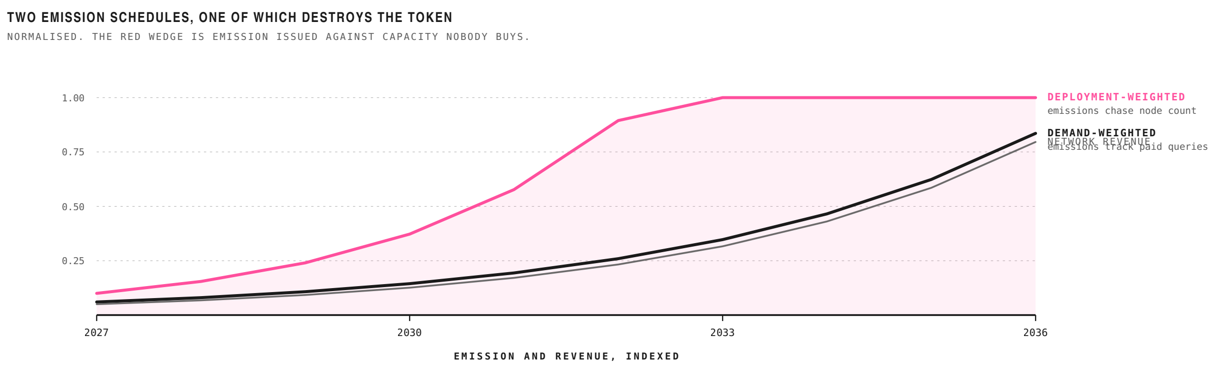
NETWORK	EMISSION DESIGN	WHAT HAPPENED	OUTCOME
Helium (IoT)	emissions per hotspot, coverage-weighted	**500k gateways against a few thousand needed**	revenue per device fell to ~$0.16/month
Helium (Mobile)	emissions per radio, boosted hexes	**denylist, boosted-hex gaming**	discretionary burn introduced to defend the token
Hivemapper	emissions per image, then per hex	**coverage saturation in mapped areas**	MIP-24 moved to hex-based freshness
Crankk	proof-of-network-participation	**no demand to meter against**	1,877-day payback
GEODNET	emissions per station, halving schedule	handled by capping stations per area	burn from real usage; nearest to net-positive

GEODNET IS THE PARTIAL EXCEPTION AND THE MECHANISM IT USES, CAPPING STATIONS PER AREA AND BURNING AGAINST REAL QUERIES, IS THE ONE TO COPY.

WHY THIS IS A SECTION AND NOT A PARAGRAPH

Emission design is usually treated as a launch decision, chosen late and tuned by governance afterwards. In this category it is the primary determinant of whether the network survives contact with its own growth, and it cannot be tuned afterwards because the operators who joined under the old schedule will vote against changing it. Helium needed a denylist, discretionary burns and eventually a sub-DAO merger to unwind decisions taken at genesis. Hivemapper needed MIP-24. **Both corrections were expensive and both were foreseeable from the saturation arithmetic before either network launched.**

FIGURE 4: TWO EMISSION SCHEDULES, ONE OF WHICH DESTROYS THE TOKEN



NORMALISED AND ILLUSTRATIVE. THE RED WEDGE IS EMISSION ISSUED AGAINST CAPACITY NOBODY BUYS, WHICH IS THE MECHANICAL FORM OF THE SATURATION RESULT IN §22.

12 Token Design Under Saturation

CONTINUED: THE DESIGN WE WOULD REQUIRE

TABLE 8: THE DESIGN WE WOULD REQUIRE

RULE	MECHANISM	WHY
Cap emissions per geographic cell	one rewarded node per H3 cell at the resolution matching sensor range	Wingbits and Hivemapper both arrived here; adopt it from the start
Weight by marginal information, not presence	a node in an already-covered cell earns near zero regardless of uptime	the direct answer to the saturation result
Meter demand, not supply	emissions scale with paid queries against the cell, not with node count	inverts the DeWi default and is the core design claim
Burn on use, at a fixed fiat-denominated rate	buyers burn to query; the burn is the only sink	Helium's data-credit mechanism, which is the part that worked
No emissions before a paying buyer exists	genesis gated on first contract, not on first node	avoids subsidising supply into a market that has not appeared

THE THIRD RULE IS THE LOAD-BEARING ONE AND INVERTS THE DEWI DEFAULT. THE FIFTH IS THE ONE MOST TEAMS WILL RESIST.

WHY DEMAND-WEIGHTING IS THE WHOLE DESIGN

Under deployment weighting, an operator's rational strategy is to add nodes wherever emissions are highest, which is wherever coverage is thinnest, which is often where no buyer exists. The network grows, the token inflates, and revenue per node falls monotonically. Helium reached 376,000 hotspots and roughly $29 of annual revenue per device; that is not a failure of execution, it is the emission schedule working as designed.

Under demand weighting, emissions for a geographic cell scale with paid queries against that cell. An operator in an uncovered area with buyers earns; an operator in a covered area earns almost nothing regardless of uptime; an operator in an area with no buyer earns nothing at all until one appears. Supply follows revenue instead of preceding it. The cost is a slower and less exciting growth curve, which is precisely why no network has shipped it.

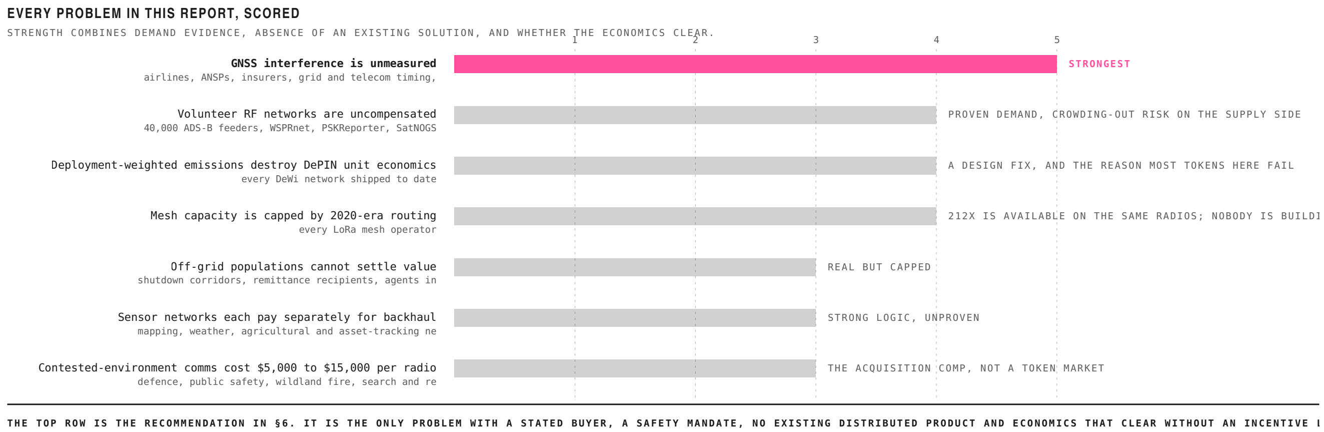
> **THE GENESIS RULE**
>
> No emissions before a paying buyer exists. Every network in Table 7 launched emissions to bootstrap supply on the assumption that demand would follow, and in four of five cases it did not follow fast enough to matter. For an observation network with a stated regulatory buyer, the sequencing can be inverted: sign one national PNT office or aviation authority, then emit against that contract's coverage requirement. This is slower, it is less fundable by conventional DePIN standards, and it is the single design decision most likely to determine whether the token accrues value.

13 The Problem Map

EVERYTHING THIS REPORT IDENTIFIES, RANKED

A thesis is only as good as the problem it attaches to. This section lists every problem surfaced in this report, who has it, what they do about it today, and what could be built. Scoring combines demand evidence, absence of an existing product, and whether the economics clear without an incentive layer.

FIGURE 7: EVERY PROBLEM, SCORED



ONE PROBLEM SCORES FIVE. IT IS THE ONLY ONE WITH A STATED BUYER, A SAFETY MANDATE, NO DISTRIBUTED PRODUCT IN MARKET, AND ECONOMICS THAT CLEAR WITHOUT SUBSIDY.

TABLE 9: PROBLEMS AND CANDIDATE SOLUTIONS

PROBLEM	WHO HAS IT	WHAT THEY DO TODAY	CANDIDATE SOLUTIONS	STRENGTH
GNSS interference is unmeasured	airlines, ANSPs, insurers, grid and telecom timing, defence	pilot reports; interference inferred from ADS-B anomalies after the fact	distributed receive-only measurement network; LEO PNT replacement signal; per-site mitigation hardware	5/5
Volunteer RF networks are uncompensated	40,000 ADS-B feeders, WSPRnet, PSKReporter, SatNOGS operators	a free premium account; two acquisitions produced nothing for them	token-incentivised observation networks	4/5
Deployment-weighted emissions destroy DePIN unit economics	every DeWi network shipped to date	Helium: 376,000 hotspots against $29 of revenue each	demand-weighted emission with a genesis gate	4/5
Mesh capacity is capped by 2020-era routing	every LoRa mesh operator	163 delivered messages per hour per collision domain	source routing, scheduled access, multi-channel reuse, network coding	4/5
Off-grid populations cannot settle value	shutdown corridors, remittance recipients, agents in low-connectivity markets	cash, informal agents, or nothing during a shutdown	LoRa mesh settlement rail; satellite direct-to-device; Bluetooth mesh relay	3/5
Sensor networks each pay separately for backhaul	mapping, weather, agricultural and asset-tracking networks	$19/month per device for LTE, or a gateway and a contract	shared mesh substrate carrying multiple payloads	3/5
Contested-environment comms cost $5,000 to $15,000 per radio	defence, public safety, wildland fire, search and rescue	tactical MANET radios, or nothing at the low end	commodity LoRa mesh as a supplement; goTenna Pro class hardware	3/5

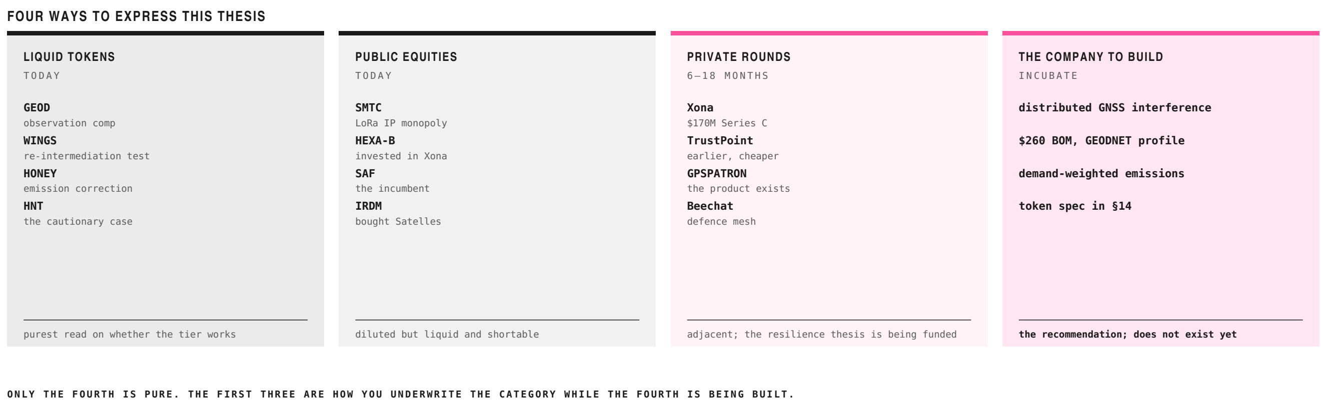
SCORES ARE OURS AND ARE THE FIRST THING TO ARGUE WITH. THE COLUMNS BEFORE THEM ARE FACTUAL AND SHOULD BE CHECKED SEPARATELY.

14 What to Buy Today

LIQUID TOKENS AND PUBLIC EQUITIES

The recommendation in §6 is a company that does not exist. Between now and its existence there are four ways to hold this thesis, and two of them are liquid.

FIGURE 8: FOUR WAYS TO EXPRESS THE THESIS



ONLY THE FOURTH IS PURE EXPOSURE. THE FIRST THREE ARE HOW THE CATEGORY IS UNDERWRITTEN WHILE THE FOURTH IS BUILT.

TABLE 10: LIQUID TOKENS

TICKER	NAME	TIER	CURRENT STATE	WHY HOLD IT	CONVICTION
GEOD	GEODNET	observation tier, receive-only GNSS	~$10M revenue, ~20,500 stations, ~$85M cap	the closest listed comparable to the recommendation	highest
HNT	Helium	coverage tier, pivoting to wholesale offload	~$11M revenue, 376,000 hotspots, ~$177M cap	the cautionary case; the offload pivot is the delivery-tier trade	medium
HONEY	Hivemapper	observation tier, street imagery	100,000 devices, hex-based emission after MIP-24	the emission-design correction already executed	medium
WINGS	Wingbits	observation tier, ADS-B	$9.2M raised, TGE April 2026	direct read on whether re-intermediating volunteers works	high
2Z	DoubleZero	transit layer, chain agnostic	live since Oct 2025, SEC no-action letter, 70+ links, 25+ cities	exposure to every network that adopts the stack, not to one of them	high
APT	Aptos	settlement layer	the chain this report recommends for the permissionless leg	indirect and diluted; not a way to own this thesis	low

CONVICTION IS OURS. GEOD IS THE ONLY LISTED ASSET THAT IS ESSENTIALLY THE SAME BUSINESS AS THE RECOMMENDATION WITH A DIFFERENT DATA PRODUCT.

GEOD is the highest-conviction liquid position. Same L-band receivers, same rooftop operator, same receive-only regulatory posture, same requirement for geographic density. It trades near $85 million on roughly $10 million of revenue. If the observation-tier thesis is right, GEOD re-rates before anything we fund does, and if it does not re-rate that is early evidence against the framework.

WINGS is the experiment. Wingbits is the first network to attempt paying volunteers who were previously uncompensated in an observation market. Whether that works is condition 4 in §35 and the largest unknown in this report. A small position is the cheapest way to buy information about it.

14 What to Buy Today

CONTINUED: PUBLIC EQUITIES

TABLE 11: PUBLIC EQUITIES

TICKER	COMPANY	EXPOSURE	HOW IT MAPS	LISTING	NOTE
SMTC	Semtech	LoRa IP owner and chipset monopoly	every LoRa mesh node ships an SX126x; FY26 revenue $1.05B, LoRa growing ~20%/yr	NASDAQ	highest-purity listed exposure
HEXA-B	Hexagon AB	NovAtel receivers, resilient PNT	sells the mitigation layer and invested in Xona's Series C	Stockholm	strategic, already positioned
SAF	Safran	IDM interference detection suite, PNT over fibre	the incumbent this thesis competes with on measurement	Euronext Paris	the acquirer, not the disruptee
TRMB	Trimble	precision positioning, correction services	GEODNET's incumbent competitor; exposed to RTK displacement	NASDAQ	short-side expression of the DePIN thesis
IRDM	Iridium	acquired Satelles for LEO secure timing	$39M into exactly the PNT-resilience problem	NASDAQ	already executing the adjacent trade
RTX	RTX / Collins Aerospace	owns FlightAware	the buyer in the observation-tier acquisition precedent	NYSE	too diversified for clean exposure
U-BLOX	u-blox	GNSS receiver modules	the $38 line item in the node bill of materials	SIX Swiss	component supplier to whatever gets built
ASTS	AST SpaceMobile	direct-to-device satellite	the bear case in §35 expressed long	NASDAQ	hedge against the mesh consumer case

EXPOSURE HERE IS DILUTED BY DEFINITION: NONE OF THESE COMPANIES IS A PURE PLAY. THE LIST IS ORDERED BY PURITY, NOT BY CONVICTION.

Semtech is the highest-purity listed exposure to LoRa mesh. It owns the intellectual property, and every node in every network in this report ships an SX126x. Fiscal 2026 revenue was $1.05 billion against $909 million the prior year, with management guiding LoRa growth near 20 percent annually on Amazon Sidewalk and industrial expansion. The caveat is that LoRa is one of three segments and the stock now trades substantially on a data-centre interconnect story that has nothing to do with this thesis.

Trimble is the short-side expression. If decentralised correction networks take share in precision positioning, the incumbent selling RTK subscriptions is where it comes from. We would not put this on, but a fund that believes the GEODNET model generalises should know where the other side of it sits.

15 Private Markets

WHO IS ALREADY FUNDED, AND WHAT IS MISSING

The resilience half of this problem is being funded aggressively. The measurement half is not being funded at all. That asymmetry is the opportunity and it is also the risk.

TABLE 12: THE LEO PNT CONTEXT

METRIC	VALUE	SOURCE
Market size 2026	**$154M**	Meticulous Research
CAGR to 2036	**32.4%**	Meticulous Research
Alternative estimate	**$70M 2025 to $570M 2030**	53.9% CAGR
Xona Series C	**$170M**	March 2026, led by Mohari
Satelles acquisition	**$39M into Iridium**	LEO secure timing
Our GNSS interference estimate	**$26M**	§8, subscription revenue only

OUR ESTIMATE IN THE FINAL ROW IS SUBSCRIPTION REVENUE FOR INTERFERENCE DATA ONLY. IT SITS INSIDE A MARKET THAT VENTURE INVESTORS ARE FUNDING AT $170M A ROUND.

Xona Space Systems raised a $170 million Series C in March 2026, taking total funding past $320 million, with Hexagon and Samsung Next on the cap table alongside Craft and ICONIQ. It is building a 258-satellite constellation broadcasting a signal roughly a hundred times stronger than GPS, and its first demonstrator reported 42-millimetre on-orbit accuracy with anti-spoofing. TrustPoint is doing the same in C-band with far less capital. Iridium acquired Satelles for LEO secure timing.

SUBSTITUTE OR COMPLEMENT

The obvious objection: if Xona ships an unjammable signal, who needs a map of the jamming? The answer is that resilience and situational awareness are different products with different buyers. A resilient receiver protects one platform. It does not tell an air navigation service provider which sectors are affected, does not support an insurer's underwriting, does not attribute an emitter, and does not exist on the hundreds of millions of legacy receivers already deployed. **Xona is the strongest evidence that this problem is real and fundable, and it does not address the same buyer.** If anything a LEO PNT constellation increases demand for interference data, because it creates a commercial reason to know exactly where the legacy signal fails.

15 Private Markets

CONTINUED: WHERE WE WOULD GO

TABLE 13: PRIVATE OPPORTUNITIES

COMPANY	WHAT IT DOES	STATE	RELEVANCE	ACCESS
Xona Space Systems	LEO PNT constellation, 258 satellites	$170M Series C, March 2026; total >$320M; Hexagon and Samsung Next on the cap table	adjacent and complementary: a resilient signal still needs interference mapping	hard to enter, priced
TrustPoint	C-band LEO PNT, up to 300 satellites	$6.5M raised; 2027 soft launch	earlier and cheaper entry into the same resilience thesis	accessible, higher risk
GPSPATRON	GP-Probe and GP-Cloud interference detection	centralised enterprise probe network	the closest existing product; the decentralisation is the missing piece	acquire or partner, not venture-shaped
oneNav	L5-band GNSS receiver	next-generation receiver architecture	component-level exposure to the same problem	component play
Xairos	secure time distribution	quantum time transfer for PNT resilience	the timing half of the same threat	early, technical
Beechat Network Systems	Reticulum mesh, NATO DIANA 2026	post-quantum hardened tactical mesh, MAVLink over Reticulum	the defence expression of the delivery tier	defence-procurement timelines
The company that does not exist	distributed GNSS interference measurement	$260 BOM, GEODNET operator profile, demand-weighted emissions	the recommendation in §6; we would incubate	**to be built**

THE FINAL ROW IS THE RECOMMENDATION. THE ROWS ABOVE IT ARE HOW THE THESIS IS HELD IF THAT COMPANY CANNOT BE FOUND OR BUILT.

Our ranking within the private set: **TrustPoint before Xona** on entry price for the same thesis, **GPSPATRON as an acquisition** because the product exists and the decentralisation is the missing piece, and **Beechat** as the defence expression of the delivery tier if the NATO DIANA programme converts. None of these is the recommendation. All of them are ways to be in the room while the recommendation is built.

16 What the Token Has to Look Like

A LAUNCHABLE SPECIFICATION

§10 states the emission rules the saturation result forces. This section turns them into a specification a team could build against and an investor could underwrite, because the difference between the two is where most DePIN diligence fails.

TABLE 14: THE SPECIFICATION

PARAMETER	DESIGN	EFFECT	WHY
Emission rate	fixed per epoch, ORE-style	issuance does not scale with node count, so growth dilutes rather than inflates	directly answers the saturation result in §28
Emission distribution	by marginal information value per H3 cell	a node in a covered cell earns near zero regardless of uptime	coverage of covered ground is worth nothing
Demand coupling	emission per cell scales with paid queries against that cell	supply follows revenue instead of preceding it	inverts the DeWi default; the core design claim
Sink	burn-to-query at a fiat-denominated rate, HNT-style	buyers burn for non-transferable credits; they never hold a volatile asset	the one Helium mechanism that worked
Genesis gate	no emission before the first paying contract	the network starts when a buyer exists, not when a node does	the rule most teams will resist
Node identity	Move resource, non-copyable, tied to a hardware key	registration cannot be duplicated or discarded	Sybil resistance at the type level
Verification	ephemeris cross-check plus inter-node agreement	a fabricated observation must be consistent with the constellation and neighbours	the tier-3 property from §3
Supply	fixed cap, no premine, no VC allocation at genesis	investor exposure via equity in the operating company, not a token allocation	avoids the tri-net problem where 0% VC made it uninvestable

ROWS ONE THROUGH THREE ARE THE ECONOMIC CORE AND ARE NON-NEGOTIABLE. ROWS FOUR AND FIVE ARE THE SEQUENCING. ROWS SIX THROUGH EIGHT ARE STRUCTURE.

READING THE TABLE

Rows one through three are the economic core and we would not fund a team that negotiated on them. Rows four and five are sequencing, and they are where most argument will land, because both delay the token. Rows six through eight are structural and are the cheapest to get right if chosen at the outset and the most expensive to retrofit.

WHERE THE MECHANISMS COME FROM

Two existing tokens each solved half of this and neither solved both. **ORE contributed the emission rate:** fixed issuance per unit time split competitively, so additional participants dilute each other. That is automatically the answer to the saturation result, and it is the mechanism Helium most conspicuously lacked while emitting per device to 376,000 hotspots.

Helium contributed the sink: buyers burn the token for non-transferable credits at a fixed fiat-denominated rate, so demand comes from usage and the buyer never holds a volatile asset. That mechanism worked, and it is the part worth copying. The coverage-weighted emission wrapped around it is the part that did not.

A network combining ORE's issuance with Helium's sink, distributed by marginal information value, and gated on a paying contract at genesis, is a design none of the five networks in Table 7 has shipped. It is slower and less exciting than the alternatives, which is precisely why.

16 What the Token Has to Look Like

CONTINUED: ANTI-PATTERNS

TABLE 15: WHAT WE WOULD REFUSE TO FUND

ANTI-PATTERN	WHO DID IT	OUTCOME
Emission per device	**Helium**	376,000 hotspots, $29 revenue each
Coverage-weighted rewards	**Helium Mobile**	denylist, boosted-hex gaming
Rewards before demand	**Crankk**	1,877-day payback
No investor allocation at all	**tri-net**	structurally uninvestable
Token as the product	**most DePIN pitches**	the data is the product

EACH ROW IS A DESIGN THAT HAS ALREADY BEEN RUN AND HAS ALREADY PRODUCED THE OUTCOME IN THE FINAL COLUMN.

WHY THESE FIVE AND NOT OTHERS

Each row is a design that has already been run by a network with real capital and real operators, and each produced the outcome in the final column within three years. We list no hypothetical failures. A team proposing any of the five should be asked which of the precedents they believe does not apply to them, and the answer is diagnostic: the good answer is specific and mechanical, the bad answer is that their community is different.

ON THE INVESTOR ALLOCATION

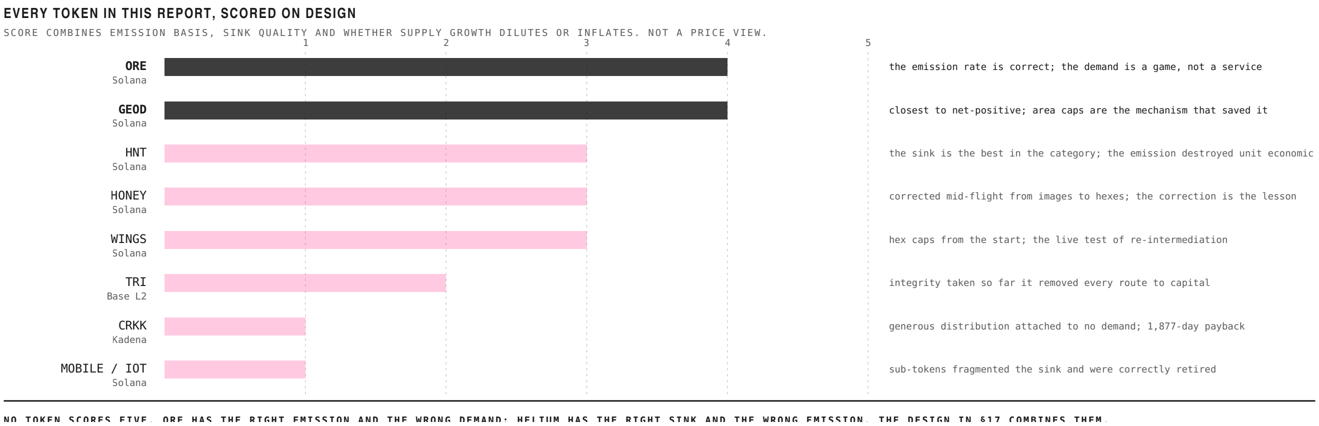
The last row of the specification is the one that will generate the most argument. We would take exposure through equity in the operating company allocation at genesis, for two reasons. A venture allocation in a network whose entire thesis is that supply must follow demand creates precisely the pressure to emit early that the design exists to prevent. And §31 documents a project whose 0 percent venture allocation made it structurally uninvestable despite being the most complete implementation of this thesis in existence. **The answer is neither a large token allocation nor none: it is equity in the company that operates the network, with token exposure earned on the same schedule as every other operator.**

17 Tokenomics of Every Network Here

WHAT EACH GOT RIGHT AND WRONG

Eight tokens appear in this report. None of them has a design we would copy whole, and each has exactly one mechanism worth taking. This section separates the mechanisms from the networks, because the proposal in §17 is an assembly of parts rather than an invention.

FIGURE 9: EVERY TOKEN, SCORED ON DESIGN



SCORE COMBINES EMISSION BASIS, SINK QUALITY AND WHETHER PARTICIPANT GROWTH DILUTES OR INFLATES. IT IS NOT A PRICE VIEW AND SHOULD NOT BE READ AS ONE.

TABLE 16: THE EIGHT DESIGNS

TOKEN	CHAIN	MAX SUPPLY	EMISSION BASIS	SINK	INSIDER ALLOCATION	SCORE
ORE	Solana	3,000,000	fixed 1 per minute, split competitively	90% of protocol revenue buys and burns	none: no premine, no insiders	4/5
GEOD	Solana	1,000,000,000	per station, capped per area	RTK query fees buy and burn	foundation and investor allocations	4/5
HNT	Solana	223,000,000	per hotspot, coverage-weighted	burn to Data Credits at $0.00001 fixed	investor and team allocations	3/5
HONEY	Solana	10,000,000,000	per hex after MIP-24, was per image	enterprises burn for map access	20% investors, 20% employees, 15% company	3/5
WINGS	Solana	undisclosed	24 per day per H3 hex, one station each	data buyers	$9.2M raised across two rounds	3/5
TRI	Base L2	3^27 = 7.6 trillion	1,000 per proof, four arms	none specified	0% premine, 0% VC, 0% treasury	2/5
CRKK	Kadena	undisclosed	proof of network participation	none material	80% miners, 1% project	1/5
MOBILE / IOT	Solana	merged into HNT	sub-DAO emission per radio	converted to HNT	inherited from Helium	1/5

SUPPLY AND ALLOCATION FIGURES ARE FROM PROJECT DOCUMENTATION AND MARKET DATA PROVIDERS. SOURCES DISAGREE ON ORE'S CAP BETWEEN 3M AND 5M; WE USE THE LOWER FIGURE, WHICH IS WHAT DEFILLAMA AND COINMARKETCAP CARRY.

ORE IS THE EMISSION MODEL

Roughly one token per minute globally, split competitively among whoever is participating, against a capped supply with no premine and no insider allocation, on a frozen contract. Additional participants take a smaller share each; they never cause more tokens to exist. **That is structurally the answer to the saturation result in §32**, and it is the property Helium most conspicuously lacked while emitting per device to 376,000 hotspots.

ORE's weakness is the other half. Its demand is a mining game rather than a service, and roughly 90 percent of protocol revenue buying and burning is impressive mechanically while telling you nothing about whether anyone needs the output. It reports

17 Tokenomics of Every Network Here

CONTINUED: MECHANISMS TO ADOPT AND REJECT

TABLE 17: THE MECHANISMS, ISOLATED

MECHANISM	WHO PROVED IT	WHAT IT DOES	VERDICT
Fixed emission per unit time	ORE	issuance is independent of participant count, so growth dilutes rather than inflates	adopt
Competitive split of a fixed pot	ORE	more nodes means a smaller share each, never more tokens	adopt
Burn to a fiat-denominated credit	HNT	buyers never hold the volatile asset; demand is usage, not speculation	adopt
Cap rewarded nodes per geographic cell	GEODNET, Wingbits	removes the incentive to pile into already-covered ground	adopt
Halving schedule	HNT, GEODNET	predictable disinflation the supply side can underwrite against	adopt
Emission per device	Helium	rewards presence rather than contribution; the root failure	reject
Sub-tokens per network arm	Helium IOT and MOBILE	fragments the sink and confuses value accrual	reject
Zero investor allocation	tri-net	removes the capital that builds the thing	reject
Emission before demand	Crankk, Helium	subsidises supply into a market that has not appeared	reject

FIVE MECHANISMS TO ADOPT AND FOUR TO REJECT. EVERY ROW HAS ALREADY BEEN RUN BY A LIVE NETWORK, SO NONE OF THIS IS THEORETICAL.

THE TWO CORRECTIONS ALREADY MADE IN PUBLIC

Hivemapper is instructive because it corrected mid-flight. It began emitting per image, which rewarded driving the same well-mapped streets repeatedly, and moved via MIP-24 to a hex-based model rewarding coverage of geographic cells and freshness. Wingbits launched with H3 resolution-6 cells and one rewarded station per cell from the start, which is the same correction applied before. **Both arrived at geographic capping independently, which is strong evidence it is the right primitive.**

GEODNET reached the same place by capping stations per area and burning against real RTK queries, and it is the closest network in this report to net-positive issuance. That it did so while selling a single data product to a competitive market is the reason it is the comparable in §8 rather than merely an analogy.

THE TWO FAILURES AT THE EXTREMES

Crankk distributed 80 percent of emission to miners, which is generous, against demand that did not exist, which is fatal: a 1,877-day payback at published rates. tri-net went further, with zero premine, zero venture allocation and zero treasury, and the result is the most complete implementation of this thesis in existence with no route to the capital that would finish it. **Generosity to the supply side and integrity about allocation are both good instincts that become failures when they replace a demand analysis.**

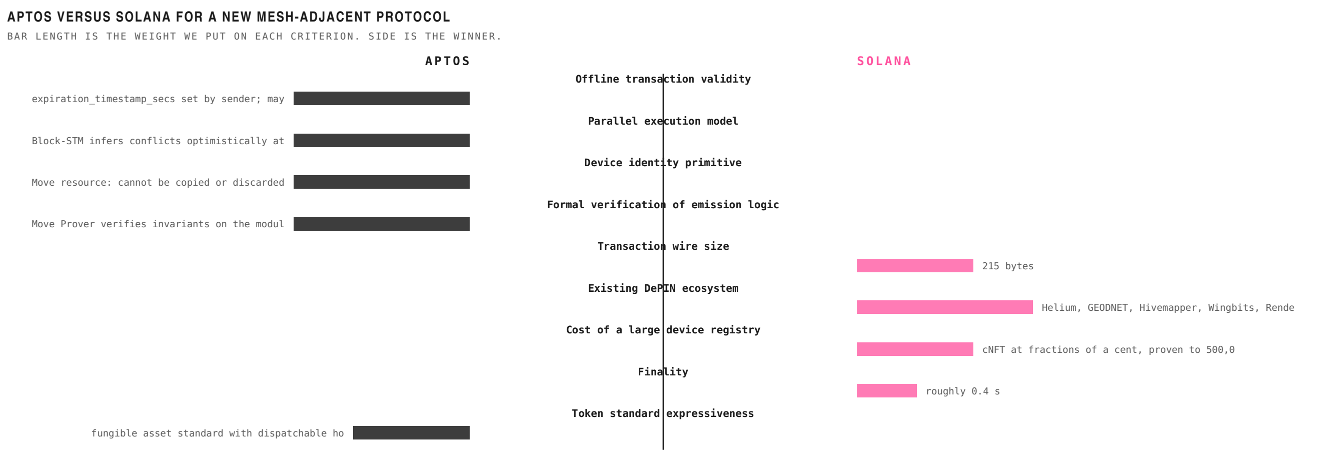
> **WHAT NOBODY HAS SHIPPED**
>
> A network with ORE's fixed competitive emission, Helium's fiat-denominated burn sink, GEODNET and Wingbits' geographic capping, and a genesis gate on a paying contract. Each part is proven separately by a live network. The assembly does not exist, and assembling it is most of what makes the proposal in §17 differentiated.

18 Why Aptos and Not Solana

FOR A NEW PROTOCOL, NOT AN EXISTING ONE

Every DePIN network in this report runs on Solana. Helium migrated there from its own L1, GEODNET from Polygon, Render from Ethereum. Every observed migration in this category has gone toward Solana and none away from it. Arguing for a different chain therefore requires arguing against a real gradient, and we do not make the argument for existing networks. **We make it for a protocol that does not exist yet, where the switching cost is zero and the design constraints are different.**

FIGURE 10: CRITERION BY CRITERION, WEIGHTED



BAR LENGTH IS THE WEIGHT WE ASSIGN EACH CRITERION AND SIDE IS THE WINNER. SOLANA TAKES THE TWO HEAVIEST PRACTICAL OBJECTIONS.

TABLE 18: THE COMPARISON

CRITERION	APTOS	SOLANA	WINNER	WEIGHT
Offline transaction validity	expiration_timestamp_secs set by sender; may be set never to expire	durable nonce account, extra account and instruction, must be advanced	**Aptos**	high
Parallel execution model	Block-STM infers conflicts optimistically at runtime	Sealevel requires declaring account access upfront	**Aptos**	high
Device identity primitive	Move resource: cannot be copied or discarded, enforced by the type system	compressed NFT: a Merkle-tree optimisation, not a type guarantee	**Aptos**	high
Formal verification of emission logic	Move Prover verifies invariants on the module itself	no equivalent in the standard toolchain	**Aptos**	high
Transaction wire size	264 bytes for a native transfer	215 bytes	**Solana**	medium
Existing DePIN ecosystem	thin: no comparable network deployed	Helium, GEODNET, Hivemapper, Wingbits, Render all live	**Solana**	high
Cost of a large device registry	resource per account, predictable storage cost	cNFT at fractions of a cent, proven to 500,000 devices	**Solana**	medium
Finality	roughly 0.9 s	roughly 0.4 s	**Solana**	low
Token standard expressiveness	fungible asset standard with dispatchable hooks at the token level	SPL with Token-2022 extensions	**Aptos**	medium
Weighted total	—	—	**14 : 8**	–

WEIGHTS ARE OURS. THE APTOS COLUMN IS READ FROM APTOS-CORE; THE SOLANA COLUMN FROM PUBLISHED DOCUMENTATION AND THE HELIUM MIGRATION RECORD.

THE FOUR THAT MATTER

Offline validity. A transaction signed on a disconnected device and carried by relays must still be valid hours later. `expiration_timestamp_secs` is chosen by the sender and the source comment states it may be set so the transaction does

19 The Protocol We Would Fund

SPECIFICATION

A verified RF observation network on Aptos. Nodes measure GNSS interference and spectrum occupancy, submit signed observations, and are paid from a fixed emission split by the marginal information value of what they contributed. Buyers burn tokens for non-transferable credits to query the resulting picture. Working name **MERIDIAN**, which should be replaced before anything is filed.

TABLE 19: PROTOCOL SPECIFICATION

PARAMETER	DESIGN	RATIONALE
Supply	42,048,000 maximum, no premine, no genesis VC allocation	geometric sum of the halving schedule; nothing minted outside it
Emission	1,200 per hour at genesis, halving every 2 years	ORE's fixed rate: participant growth dilutes, it does not inflate
Distribution	competitive split across H3 cells, weighted by marginal information value	a node in a covered cell earns near zero regardless of uptime
Demand coupling	per-cell emission scales with paid queries against that cell	supply follows revenue; the inversion of the DeWi default
Sink	burn to non-transferable Observation Credits at a fixed USD rate	Helium's data-credit mechanism, the part that worked
Genesis gate	no emission until the first paying contract is signed	the network starts when a buyer exists, not when a node does
Node identity	Move resource bound to a hardware attestation key	non-copyable and non-droppable; Sybil resistance in the type system
Verification	ephemeris cross-check plus inter-node agreement per epoch	a fabricated observation must satisfy orbital geometry and its neighbours
Investor exposure	equity in the operating company; token earned on the operator schedule	avoids both the Helium pressure to emit early and the tri-net dead end

EVERY PARAMETER IS EITHER TAKEN FROM A LIVE NETWORK IN §15 OR DERIVED FROM THE SATURATION RESULT IN §32. NONE OF IT IS NOVEL IN ISOLATION.

It does not build its own storage or transit. Observation records are written to **Shelby**, the Aptos Labs and Jump Crypto hot-storage protocol that names DePIN data feeds as a target workload and attaches cryptographic verification to each request. Gateway-to-gateway backhaul runs over **DoubleZero**. Both are chain agnostic, so the Aptos decision costs nothing on either layer, and Shelby is Aptos-native besides. §19 sets both out.

HOW IT WORKS IN ONE PARAGRAPH

A node is a $260 receiver package registered as a Move resource bound to a hardware attestation key. Each epoch it submits observations: per-satellite carrier-to-noise ratios, automatic gain control trend, an RF spectrum snapshot, and carrier-phase double differences. The protocol cross-checks each submission against published ephemeris and against every other node observing the same satellites in the same window. Surviving observations earn a share of that epoch's fixed emission, weighted by how much information the node added to its H3 cell. Buyers, meaning aviation authorities, insurers, maritime operators and defence, burn tokens for credits at a fixed dollar rate to query the interference picture.

> **WHAT MAKES THIS DIFFERENT FROM EVERY NETWORK IN TABLE 16**
>
> Emission does not scale with node count, so recruiting operators cannot destroy unit economics. Emission per cell scales with paid demand against that cell, so supply follows revenue instead of preceding it. And nothing is emitted at all until a buyer has signed, which inverts the sequencing every network in this report used. The cost is a slower and less exciting launch. The slower launch is the point.

19 The Protocol We Would Fund

WHY THIS NEEDS MOVE SPECIFICALLY

INVARIANT	SPECIFICATION	WHAT IT REMOVES
Emission ceiling	sum of all mints in any epoch <= EMISSION_PER_EPOCH	no path, including governance, can exceed the schedule
Genesis gate	total_minted == 0 while first_contract_at == none	the sequencing rule is enforced by the module, not by intent
Supply cap	total_minted <= MAX_SUPPLY for all reachable states	the cap is a theorem, not a constant in a comment
Registry uniqueness	each attestation key maps to at most one NodeCapability	Sybil resistance proved rather than monitored
Credit conservation	credits issued == tokens burned x rate, always	the sink cannot leak
No admin mint	no public entry function mints outside distribute_epoch	removes the discretionary-mint risk that every reviewer asks about

A MOVE PROVER SPECIFICATION CARRYING THESE INVARIANTS TURNS THE TOKENOMICS FROM A DOCUMENT INTO A CHECKED ARTEFACT. NO OTHER CHAIN IN §27 OFFERS THIS IN ITS STANDARD TOOLCHAIN.

Each row is a claim a token buyer currently has to take on trust from a whitepaper. On Move they are machine-checked properties of the deployed module. **An investor underwriting emission risk can read the proof** , and a protocol that ships this becomes very difficult for a competitor with a PDF to argue against.

WHAT THIS COSTS AND WHAT IT DOES NOT

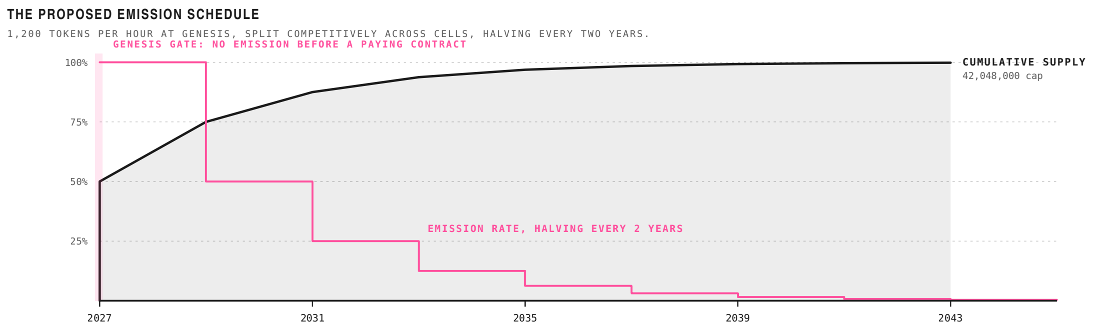
Writing a Prover specification is real work, roughly comparable to writing a thorough test suite, and it is work most teams will resist because it produces no visible feature. It does not prove the economics are wise, only that the module does what the specification says: a badly chosen emission rate can be proved correct. What it removes is the entire class of risk that dominates DePIN diligence, which is not whether the schedule is optimal but whether some path through the code can mint outside it.

It also changes what a security review is for. Instead of an auditor reasoning about reachable states by inspection, the invariants are discharged mechanically and the audit can concentrate on whether the specification captures the intent. **That is a better use of an expensive reviewer and a materially stronger artefact to hand a token buyer.**

20 Emission Schedule and Launch

THE NUMBERS

FIGURE 11: THE PROPOSED EMISSION SCHEDULE



FIXED HOURLY EMISSION HALVING EVERY TWO YEARS, WITH NO ISSUANCE BEFORE THE FIRST PAYING CONTRACT. CUMULATIVE SUPPLY CONVERGES ON THE CAP.

TABLE 20: EMISSION BY ERA

ERA	TOKENS / HOUR	ISSUED IN ERA	CUMULATIVE	% OF CAP
2027–2029	1,200	21,024,000	21,024,000	50.0%
2029–2031	600	10,512,000	31,536,000	75.0%
2031–2033	300	5,256,000	36,792,000	87.5%
2033–2035	150	2,628,000	39,420,000	93.8%
2035–2037	75	1,314,000	40,734,000	96.9%
2037–2039	38	657,000	41,391,000	98.4%
2039–2041	19	328,500	41,719,500	99.2%

MAXIMUM SUPPLY OF 42,048,000 IS THE GEOMETRIC SUM OF THE SCHEDULE, NOT A CHOSEN ROUND NUMBER. HALF THE SUPPLY IS ISSUED IN THE FIRST TWO YEARS, WHICH IS AGGRESSIVE AND DELIBERATE: THE GENESIS GATE MEANS THOSE TWO YEARS ONLY BEGIN ONCE DEMAND EXISTS.

LAUNCH SEQUENCE

Pre-genesis. Build the node, deploy 50 to 200 units in a known interference corridor at the company's expense, and sell the resulting data to one aviation authority or defence buyer as a conventional subscription. No token. This is the phase most DePIN teams skip and it is the phase that determines whether the rest works.

Genesis. On signature of that first contract, the module unlocks emission and the network opens to third-party operators in the cells the contract requires coverage of. Emission is scoped to demand from the first block.

Expansion. Each subsequent contract widens the set of cells earning emission. Operators can see which cells are unlocked and what they pay, so capital allocation on the supply side is informed. The network grows along the demand curve by construction.

WHAT THE INVESTOR OWNS

Equity in the operating company, which holds the buyer relationships, the node design and the data platform, plus token exposure earned on the same schedule as any other operator. We would not take a genesis token allocation, for the reason in §14: a venture allocation in a network whose thesis is that supply must follow demand creates exactly the pressure to emit early that the design exists to prevent.

THE HONEST WEAKNESSES

- The genesis gate makes this unfundable by conventional DePIN standards, because there is no token to sell before revenue exists.
- Half the supply issues in two years, which is fast; the defence is that the clock starts at demand.
- Building on Aptos means building the oracle, indexer and device-registry tooling that would be off the shelf on Solana.

21 Shelby and DoubleZero

TWO LAYERS THIS PROPOSAL DOES NOT HAVE TO BUILD

The specification in §17 assumes storage and transit as though they were solved. They largely are, by two protocols that are chain agnostic by design, which means the Aptos decision in §16 forfeits neither. One of them is not yet in production, and we name the dependency risk.

FIGURE 12: THE STACK, AND WHAT ALREADY EXISTS

THE STACK, AND WHAT ALREADY EXISTS

THREE OF THE FIVE LAYERS ARE LIVE OR IN TESTNET AND TWO OF THEM ARE DELIBERATELY CHAIN AGNOSTIC.

OBSERVATION $260 receiver, Move resource identity	**MERIDIAN nodes** the recommendation in §6	**TO BE BUILT**
SETTLEMENT AND COORDINATION emission, registry, provable invariants	**Aptos** §16	**LIVE**
STORAGE AND SERVING spectrum snapshots, time series, query serving	**Shelby** chain agnostic; Aptos native	**EARLY ACCESS, PRODUCTION 2026**
TRANSIT gateway-to-gateway backhaul for observation data	**DoubleZero** chain agnostic; 2Z on Solana	**MAINNET-BETA SINCE OCT 2025**
EDGE TRANSPORT last kilometres where backhaul is absent	**LoRa mesh** §23 to §27	**DEPLOYED, UNPAID**

FIVE LAYERS. THREE ARE LIVE OR IN TESTNET. THE PROPOSAL IS THE TOP ONE AND SITS ON INFRASTRUCTURE IT DOES NOT HAVE TO FINANCE.

TABLE 21: SHELBY ARCHITECTURE

PROPERTY	SHELBY	NOTE
Built by	**Aptos Labs and Jump Crypto**	announced June 2025
Architecture	**four layers: client SDK, RPC nodes, storage providers, coordination**	control plane and data plane deliberately separated
Node role 1	**RPC node**	gateway; S3-compatible API, erasure-encodes on write, gathers and decodes on read, holds the client micropayment channel
Node role 2	**storage provider (SP)**	stores erasure-coded chunks, runs peer-to-peer audits, large local storage, no RAID required
Coding	**Clay codes, MSR and MDS**	under 2x replication overhead against 3x for triple replication, and minimal repair bandwidth
Auditing	**hybrid: peer-to-peer SP challenges plus on-chain audit-the-auditor**	correctness-critical operations settle on Aptos
Payment model	**reads are paid, via micropayment channels**	serving is compensated, not idle storage
Backbone	**dedicated private fibre between RPC nodes and SPs**	DoubleZero; the two protocols are one stack, not two
Coordination layer	**Aptos smart contract**	storage commitments, audit outcomes, channel metadata, participation
Reference client	**Cavalier, open source, C, by Jump Crypto**	tile-based architecture on dedicated cores, in the lineage of Firedancer
Status	**pre-mainnet; Early Access testnet since March 2026**	node onboarding managed rather than self-serve; token terms not published

FROM THE SHELBY WHITEPAPER (ARXIV 2506.19233) AND THE PROTOCOL DOCUMENTATION. SHELBY IS PRE-MAINNET AND ITS TOKEN TERMS ARE NOT PUBLISHED; NOTHING IN THIS REPORT ASSUMES THEM.

21 Shelby and DoubleZero

CONTINUED: THE SHARED DESIGN PRINCIPLE

PAID READS, WHICH IS THE SAME IDEA AGAIN

Shelby compensates *serving* rather than idle storage. A storage provider that holds data nobody requests earns comparatively little; one serving heavy read traffic earns more, and the audits ensure it is holding what it claims. That is the third independent appearance of the same principle in this report, after DoubleZero's Shapley-weighted Proof of Utility and the demand-weighted emission specified in §14. Three teams solving three different physical-infrastructure problems have each concluded that reward should track marginal delivered utility.

TABLE 23: DOUBLEZERO EDGE, A LIVE SINK AT A KNOWN PRICE

PROPERTY	DOUBLEZERO EDGE	NOTE
Launched	public beta, 16 April 2026	shred subscription over the DZ network
Priced	$30 to $100 USDC per device per epoch	by city; an epoch is roughly two days
Publishers	409 validators, about 50% of Solana stake	over 58% of validators by mid-2026
Measured advantage	6 ms average, up to 28 ms	20 ms Europe, 80 ms US, 100 ms+ Asia under congestion
Revenue split	50% fibre contributors, 32.5% publishing validators	17.5% protocol clients
Burn	10% of revenue burns 2Z	the sink, funded by real subscriptions
Adopters	Jito, Temporal, Staking Facilities, Triton, Coinbase	Coinbase connected 24 June 2026
Payment rail	USDC into a per-subscriber escrow account	escrow funds future epochs automatically
Prior model retired	5% validator block-reward fee removed at epoch 939	had generated roughly $3.5M annualised across 455 validators
Planned expansion	centralised exchange feeds, prediction markets, order-by-order data	positioning as a unified market-data layer

EDGE IS THE CLEAREST WORKING EXAMPLE IN THIS REPORT OF THE ECONOMIC MODEL §18 PROPOSES: A REAL SERVICE, PRICED IN FIAT, WITH A FIXED SHARE OF REVENUE BURNING THE TOKEN.

Edge deserves attention beyond its bandwidth. It sells low-latency Solana shreds by subscription at $30 to $100 USDC per device per epoch, and splits the proceeds: 50 percent to the contributors supplying fibre, 32.5 percent to the validators publishing the data, 17.5 percent to protocol clients, and **10 percent of revenue burns 2Z**. DoubleZero simultaneously retired its previous 5 percent validator block-reward fee, which had been generating roughly $3.5 million annualised, in favour of this.

That is a protocol replacing a tax on its participants with a product sold to outsiders, and burning a fixed share of the proceeds. **It is the burn-to-query sink specified in §18, running in production, at a published price, with Coinbase as a subscriber.** The genesis-gate argument becomes much easier to make to a sceptical team when a peer protocol has already shown that a real subscription revenue line outperforms an emission-funded one.

THE AUDITING MODEL IS PRIOR ART FOR OUR VERIFICATION DESIGN

Shelby's hybrid auditing has storage providers challenge each other to prove possession, with an on-chain audit-the-auditor step settling disputes on Aptos. Structurally that is what §7 proposes for observation: nodes are checked against each other for agreement on the same satellites in the same window, with an ephemeris cross-check settling on-chain. **The pattern of cheap adversarial peer checking backed by expensive on-chain arbitration is already implemented by a team at Jump Crypto**, which substantially de-risks the claim that it can be built.

21 Shelby and DoubleZero

CONTINUED: TRANSIT, AND A PRECEDENT THAT MATTERS

TABLE 22: DOUBLEZERO

PROPERTY	DOUBLEZERO	NOTE
Built by	Malbec Labs; founders from Solana Foundation, NASDAQ and fibre operators	Austin Federa, Mateo Ward, Greg McConnell
Funding	$28M token round, March 2025	co-led by Multicoin Capital and Dragonfly
What it is	dedicated fibre network for distributed systems	two rings: edge filtering of spam, then private transit
Token	2Z, live since 2 October 2025	SPL on Solana; Binance and Coinbase listed
Supply	10 billion genesis, 3.47 billion circulating at launch	roughly 65% on a four-year lockup
Allocation	Foundation 29%, Jump Crypto 28%, Malbec Labs 14%	disclosed and vested
Reward mechanism	Proof of Utility, Shapley-value weighted	paid for marginal performance contributed, not raw supply
Deployed	70+ fibre links across 25+ cities	roughly 22% of staked SOL connected at mainnet-beta
Measured benefit	77% of links and 82% of city-pair routes beat the public internet	epoch 24 data
Regulatory	SEC no-action letter, September 2025	enabled major exchange listings; Grayscale watchlist Q1 2026
Validator sale eligibility	Solana, Sui, Aptos, Celestia, Avalanche operators	Aptos validators are already in the holder base
DoubleZero Edge	launched Q2 2026	multicast market data; built to extend beyond Solana to exchanges and prediction markets
Contributor capex	$60,000 to $130,000 per link	jurisdiction dependent

MAINNET-BETA SINCE OCTOBER 2025. THE FINAL TWO ROWS ARE THE ONES THAT MATTER MOST TO THIS PROPOSAL AND THEY HAVE NOTHING TO DO WITH BANDWIDTH.

DoubleZero runs more than seventy contributed fibre links across twenty-five cities and carries roughly a quarter of Solana's mainnet stake weight. For this proposal it is the gateway-to-gateway transit layer, and more it is the backbone Shelby itself runs on between RPC nodes and storage providers: **the two are one stack rather than two independent choices**, and adopting Shelby inherits DoubleZero underneath it.

> **THE MECHANISM WE PROPOSED ALREADY EXISTS, AND IT WORKS**
>
> DoubleZero pays contributors by **Proof of Utility, weighted with Shapley values**: reward is a function of the marginal performance each link contributes to the network, not of raw supply offered. That is precisely the emission principle specified in §14 and §17, where a node in an already-covered cell earns near zero regardless of uptime. We arrived at it from the saturation result in §36; DoubleZero arrived at it from network economics. **Two independent derivations of the same rule, one of them already live and paying, is the strongest available evidence that marginal-contribution weighting is the correct primitive for physical infrastructure.**

21 Shelby and DoubleZero

CONTINUED: THE REGULATORY PRECEDENT

THE REGULATORY PRECEDENT

In September 2025 the SEC's Division of Corporation Finance issued a no-action letter confirming that 2Z does not require registration as a security and that programmatic flows across the network are not securities transactions. For a report recommending a new token this is the most useful external fact here: there is now a documented path by which a DePIN utility token, paying contributors for measurable physical work, has been examined and cleared.

The features that appear to have mattered are worth naming because §17 shares them: the token buys a real service, contributors are paid for measurable utility rather than for holding, allocations are disclosed and vested, and flows are programmatic rather than discretionary. **The genesis gate in §18 strengthens this position**, because it makes the utility relationship prior to the token.

THE DEPENDENCY RISK, STATED PLAINLY

Shelby is pre-mainnet. Node onboarding is managed, not self-serve, and token terms are unpublished. Building a hard dependency on it means accepting testnet-stage risk on a layer the network cannot operate without, and accepting unknown future economics on a cost line. The mitigation is that Shelby exposes an S3-compatible API, which makes the storage layer substitutable almost by definition: the same data sits on S3 during pre-genesis and migrates when Shelby reaches production, forfeiting only the verifiable-retrieval property in the interim. **We would build behind that abstraction from the first commit and treat Shelby as the target.**

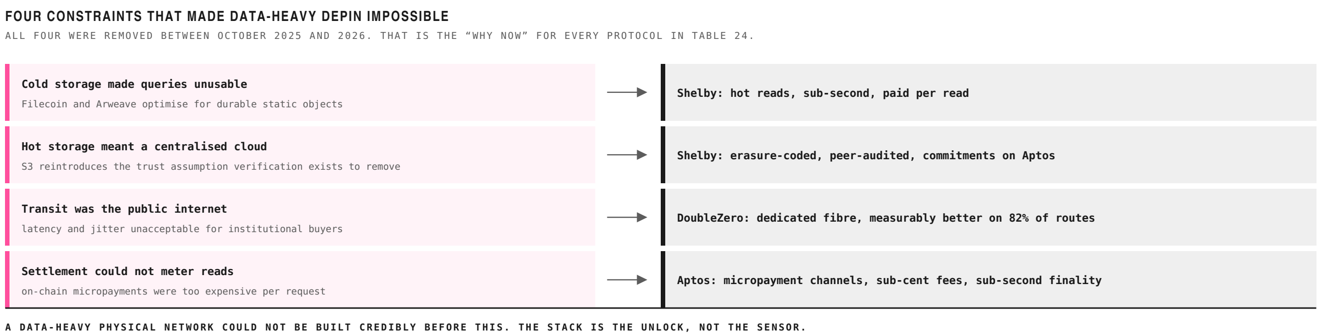
> **WHAT THIS CHANGES ABOUT THE RECOMMENDATION**
>
> The proposal is narrower than it first appears: an observation network and an emission design, on settlement, storage and transit that already exist or are being built by well-capitalised teams. That is a smaller thing to underwrite than a full stack, and it is why the $260 bill of materials in §7 is close to the whole capital story on the supply side.

22 What Gets Built On This

THE PROTOCOLS THAT PLAUSIBLY FOLLOW

The stack in §19 is more interesting than the sum of its parts, because between October 2025 and mid-2026 it removed four constraints that had made data-heavy physical networks impractical. Every protocol proposed below was buildable in principle before and uneconomic in practice.

FIGURE 13: FOUR CONSTRAINTS, NOW REMOVED



THE LEFT COLUMN IS WHAT A DATA-HEAVY DEPIN FACED UNTIL RECENTLY. THE RIGHT COLUMN IS WHAT REPLACED IT. THIS IS THE "WHY NOW" FOR EVERYTHING IN TABLE 24.

Consider what an RF observation network needs. Continuous high-rate writes from thousands of nodes. Bursty geographic reads when a buyer queries a region. Verifiable provenance from capture through storage to retrieval. Sub-second latency for institutional users. Per-read accounting cheap enough to meter. **Every one of those was either impossible or required a centralised cloud eighteen months ago**, and a centralised cloud reintroduces exactly the trust assumption that the verification argument in §3 exists to remove.

TABLE 24: PROTOCOLS THAT PLAUSIBLY FOLLOW

PROTOCOL	SHAPE	WORKLOAD	CONVICTION	WHY
RF and GNSS interference observation	the recommendation, §17	spectrum snapshots and C/N0 time series	**highest**	stated buyer, safety mandate, no distributed product in market
Verifiable sensor video	wildfire, traffic, perimeter	continuous video with provenance from capture to query	**high**	insurers and agencies already buy this from centralised vendors
Seismic and infrasound early warning	dense private arrays	high-rate waveform data, latency critical	**high**	public networks are sparse and the value is in the first ten seconds
Maritime RF and AIS integrity	sanctions and dark-fleet detection	position streams cross-checked against RF observation	**high**	the ADS-B pattern in a market with enforcement budgets
Grid and PMU timing integrity	power quality and timing-attack detection	phasor data at high rate, timing critical	medium	adjacent to the GNSS buyer; same threat model, same nodes
Autonomous vehicle sensor provenance	liability and insurance	sensor logs with verifiable chain of custody	medium	the buyer is a claims department: slow, but it pays
AI training-data provenance	capture-to-training custody	verifiable origin for datasets used in training	medium	Shelby names AI workloads explicitly; demand is real, the standard is not
Distributed radio astronomy	SatNOGS commercialised	wideband IQ capture, enormous volume	low	scientifically valuable, commercially thin

CONVICTION IS OURS. THE COMMON SHAPE ACROSS THE TOP FOUR IS A PHYSICAL SENSOR, A GEOGRAPHIC QUERY, AND A BUYER WITH A REGULATORY OR ENFORCEMENT MANDATE RATHER THAN A DISCRETIONARY BUDGET.

FIGURE 14: WHAT PLAUSIBLY GETS BUILT

22 What Gets Built On This

CONTINUED: EXISTING NETWORKS THAT SHOULD MIGRATE

The second-order opportunity is not new protocols but existing ones. Several live networks are paying centralised vendors for exactly the services this stack now provides, and their workloads are closer to Shelby's design target than most of what it was announced for.

TABLE 25: NETWORKS WHOSE ECONOMICS IMPROVE ON THIS STACK

NETWORK	SCALE	TODAY	WHAT THE STACK GIVES THEM	FIT
Hivemapper	100,000 dashcams generating continuous imagery	pays centralised cloud; imagery is read-heavy and geographic	**Shelby is close to purpose-built for this workload**	**high**
GEODNET	20,500 base stations streaming RTK corrections	latency sensitive, continuous, regional	**DoubleZero for transit, Shelby for correction archives**	**high**
Wingbits and ADS-B networks	thousands of receivers, high-rate position data	large volume, geographic queries	**the same workload shape as the recommendation**	**high**
WeatherXM	weather stations with continuous telemetry	time-series reads at modest volume	storage is a real line item at station scale	medium
DIMO	vehicle telemetry from connected cars	per-vehicle streams with privacy constraints	smart-contract access control fits the consent model	medium
Helium	IoT records and carrier offload data	already on Solana and would not move settlement	could still use Shelby for data without changing chain	low

NONE OF THESE WOULD NECESSARILY CHANGE SETTLEMENT CHAIN. SHELBY AND DOUBLEZERO ARE CHAIN AGNOSTIC, SO ADOPTING THEM IS A COST DECISION RATHER THAN AN ECOSYSTEM DECISION.

Hivemapper is the clearest case. A hundred thousand dashcams producing continuous imagery, read heavily and geographically, currently sitting on centralised cloud storage that is a material and growing line item. Shelby is close to purpose-built for it: erasure coding at under 2x overhead against triple replication, paid reads that turn a cost centre into a revenue share, and geographic placement groups that match how the data is queried.

GEODNET is the most strategically interesting. Correction streams are latency-sensitive and regional, which is DoubleZero's exact case, and the correction archive is a Shelby workload. It is also the closest comparable to the recommendation in §8, so a GEODNET that adopted this stack would be running the infrastructure experiment on our behalf.

> **WHY THIS MATTERS FOR THE INVESTMENT CASE**
>
> Every migration above increases demand for Shelby reads and DoubleZero bandwidth without requiring any of those networks to change chain, adopt a new token, or agree with anything in this report. **That makes 2Z and a future Shelby token exposure to the whole category** , which is a materially better risk profile than picking the winning sensor network. It is the same logic as owning the LoRa chipset vendor in §12, one layer up and with far better unit economics.

WHAT WE WOULD WATCH AS CONFIRMATION

The single cleanest signal that this thesis is correct is a live DePIN network with real data volume announcing Shelby integration. Not a partnership press release, but a measurable share of its reads served through Shelby RPC nodes. If that has not happened by the time Shelby reaches production, the storage layer is solving a problem the networks do not think they have, and the argument in this section is wrong.

23 Market Map

WHAT EXISTS, AND WHAT DOES NOT

The category has more built infrastructure than it has businesses. This section separates the two.

WHAT EXISTS

Protocols. Meshtastic and MeshCore, both open source, neither tokenised, both culturally resistant to commercialisation. **Attempts.** Crankk (LoRaWAN plus a Meshtastic fork on Kadena, 1,877-day payback); Nodle (smartphone-based connectivity); Althea (mesh ISP marketplace, largely dormant). **Transport bridges.** Bitchat, which already relays Bitcoin and Cashu peer-to-peer through a Bluetooth mesh until a connected device broadcasts them: a functioning off-grid settlement path in production, with no token and no payment to relays. **The professional tier.** Tactical mesh radios at $5,000–15,000 per unit from Doodle Labs, Rajant and DTC; Beechat's Reticulum-based platform selected for NATO's DIANA 2026 programme. That market explicitly notes the gap between consumer LoRa-class devices and operational systems, which is the gap this thesis argues is fillable from below.

TABLE 7: WHAT DOES NOT EXIST

OPPORTUNITY	WHAT IT IS	WHO IS BUILDING IT	CONVICTION	BASIS
Transit settlement primitive	pay-per-delivered-packet on MeshCore-class routing; operators bid airtime	nobody	**high**	§6–7 bound the revenue; architecture is the gate
Off-grid stablecoin settlement	extend the Bitchat primitive from 300 m to 15 km in shutdown-prone markets	6 transport bridges, none paid	**high**	only payload clearing the $0.10 threshold
Node-as-substrate economics	subsidise hardware against future payload revenue, not coverage emissions	nobody	medium	requires a second payload to prove
GNSS interference monitoring	receive-only network measuring jamming and spoofing directly	nobody	**high**	documented regulator gap; GEODNET comp
Network census	the deployed mesh has never been measured; public maps see opt-ins only	nobody	low cost	underwriting edge, not a business

CONVICTION IS OURS, NOT A MARKET CONSENSUS. THE BASIS COLUMN POINTS TO THE SECTION OF THIS REPORT THAT SUPPORTS OR CONSTRAINS EACH ITEM.

ON THE CENSUS

The fifth row deserves a note because it is not a business. Nobody knows how large the deployed Meshtastic network is. The public maps show only nodes that opt into MQTT uplink *and* run a default-key or unencrypted channel: a self-selected, internet-connected minority, by construction. Every published node count is a lower bound of unknown tightness.

For an operator this is a curiosity. For a fund underwriting the category it is an edge: the supply side of the delivery tier is systematically unmeasured, which means it is systematically mispriced in both directions. Roughly two weeks of packet capture against the public broker yields the empirical hop distribution, the observed channel utilisation against the prediction in §7, and a defensible undercount estimate from the locally-heard node counts that gateways already publish. We are running it; results will either confirm or correct §6.

> **WHY THE WHITESPACE IS EMPTY**
>
> Six transport bridges exist and none pays the mesh. The reason is not oversight: the demand these projects served was censorship resistance, which a free volunteer relay satisfies completely. Payment becomes necessary only when volume exceeds what volunteers will donate. §7 says that threshold arrives quickly. It has not arrived yet.

24 Comparables and the Re-rating Mechanism

HONEST SIZING

This section sizes the opportunity without the substrate argument, which is the conservative case.

TABLE 8: DEPLOYED NETWORKS BY REVENUE, UNITS AND VALUATION

NETWORK	TICKER	ANN. REVENUE	DEPLOYED UNITS	MARKET CAP	REV. MULTIPLE
Helium	HNT	~$11.0M	376,000 hotspots	~$177M	**16×**
GEODNET	GEOD	~$10.0M	20,500 base stations	~$85M	**8.5×**
Hivemapper	HONEY	undisclosed	100,000 dashcams	—	**—**
Wingbits	WINGS	pre-revenue	thousands of receivers	—	**—**
DePIN sector	—	~$72M FY25	650+ projects	$9.4–18.9B	**130–260×**

REVENUE FIGURES ARE THE MOST RECENT PUBLIC DISCLOSURES AND ARE NOT DIRECTLY COMPARABLE IN ACCOUNTING TREATMENT. SECTOR MULTIPLE IS COMPUTED ON MESSARI'S FY25 ONCHAIN REVENUE ESTIMATE AGAINST THE RANGE OF PUBLISHED SECTOR CAPS.

THE HONEST COMPARABLE

GEODNET is the sharpest and least flattering comparison. It does exactly one thing, has real paying customers in agriculture and robotics, and trades around $85 million on roughly $10 million of annualised revenue from 20,500 base stations. **That is the realistic ceiling for a single-payload network in this category in the near term, and it is a modest venture outcome.** Any pitch implying otherwise for a mesh transit network should be discounted against this row.

Helium is the cautionary one: 376,000 deployed hotspots against $11 million of revenue, which is roughly $29 of annual revenue per deployed device. Deployment count is not the metric. It never was.

THE RE-RATING MECHANISM

The sector row is the interesting one. Roughly $72 million of FY25 onchain revenue against $9.4–18.9 billion of market capitalisation puts the category at somewhere between 130× and 260× revenue, while the leading individual networks trade at 8–16×. The gap is narrative. The category is priced on a story about physical infrastructure, and individual networks are priced on cash.

> **HOW THIS THESIS PAYS EVEN IF TRANSIT REVENUE STAYS SMALL**
>
> If verifiability enue, then the correct question for every network in the sector is which tier it occupies, and the answer is currently not priced at all. A network that develops a credible, cheap verification story should re-rate against one that does not, independent of near-term revenue. The framework is therefore tradeable before any of the specific opportunities in §11 produce a dollar: it is a screen for which of the 650 tracked projects are selling something checkable.

SIZING THE RECOMMENDATION

For the GNSS interference network, GEODNET is a direct comparable on hardware class, operator profile and receive-only posture, which makes the supply-side economics a known quantity rather than a projection. The demand side is better: aviation authorities, airlines, insurers and defence, against GEODNET's agriculture and survey base.

For the settlement rail, the sizing is not transit revenue but interchange on value moved. A five-hundred-dollar transfer paying ten cents is two basis points. The addressable volume is remittance and domestic transfer in shutdown-prone corridors, which is a market measured in hundreds of billions annually and does not require the mesh to carry any material share of it to matter.

25 Consumer Versus Wholesale

THREE EXITS, ONE PATTERN

The thesis so far is analytical. This section tests it against completed transactions, which is the only evidence in this report that does not depend on our model being right.

FIGURE 9: WHAT WAS SOLD AND WHAT WAS KEPT

	CONSUMER-FACING LAYER		WHOLESALE INFRASTRUCTURE LAYER
goTenna	consumer mesh hardware ABANDONED 2019	→	tactical mesh radios ACQUIRED BY FORTERRA, OCT 2025
FlightAware	40,000 volunteer feeders NEVER COMPENSATED	→	enterprise flight data ACQUIRED BY COLLINS AEROSPACE
Helium	Helium Mobile MVNO SOLD JUN 2026, UNPROFITABLE	→	wholesale carrier offload RETAINED BY NOVA LABS

THREE COMPANIES, THREE UNRELATED TECHNOLOGIES, THREE INDEPENDENT DECISIONS. IN EVERY CASE THE CONSUMER-FACING BUSINESS WAS ABANDONED OR DIVESTED AND THE WHOLESALE LAYER BENEATH IT WAS RETAINED OR ACQUIRED.

GOTENNA

Founded in 2012 in response to communications failures during Hurricane Sandy. Built consumer mesh hardware. Powered txTenna, the first Bitcoin-over-mesh product. By 2019 it had raised a $24M Series C *explicitly* to pivot from consumer gadgets to public-sector clients, and built the Pro line: tunable VHF/UHF at five watts, 384-bit ECC key management, ATAK integration, 30-hour battery. Over 10,000 Pro X units across 16 countries within a year of release. Multiple AFWERX and AFRL awards. Acquired by Forterra, an autonomous mission systems company, in October 2025.

This is the sharpest case because goTenna had every advantage: first mover, working crypto integration, real consumer traction, and capital. It walked away from the consumer market anyway.

FLIGHTAWARE AND HELIUM

FlightAware never paid its volunteers and sold its enterprise data business to Collins Aerospace. Helium sold Helium Mobile: the consumer MVNO, roughly 600,000 cumulative signups, still unprofitable, to Andrew Yang's Noble Mobile in June 2026, while Nova Labs retained the wholesale carrier-offload business and the network itself. Nova Labs reportedly had around ten interested acquirers. Frank Mong's framing was that Helium Mobile existed to prove the hotspot network was valuable, that the point was proven, and that the company was returning to wholesale.

> **THE PATTERN**
>
> The consumer-facing layer of decentralized wireless does not monetise. The wholesale infrastructure layer beneath it does. Three independent instances, supported by transactions.

This has a direct consequence for how the delivery and observation tiers should be underwritten. Both should be sold to enterprises and neither to people. It also implies that the natural exit for a mesh protocol may be a defence acquisition rather than a liquid token: a good venture outcome and a poor fit for a fund underwriting token liquidity. We flag this as an unresolved tension in the delivery-tier case.

26 Starlink Is a Mesh

BUT NOT THE MESH THAT COMPETES

A recurring claim in this category holds that Starlink's direct-to-device service makes it a giant radio mesh, and therefore that terrestrial mesh is already obsolete. The first half is true and more interesting than the people making the claim usually realise. The second half does not follow, and the reason it does not follow is the most useful thing in this section.

THE MESH IS REAL, AND IT IS OPTICAL

PARAMETER	VALUE	NOTE
Active satellites	10,000+	crossed in 2026
Optical terminals per satellite	up to 3	laser inter-satellite links
Per-link capacity	100–200 Gbps	1550 nm, same window as subsea fibre
Laser mesh uptime	>99%	operator-reported
Daily constellation throughput	42+ PB	aggregate
Spectrum acquired from EchoStar	~65 MHz	AWS-3, AWS-4, H-block; FCC approved 12 May 2026
Direct-to-cell filing	15,000 satellites	VLEO, 326–335 km
Stated D2C target	150 Mbps	ITU Space Connect, Feb 2026
Orbital data centre filing	up to 1,000,000 satellites	FCC accepted 4 Feb 2026
Mini laser terminal	25 Gbps at 4,000 km	for third-party satellites

OPERATOR AND REGULATORY DISCLOSURES, 2026. THE FINAL TWO ROWS ARE FILINGS AND DEMONSTRATIONS RATHER THAN DEPLOYED CAPABILITY.

Each Starlink satellite carries up to three optical terminals operating near 1550 nm, the same window as subsea fibre, at 100 to 200 Gbps per link. SpaceX has flown tens of thousands of these terminals. The constellation routes traffic satellite to satellite without touching the ground, at better than 99 percent uptime. By any reasonable definition that is a mesh network, and it is the largest one ever built.

FIGURE 13: TWO MESHES, DIFFERENT LAYERS

STARLINK IS A MESH. IT IS NOT AN EDGE MESH.

AN INDEPENDENT MEASUREMENT STUDY FOUND STARLINK USUALLY TAKES A SINGLE HOP TO THE NEAREST GROUND STATION; LASERS CARRY TRAFFIC MAINLY WHERE NO GATEWAY IS IN RANGE.

ORBITAL MESH — MESHED
Starlink optical inter-satellite links
10,000+ satellites · 3 terminals each · 100–200 Gbps per link · >99% uptime

ACCESS LAYER — NOT MESHED
Direct-to-cell and terrestrial cellular
bent pipe to the nearest ground station · ~65 MHz exclusive spectrum · 150 Mbps target

EDGE MESH — MESHED
LoRa: Meshtastic and MeshCore
$15–50 nodes · 2–15 km per hop · unlicensed · no backhaul required

THE TWO MESHES SOLVE DIFFERENT PROBLEMS AT DIFFERENT LAYERS. THEY COMPETE ONLY AT CONSUMER MESSAGING, WHICH §8 ALREADY SHOWS CANNOT PAY FOR ITSELF.

THE ORBITAL LAYER IS MESHED AND THE ACCESS LAYER IS NOT. LORA MESHES THE EDGE, WHICH IS THE LAYER NEITHER OF THE OTHER TWO REACHES.

WHAT THE CLAIM GETS WRONG

An independent academic measurement study tracking real Starlink connections found that the network usually takes a *single hop to the nearest ground station* rather than routing through the constellation. The laser mesh carries traffic mainly where no gateway is in range: open ocean, polar regions, and remote terrain. For a typical user the architecture is a bent pipe, not a mesh.

More importantly, the meshing happens 550 km up, between assets owned by one company, under one jurisdiction, on spectrum that company purchased. **Starlink meshes the core. It does not mesh the edge.** A phone talking to a satellite is a star topology with a very tall centre. Nothing in the architecture lets two handsets three kilometres apart reach each other when the

27 The Delivery Tier

WHY TOPOLOGY INVERTS THE PROBLEM

Change the network topology from star to mesh and both properties invert. The unit of economic work stops being a radio's claimed position and becomes a packet that arrived. That single substitution does two things at once.

DELIVERY IS SELF-VERIFYING

The destination of a relayed packet is a second party with its own interest in the transaction. A relay cannot fabricate a delivery to a counterparty that received nothing, because the counterparty's signed acknowledgement is the receipt. Verification collapses from proof of location, which is unsolved, to proof of delivery, which is a signature over a path attestation. This is ordinary cryptography and it is cheap.

AIRTIME IS SCARCE

Mesh relay consumes a shared channel. Under duty-cycle limits and carrier-sense contention, channel time is rivalrous and congestible: the resource class that token markets price well, and the one coverage never was. The scarcity is computable from the waveform, which means the ceiling on the entire category can be derived.

TIME ON AIR

Time on air follows the Semtech modem design guide for a 48-byte payload, 16-symbol preamble, explicit header, CRC enabled, coding rate 4:5, with low-data-rate optimisation applied where the symbol period exceeds 16 ms. Receiver sensitivity is derived from the thermal noise floor consistent when bandwidth or spreading factor changes.

$$S = -174 + 10\log_{10}(BW) + NF + SNR_{\min}(SF) \tag{1}$$

$$T_{sym} = \frac{2^{SF}}{BW}, \qquad T_{oa} = (n_{pre} + 4.25)\,T_{sym} + n_{pay}\,T_{sym} \tag{2}$$

where

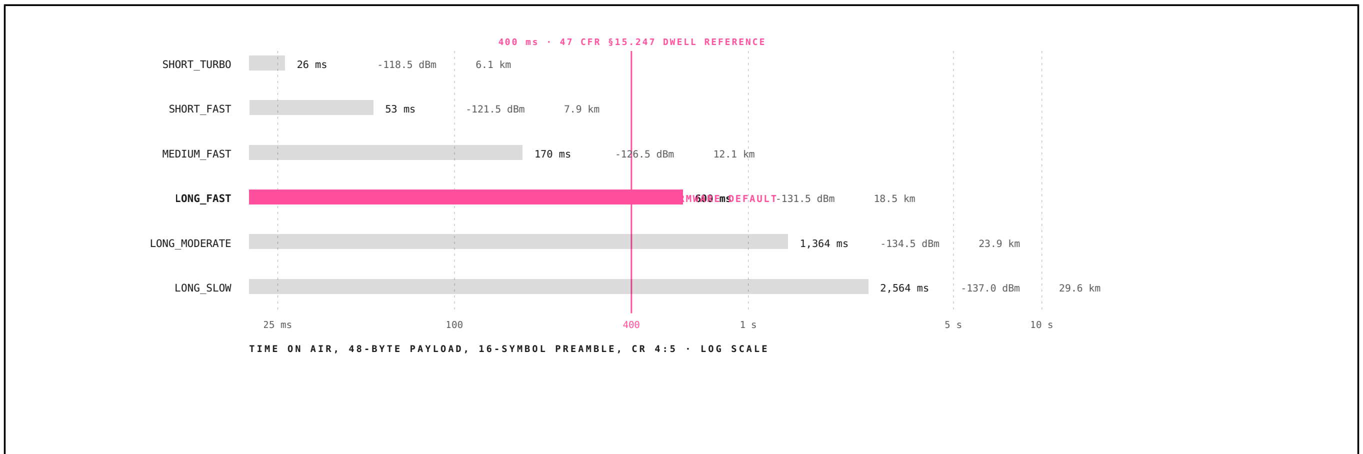
- S receiver sensitivity, dBm
- NF receiver noise figure, 6.0 dB for the SX126x
- n_{pay} payload symbol count, from the coding rate and header mode

TABLE 2: COMPUTED PERFORMANCE OF THE STANDARD MODEM CONFIGURATIONS

PRESET	SF	BW KHZ	SENS. DBM	BIT/S	TOA MS	RANGE KM	≤ 400 MS
SHORT_TURBO	7	500	-118.5	21875	**26**	6.1	yes
SHORT_FAST	7	250	-121.5	10938	**53**	7.9	yes
MEDIUM_FAST	9	250	-126.5	3516	**170**	12.1	yes
LONG_FAST	11	250	-131.5	1074	**600**	18.5	no
LONG_MODERATE	11	125	-134.5	537	**1364**	23.9	no
LONG_SLOW	12	125	-137.0	293	**2564**	29.6	no

SENSITIVITY FROM EQ. (1); TIME ON AIR FROM EQ. (2); RANGE AT n = 2.7 WITH A 10 dB MARGIN. THE FINAL COLUMN FLAGS THE 400 ms DWELL REFERENCE IN 47 CFR §15.247; WHICH PROVISION APPLIES DEPENDS ON THE EQUIPMENT AUTHORISATION.

FIGURE 2: TIME ON AIR AND THE DWELL BUDGET



28 Meshtastic and MeshCore

THE TWO FIRMWARES THE THESIS TURNS ON

Almost every deployed LoRa mesh runs one of two open-source firmwares, and the difference between them is the difference between a hobby network and a settlement layer. Any judgement about the delivery tier is a judgement about which one an operator is running.

MESHTASTIC

Started by Kevin Hester in 2019, Meshtastic is the incumbent by an enormous margin: roughly 40,000 GitHub stars, an 80,000-member subreddit, more than 100 supported hardware variants, and active meshes in most major American cities. It runs on $15 boards and requires no configuration to join a local network. That distribution is the reason the physical layer for this thesis already exists.

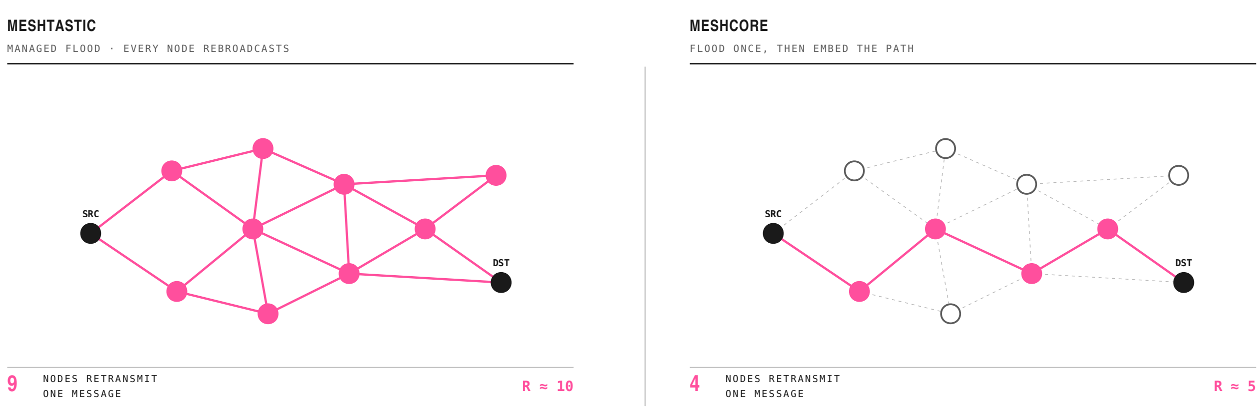
It routes by *managed flood*. A node that hears a packet it has not seen rebroadcasts it, after waiting a random number of slot times drawn from a contention window sized by observed channel utilisation, and suppresses its own transmission if it hears a neighbour rebroadcast first. The hop limit defaults to 3 and cannot exceed 7. Version 2.6 added next-hop routing for direct messages, which floods the first message to learn a path and then unicasts along it, but broadcast traffic remains flood-based.

This is an excellent design for its purpose. It self-organises, needs no planning, and has no operator role to misconfigure. It is also the direct cause of the capacity result in §8 and the reach collapse in §20.

MESHCORE

Released in early 2025, MeshCore was written by people who looked at flood routing, concluded it does not scale, and rebuilt the stack around structured routing. It floods once to discover a path, embeds that path in every subsequent packet, and falls back to flooding after roughly four failed retries. Node roles are explicit: companions are end devices that never relay, repeaters are dedicated infrastructure, and room servers hold store-and-forward state. Paths are observable and manually configurable by repeater operators. The hop ceiling is 64.

FIGURE 3: HOW ONE MESSAGE PROPAGATES UNDER EACH FIRMWARE



IDENTICAL ELEVEN-NODE TOPOLOGY, IDENTICAL SOURCE AND DESTINATION. UNDER MANAGED FLOOD EVERY INTERMEDIATE NODE RETRANSMITS. UNDER SOURCE ROUTING ONLY THE FOUR NODES ON THE DISCOVERED PATH DO. R IS THE RATIO THIS PRODUCES.

28 Meshtastic and MeshCore

CONTINUED: WHY THE DISTINCTION IS ECONOMIC

FEATURE BY FEATURE

TABLE 1: THE TWO FIRMWARES COMPARED

PROPERTY	MESHTASTIC	MESHCORE	REQUIRED FOR
Broadcast routing	managed flood, every node relays	**flood once, then source-routed**	settlement
Role separation	none, every node is a repeater	**companion / repeater / room server**	settlement
Hop ceiling	7, default 3	**64**	settlement
Delivery confirmation	ambiguous acknowledgement	**deterministic, per path**	settlement
Path visibility	opaque to operators	**observable and configurable**	settlement
Community	40k stars, 80k subreddit	smaller, growing	–
Hardware support	100+ variants	most of the same boards	–

THE FINAL COLUMN MARKS PROPERTIES A SETTLEMENT LAYER REQUIRES. MESHTASTIC WINS ON DISTRIBUTION; MESHCORE WINS ON EVERY PROPERTY A PAYMENT SYSTEM NEEDS.

Read MeshCore's feature list as a network engineer and it is source routing adapted to a low-bandwidth channel. Read it as an economist and it is a settlement-layer specification: identified counterparties, an auditable route, a deterministic delivery receipt, and a clean separation between the parties who supply infrastructure and the parties who consume it. Those are the four things a payment system needs, and the four things managed flood does not provide.

WHAT EACH FIRMWARE MAKES IMPOSSIBLE

Under managed flood there is no stable notion of *who carried this packet*. A message that arrives may have traversed any of several concurrent paths, and the nodes that rebroadcast it did so without knowing whether their transmission was the one that mattered. Rewarding that is rewarding participation in a broadcast, not the provision of a service. It is the coverage problem from §4 reappearing one layer up.

Under source routing the path is a first-class object. The originator knows which nodes it selected, each relay knows it was selected, and the destination can attest to the sequence. That is the difference between a network where a receipt can be constructed and one where it cannot.

THE DISTRIBUTION PROBLEM

The uncomfortable fact is that the architecture and the users are in different places. Meshtastic has an installed base measured in the tens of thousands of active nodes and a community that has been building since 2019. MeshCore has the routing model a payment system needs and a fraction of the deployment. A protocol betting on MeshCore is betting that operators migrate, or that a paid network is a sufficient reason to run second firmware on the same hardware. Technically trivial, socially unproven.

The two firmwares run on largely the same boards, which is the reason to think migration is possible at all. A Heltec V3 or RAK4631 flashed with MeshCore is the same $15–50 of hardware from §12. The switching cost is a firmware flash, not a capital expense.

THE INVESTMENT CONSEQUENCE

Meshtastic has the users. MeshCore has the architecture. A protocol that pays for relay on managed-flood routing pays operators to congest each other, because every node rebroadcasts everything and the reward scales with retransmission delivery. We regard MeshCore-class routing as the minimum viable substrate for an incentive layer, and would decline a team building on Meshtastic's broadcast path regardless of how large its community is.

29 Measuring the Rebroadcast Factor

PRINCIPAL CONTRIBUTION

Network capacity is time on air divided by R, the number of transmissions one originated message costs under flood routing. R is the single most consequential parameter in mesh economics and it is almost universally assumed. Our own earlier work assumed 3–5.

METHOD

The Meshtastic project publishes a discrete-event simulator that models the production firmware's managed-flood logic, LoRa airtime, channel-activity detection, contention windows and collision behaviour. It is maintained by the project itself, which makes it the most defensible source short of field measurement. We wrote a headless driver and ran three repetitions per configuration across node counts from 5 to 75 and hop limits of 3, 5 and 7, with uniform random node placement in a fixed area.

$$R = \frac{\sum_{i \in N} \mathrm{tx}_i}{|\mathcal{M}_{\mathrm{orig}}|} \tag{3}$$

where

- tx_i transmissions emitted by node i over the run
- $\mathcal{M}_{\mathrm{orig}}$ the set of messages originated, not relayed

TABLE 3: SIMULATION RESULTS, HOP LIMIT 3 (FIRMWARE DEFAULT)

NODES	R	REACH	USEFUL PKTS	COLLISIONS	TX AIRTIME
5	3.12	68.6%	50.0%	7.4%	3.46%
20	5.88	49.8%	51.6%	17.1%	5.85%
30	7.19	47.7%	47.4%	22.9%	6.83%
50	7.36	32.2%	45.3%	22.2%	6.87%
75	7.19	25.8%	43.2%	19.2%	6.08%

THREE REPETITIONS PER ROW. REACH IS THE SHARE OF THE NETWORK RECEIVING A GIVEN MESSAGE. USEFUL PACKETS IS THE SHARE OF RECEPTIONS CARRYING A NON-DUPLICATE. ROWS BELOW 35% REACH ARE FLAGGED.

TABLE 4: SENSITIVITY TO HOP LIMIT

HOP LIMIT	NODES	R	REACH	COLLISIONS
3	10	4.07	57.0%	10.7%
5	10	5.26	76.7%	11.2%
5	20	7.30	58.5%	21.1%
5	30	7.82	43.6%	21.8%
7	20	6.75	57.5%	20.4%
7	30	9.16	44.2%	22.3%

HOP LIMIT RAISES R FASTER THAN IT RAISES REACH: AT 30 NODES, MOVING FROM HOP 3 TO HOP 7 COSTS 27% MORE TRANSMISSIONS FOR 7% MORE OF THE NETWORK.

RESULT

At realistic density **R = 7.36**, roughly 47 percent above the upper bound of the assumed range. R rises steeply to about 30 nodes and then plateaus near 7.2, because the hop limit truncates propagation before density can push it further. Below about 10 nodes the assumed range is defensible; above 20 it is not.

> **CORRECTION TO OUR OWN PRIOR WORK**
>
> Earlier drafts of this thesis used R = 3–5 and reported a revenue ceiling $21,022 per collision domain per year at $0.01 per message. The measured value reduces that to $14,291, a fall of 32 percent. The measurement was made after the thesis was drafted and moved the

30 The Capacity Ceiling

AND WHAT IT IS WORTH

One collision domain on one frequency slot admits 5,999 transmissions per hour at full occupancy. Past roughly 20 percent, collision and retransmission degrade the network faster than offered load rises, so 20 percent is taken as the working ceiling throughout. Delivered messages follow directly:

$$C = \frac{u \cdot 3600}{T_{oa} \cdot R} \quad \Longrightarrow \quad \Pi = 8760\, p\, C \tag{4}$$

where

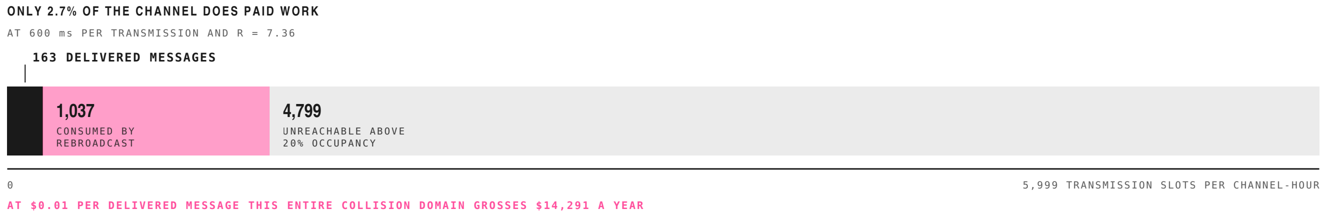
- u target channel occupancy, taken as 0.20
- C delivered messages per hour, whole network
- p price per delivered message
- Π gross annual revenue, one collision domain

TABLE 5: NETWORK CAPACITY AND GROSS REVENUE BY CONFIGURATION

PRESET	TOA MS	AIRTIME/MSG S	MSG/H 10%	MSG/H 20%	MSG/H 30%	$/YR $0.01
SHORT_TURBO	26	0.19	1851	**3703**	5555	$324,432
SHORT_FAST	53	0.39	925	**1851**	2777	$162,216
MEDIUM_FAST	170	1.25	287	**574**	861	$50,297
LONG_FAST	600	4.41	81	**163**	244	$14,291
LONG_MODERATE	1364	10.03	35	**71**	107	$6,287
LONG_SLOW	2564	18.86	19	**38**	57	$3,344

COMPUTED AT R = 7.36 FROM TABLE 2. THE FINAL COLUMN IS THE THEORETICAL GROSS REVENUE OF AN ENTIRE COLLISION DOMAIN AT ONE CENT PER DELIVERED MESSAGE, BEFORE ANY AUTOMATIC HOUSEKEEPING TRAFFIC IS SUBTRACTED.

FIGURE 4: WHERE A CHANNEL-HOUR GOES



EVERY SQUARE IS ONE TRANSMISSION OPPORTUNITY. THE DARK BLOCK AT TOP IS THE ENTIRE DELIVERED MESSAGE VOLUME; THE RED BLOCK IS AIRTIME CONSUMED BY REBROADCAST; THE REMAINDER IS UNREACHABLE ABOVE 20 PERCENT OCCUPANCY.

TABLE 6: ANNUAL REVENUE AGAINST R AND PRICE

R	BASIS	MSG/H	$0.001	$0.01	$0.10	$1.00
3.00	assumed, low	400	$3,504	$35,036	$350,363	$3,503,626
5.00	assumed, high	240	$2,102	$21,022	$210,218	$2,102,176
4.07	measured, 10 nodes	295	$2,583	$25,825	**$258,253**	$2,582,526
5.88	measured, 20 nodes	204	$1,788	$17,876	**$178,756**	$1,787,564
7.19	measured, 30 nodes	167	$1,462	$14,619	**$146,187**	$1,461,875
7.36	measured, 50 nodes	163	$1,429	$14,291	**$142,908**	$1,429,079
9.16	measured, 30 nodes, hop 7	131	$1,148	$11,480	**$114,798**	$1,147,977

BOLD ROWS ARE MEASURED.

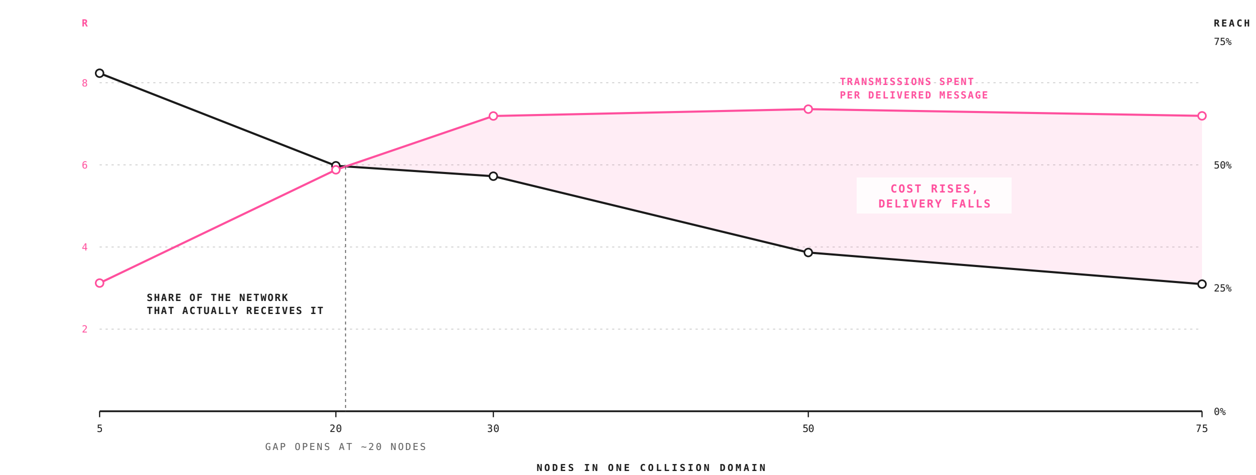
FIGURE 5: ANNUAL REVENUE AGAINST PRICE, AT ASSUMED AND MEASURED R

31 Reach Collapse

AN UNANTICIPATED RESULT

The simulation produced a second result we did not expect and did not design for. Reach: the share of the network that receives a given message, falls monotonically as node count rises, from 68.6 percent at 5 nodes to 25.8 percent at 75.

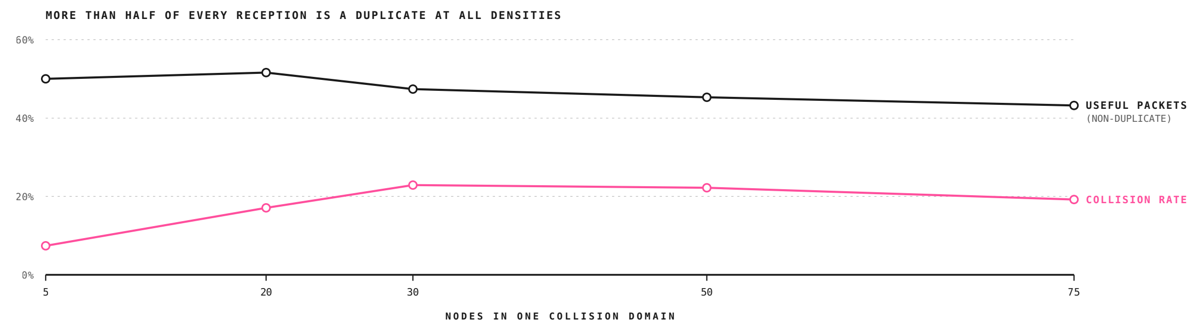
FIGURE 6: COST AND DELIVERY DIVERGE ABOVE ~20 NODES



R AND REACH PLOTTED ON A COMMON HORIZONTAL AXIS AGAINST NODE COUNT. THE SHADED REGION IS THE WIDENING GAP BETWEEN WHAT A MESSAGE COSTS AND WHAT IT DELIVERS. HOP LIMIT 3, THREE REPETITIONS PER POINT.

At the firmware default hop limit, a 75-node mesh delivers each message to roughly a quarter of the network while spending 7.2 transmissions to do it. **Adding nodes makes the mesh worse at its core function, not merely less efficient.** Raising the hop limit does not rescue this: R climbs to 9.16 at hop 7 for modest gains in reach.

FIGURE 7: COLLISIONS RISE, USEFUL PACKETS DO NOT



MORE THAN HALF OF EVERY RECEPTION IS A DUPLICATE AT ALL TESTED DENSITIES. THE COLLISION RATE MORE THAN TRIPLES BETWEEN 5 AND 30 NODES.

CORROBORATION IN THE FIELD

The result is consistent with two documented incidents. At Hamvention 2024 a regional Meshtastic network collapsed when a single user's MQTT bridge overwhelmed it. DEF CON has required specialised event firmware, short-turbo modulation with rebroadcasting restricted, to support 2,000 to 2,500 nodes in one venue. Both are the predicted failure, at the predicted scale, in production.

IMPLICATION FOR INCENTIVE DESIGN

This is sharper than the airtime ceiling and it is the most important paragraph in this report. A flooded mesh degrades before it monetises. An incentive layer that pays per rebroadcast, in a system where every node rebroadcasts everything, is a subsidy for congestion, and it pays operators to make the network worse for each other. Routing architecture is therefore a precondition.

MeshCore, released in 2025, is the relevant counterexample. It floods once to discover a path, embeds the discovered path in subsequent packets, separates repeater roles from client roles so that end devices do not relay, and supports 64 hops against Meshtastic's 7. Read as a network engineer, that is source routing adapted to a low-bandwidth channel. Read as an economist, it is a settlement-layer specification: identified counterparties, an auditable route, and a deterministic delivery receipt. We regard

32 Regulatory Ceilings

CONSTRAINTS PROTOCOL DESIGN CANNOT LIFT

Two of the constraints on this category are legal can be engineered around. They are stated here because they bound the terminal value of any investment in the delivery tier.

FIGURE 8: WHERE AIRTIME SCARCITY COMES FROM

	UNITED STATES	EUROPEAN UNION	AMATEUR BANDS
	47 CFR §15.247	EN 300 220 SRD	47 CFR §97.113
POWER CEILING	1 W conducted	+27 dBm ERP	1.5 kW
DUTY CYCLE	no duty cycle	10% rolling hourly	no duty cycle
AIRTIME IS SCARCE BECAUSE OF	PHYSICS AND CONTENTION	REGULATION	CLOSED TO PAID NETWORKS

THE SAME PROTOCOL FACES THREE DIFFERENT BINDING CONSTRAINTS. A DEPLOYMENT PLAN THAT TREATS THEM AS ONE MARKET IS WRONG.

THE UNITED STATES CONSTRAINS BY PHYSICS, THE EUROPEAN UNION BY REGULATION, AND THE AMATEUR ALLOCATION IS CLOSED TO PAID NETWORKS ENTIRELY.

THE FINAL COLUMN IS WHAT ACTUALLY LIMITS AIRTIME IN EACH REGIME. THE US AND EU CASES ARE STRUCTURALLY DIFFERENT MARKETS FOR THE SAME PROTOCOL.

THE TRANSATLANTIC ASYMMETRY

In the United States, LoRa operates under 47 CFR §15.247 as a digital transmission system: one watt conducted, and *no duty-cycle limit*. Airtime is scarce because of physics and contention. In the European Union the same protocol runs in a narrow allocation at a 10 percent duty cycle enforced on a rolling hourly basis. Airtime is scarce because of regulation, and the cap binds regardless of how good the routing gets.

The consequence is that European deployments have a hard regulatory ceiling on revenue per node that no protocol improvement can lift, and that the marginal value of airtime is structurally higher in Europe than in the United States. Any deployment plan treating them as one market is wrong. Meshtastic exposes an override for the duty-cycle limiter; a paid network would give operators an economic reason to enable a setting whose use is straightforwardly unlawful in the EU.

THE AMATEUR-BAND TRAP

Meshtastic offers a licensed mode that lifts power limits by asserting operation under an amateur licence. **No incentivised network may use it.** 47 CFR §97.113(a) prohibits an amateur station from transmitting communications for hire or material compensation, direct or indirect, paid or promised; communications in which the licensee or control operator holds a pecuniary interest; or messages encoded for the purpose of obscuring their meaning.

Token rewards are compensation under (a)(2). An operator holding the network's token has a pecuniary interest under (a)(3). Meshtastic encrypts by default, which conflicts with (a)(4) irrespective of payment. Nor is this a rule that could plausibly be amended: non-commerciality is the consideration amateurs pay for extraordinarily valuable spectrum allocated at zero cost, and a petition to permit paid amateur operation is a petition to hand a commercial operator free prime spectrum.

> **TERMINAL CONSTRAINT**
>
> Any incentivised mesh in the United States is confined to Part 15 unlicensed operation permanently. There is no licensed fallback, no power uplift, and no regulatory path to one. This is a hard ceiling on link budget and it is not a protocol-design problem.

A practical filter follows: any team pitching an incentivised mesh that has not read Part 97 §113 will propose the licensed-mode power uplift within two meetings. It is a fast and cheap diligence screen.

33 Adversarial Behaviour Under Payment

THE RUSHING PROBLEM

The delivery tier inherits twenty-five years of mobile ad-hoc network security literature, most of which was written for unpaid networks. Introducing payment changes which attacks matter, and in one case converts vandalism into a business model.

RUSHING

In on-demand routing only the first copy of a route request to arrive is forwarded; later copies are suppressed as duplicates. An attacker who propagates route requests faster than everyone else, by skipping the contention backoff, transmitting at higher power, or tunnelling out of band, captures a position on essentially every route the network discovers. Hu, Johnson and Perrig described this in 2003 and showed that existing on-demand protocols cannot deliver over paths longer than two hops in its presence.

In an unpaid mesh this is vandalism. **In a paid mesh it is a business model.** The rushing attacker sits on every route and collects transit fees across all of it while *behaving correctly by the protocol's own definition*, actually forwarding packets, earning. There is no fraud to detect. The attacker simply out-competes polite nodes at the MAC layer.

Worse, this is precisely the incentive gradient a naive reward creates. Pay per delivered packet and every rational operator disables backoff and raises power. The polite-rebroadcast behaviour that keeps Meshtastic functional today survives only because nobody is paid to defect from it. The remedy is that route selection must not be first-arrival: secure route discovery, randomised selection among several valid paths, and reputation weighting all break the rushing advantage, at a cost in discovery latency.

WORMHOLES

An attacker records packets at one location, tunnels them out of band, and replays them elsewhere, creating a phantom link. This works *even if all traffic is authenticated and encrypted and no host is compromised*. The standard defences are geographic leashes, in which packets carry signed position and timestamp, and temporal leashes requiring microsecond clock synchronisation.

Both reintroduce a dependency on trusted location or trusted time, exactly the dependency the delivery tier claims to escape. This is a genuine concession and we make it plainly. LoRa's low bit rate makes temporal leashes harder, since propagation delay is negligible against a 600 ms packet, while making wormholes more expensive to operate at scale.

COLLUSION AND SYBIL RESISTANCE

The argument that collusion is self-limiting, that synthetic traffic burns real airtime and battery, holds only while the reward per delivered message sits below the marginal cost of a fabricated delivery. From §6 the viable price point is around ten cents. At ten cents a message, the cost of faking a delivery is a $30 radio and 600 milliseconds of airtime. **The self-limiting argument is weakest at exactly the price point where the business works.**

The mitigation is that synthetic traffic must pass through another party's metering to be paid, and that stake-weighted routing raises the cost of fabricating counterparties. We state this as bounded by staking economics and regard calibrating that bound as the central unsolved protocol-design risk in the delivery tier.

GRAY HOLES

An operator that drops ten percent of traffic while forwarding the rest stays beneath most statistical detectors and collects nearly full fees. Detection requires end-to-end delivery accounting, which is a further argument for receipts as the settlement primitive.

34 Settlement Over a Constrained Link

WHAT A TRANSACTION COSTS IN AIRTIME

Section 8 concluded that only payloads worth more than roughly ten cents a message clear on mesh transit, and named financial settlement as the leading candidate. That conclusion is worth testing against the actual wire cost of a transaction, because a signed transaction is far larger than a text message and the mesh charges by airtime.

Meshtastic's usable single-packet payload after headers and encryption is about 200 bytes. A transaction longer than that is chunked, and every chunk pays the rebroadcast multiplier from §17, because a relayed fragment floods exactly like any other message. The deployed bridges make this worse: BTC Mesh and the MeshtasticBitcoinCore bridge both transmit the transaction as hexadecimal text, which doubles the byte count. The figures below assume binary encoding, and are therefore optimistic by a factor of two against what is running today.

$$A_{tx} = R[\lfloor b/P \rfloor \, T_{oa}(P) + T_{oa}(b \bmod P)] \tag{5}$$

where

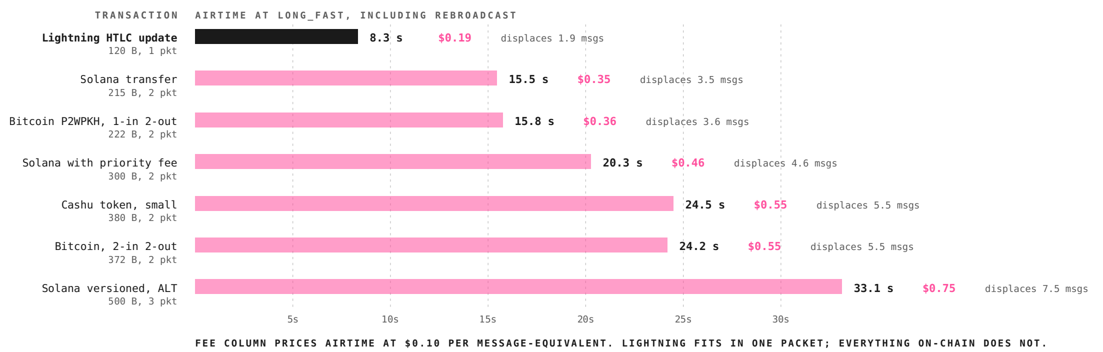
- b transaction size in bytes; P the usable payload, 200 B
- A_{tx} total airtime consumed, including rebroadcast

TABLE 11: AIRTIME COST OF ONE SETTLED TRANSACTION

SIGNED TRANSACTION	BYTES	PACKETS	AIRTIME S	CHANNEL-HOUR	MSGS DISPLACED	FEE $0.10
Lightning HTLC update	120	1	8.3	1.16%	1.9	**$0.19**
Solana transfer	215	2	15.5	2.15%	3.5	**$0.35**
Bitcoin P2WPKH, 1-in 2-out	222	2	15.8	2.19%	3.6	**$0.36**
Solana with priority fee	300	2	20.3	2.82%	4.6	**$0.46**
Cashu token, small	380	2	24.5	3.40%	5.5	**$0.55**
Bitcoin, 2-in 2-out	372	2	24.2	3.36%	5.5	**$0.55**
Solana versioned, ALT	500	3	33.1	4.60%	7.5	**$0.75**

COMPUTED AT LONG_FAST AND THE MEASURED R. THE FEE COLUMN PRICES AIRTIME AT $0.10 PER MESSAGE-EQUIVALENT, WHICH IS THE CLEARING PRICE DERIVED IN §8.

FIGURE 11: WHAT ONE TRANSACTION COSTS THE NETWORK



LIGHTNING IS THE ONLY INSTRUMENT THAT FITS IN A SINGLE PACKET. EVERY ON-CHAIN TRANSACTION SPANS TWO OR THREE, AND PAYS THE REBROADCAST MULTIPLIER ON EACH.

THE NUMBER THAT MATTERS

A Solana transfer costs 15.5 seconds of channel time and displaces 3.5 ordinary messages, which prices it at $0.35. On a $200 remittance that is 18 basis points, against 5 to 7 percent for incumbent corridors. The economics clear. They clear by a wide margin. What they do not do is clear at volume, which is §25.

35 What the Settlement Layer Must Look Like

REQUIREMENTS ON THE L1

A mesh is a delay-tolerant, low-bandwidth, intermittently-partitioned network. Most blockchain designs assume none of those conditions. This section states what a chain must support to settle over one, because the choice is not neutral and it narrows the field considerably.

TABLE 12: SETTLEMENT CAPACITY BY MODEM PRESET

PRESET	TOA MS	PACKETS	AIRTIME S	SOLANA TX / HOUR, ONE DOMAIN
SHORT_TURBO	26	2	0.7	**1,029**
SHORT_FAST	53	2	1.4	**515**
MEDIUM_FAST	170	2	4.5	**159**
LONG_FAST	600	2	15.5	**47**
LONG_MODERATE	1364	2	36.3	**20**
LONG_SLOW	2564	2	65.4	**11**

SOLANA TRANSFERS PER HOUR THAT ONE COLLISION DOMAIN CAN CARRY AT 20 PERCENT OCCUPANCY. THE LONGEST-RANGE PRESETS ARE UNUSABLE FOR SETTLEMENT AT ANY MEANINGFUL RATE.

FOUR HARD REQUIREMENTS

1. Offline construction and deferred broadcast. The signing device is not connected. The chain must permit a transaction to be built, signed and held, then broadcast minutes or hours later by an unrelated party. Bitcoin and Solana both allow this; chains with short-lived nonces or tight recent-blockhash windows do not. Solana's blockhash expiry is the specific constraint here and durable nonces are the specific answer.

2. Transaction size as a first-order design variable. At 200 bytes per packet and a rebroadcast multiplier near seven, every additional 200 bytes costs roughly 8 seconds of shared channel time. This inverts the usual ordering of chain selection criteria: throughput and fees matter less than wire size. A chain whose typical transaction is 1,200 bytes is six packets, not one, and is effectively unusable.

3. Sub-cent fees, denominated stably. A ten-cent transit fee cannot sit on top of a five-cent gas fee and remain interesting against remittance incumbents. This is the requirement that most cleanly separates candidate chains, and it is the reason the category is Solana-adjacent in practice.

4. Light verification over a constrained link. A node that cannot fetch state cannot verify anything. Either the mesh carries a compact proof, or the edge trusts the gateway. The honest position today is that off-grid settlement is *broadcast-assisted*, not trustless at the edge: the sender signs offline and trusts that some connected node will relay honestly, which it is incentivised to do only if the receipt in §17 is enforceable.

> **WHERE THIS POINTS**
>
> Payment channels dominate on wire size. A Lightning update is 120 bytes and fits in one packet, against 215 for a Solana transfer and 500 for a versioned transaction with lookup tables. The architecture the physics prefers is a channel or ecash instrument settled off-mesh, not on-chain transfers pushed through a 1 kbps link. Bitchat already relays Cashu tokens for this reason, probably by instinct.

35 What the Settlement Layer Must Look Like

CONTINUED: CHAIN SELECTION

The four requirements above narrow the field sharply, and they narrow it on an axis almost nobody uses to choose a chain.

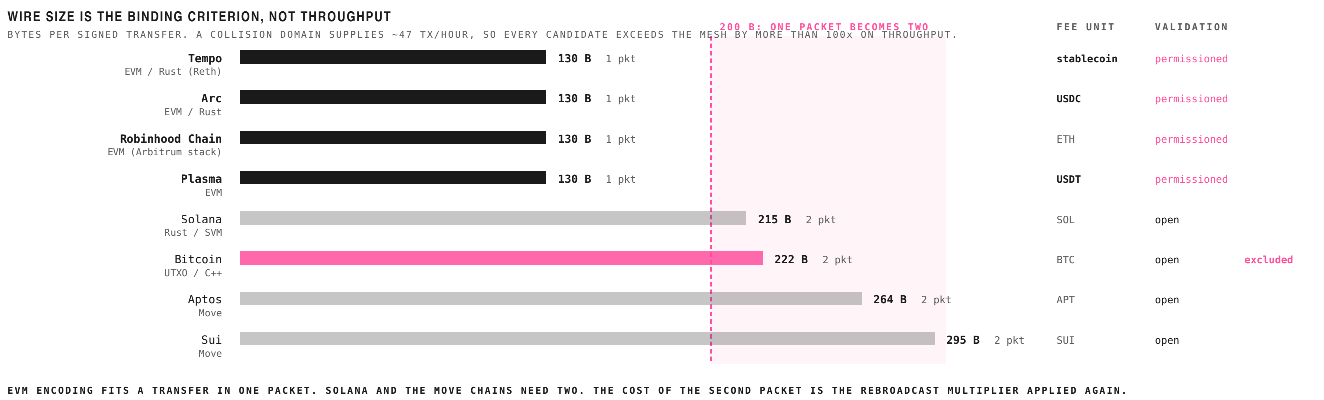
TABLE 19: SELECTION CRITERIA, WEIGHTED

CRITERION	WEIGHT	WHY
Wire size	**binding**	every 200 bytes is another packet and another R multiple
Fee denominated stably	**binding**	the transit fee sits on top of it
Offline construction validity	**binding**	the signer is not connected
Permissionless validation	conditional	required where the demand driver is censorship, optional where it is coverage
Throughput	not binding	a collision domain supplies ~47 tx/hour; every candidate exceeds this by 100x or more
Smart-contract expressiveness	secondary	receipts and attestations are simple; Move resources are a convenience, not a requirement

THE FIFTH ROW IS THE COUNTERINTUITIVE ONE AND IT DISPOSES OF THE USUAL ARGUMENT ENTIRELY.

Throughput does not matter here. Table 12 shows one collision domain can feed a chain roughly 47 transactions an hour at the US default preset. Bitcoin's 7 transactions per second is 25,200 an hour, which is more than five hundred times what the mesh can supply. Every candidate chain exceeds the mesh by two orders of magnitude or more. Selecting a settlement layer for a mesh on throughput grounds is selecting on the one dimension that is not binding.

FIGURE 15: WIRE SIZE IS THE BINDING CRITERION



BYTES PER SIGNED TRANSFER AGAINST THE 200-BYTE PACKET BOUNDARY. CROSSING IT DOUBLES THE PACKET COUNT AND APPLIES THE REBROADCAST MULTIPLIER AGAIN.

TABLE 20: CANDIDATE CHAINS ON WIRE SIZE

CHAIN	FAMILY	BYTES	PACKETS	AIRTIME S	FEE UNIT	VALIDATION	NOTE
Tempo	EVM / Rust (Reth)	**130**	**1**	8.9	stablecoin	**permissioned**	Stripe and Paradigm; no native token; fee AMM in stablecoins
Arc	EVM / Rust	**130**	**1**	8.9	USDC	**permissioned**	Circle; USDC-native gas; not live as of mid-2026
Robinhood Chain	EVM (Arbitrum stack)	**130**	**1**	8.9	ETH	**permissioned**	launched July 2026; tokenised equities focus
Plasma	EVM	**130**	**1**	8.9	USDT	**permissioned**	Tether; zero-fee USDT transfers
Solana	Rust / SVM	215	2	15.5	SOL	open	durable nonces solve deferred broadcast explicitly
Bitcoin	UTXO / C++	222	2	15.8	BTC	open	

36 The Move Chains

READ FROM THE REPOSITORIES

Wire size ranks Aptos and Sui last among the permissionless candidates. On the requirement that decides this, deferred broadcast, the ordering reverses. Both claims are checked against the source.

TABLE 21: DEFERRED BROADCAST, CHAIN BY CHAIN

CHAIN	MECHANISM	FROM THE SOURCE	SUPPORT	CAVEAT
Aptos	expiration_timestamp_secs	"This can be set to a large value far in the future to indicate that a transaction does not expire."	native	sequence number orders it; nothing else can stale it
Solana	durable nonce account	a normal blockhash expires in roughly 60 seconds	workaround	requires a pre-created nonce account and an extra instruction
Sui	TransactionExpiration::None	"The transaction has no expiration"	partial	but GasData.payment carries an ObjectRef with version and digest
Bitcoin	none	mempool eviction, fee market movement, RBF races	**absent**	and UTXO state cannot be checked offline at all

QUOTED TEXT IS FROM APTOS-LABS/APTOS-CORE AND MYSTENLABS/SUI.

Aptos is the cleanest design available for this use case. Its `RawTransaction` carries an `expiration_timestamp_secs` chosen by the sender, and the source comment says outright it can be set far enough ahead that the transaction does not expire. Ordering falls to a monotonic sequence number. A courier carrying a signed transaction across a mesh for six hours is the normal case the chain tolerates; it is what the field was written for. Solana reaches the same place with a durable nonce account, which works but exists to defeat a 60-second blockhash window design. Sui declares `TransactionExpiration::None`, but `GasData.payment` is a `Vec<ObjectRef>`, and an `ObjectRef` carries a version and a digest: if the gas coin moves, the transaction is stale. Sui's newer `ValidDuring` variant removes that dependency by paying from address balances and in exchange *bounds* expiration, which is the opposite direction from what a partitioned network wants.

37 Why Not Bitcoin

FIVE FAILURE MODES, AND A NOTE ON BITCHAT

Bitcoin is the chain every prior mesh experiment chose. All six transport bridges in §29 relay Bitcoin, and Bitchat relays Bitcoin and Cashu. It is also, on the analysis in §23, the wrong instrument, and the reasons are specific.

TABLE 21: WHY THE BASE CHAIN FAILS HERE

FAILURE MODE	MECHANISM	CONSEQUENCE
Fee volatility	a $0.10 transit fee cannot sit beneath a network fee that has repeatedly spiked above $5 and occasionally above $50	the transit price in §18 becomes non-deterministic and the corridor economics stop clearing
Deferred broadcast is unsafe	a transaction signed offline may face mempool eviction, a fee market that moved, or an RBF race by the time a gateway sees it hours later	fee bumping and CPFP both require connectivity the sender does not have
UTXO state cannot be verified offline	signing requires knowing your outputs are unspent; a partitioned node cannot check this	double-spend exposure sits with the sender, which is the wrong party
No stable unit of account	corridors price in dollars; BTC-denominated value moves against the remittance while it is in flight	the settlement instrument has to be a stablecoin or ecash, not the base asset
Lightning punishes exactly this failure mode	channels require liveness to monitor for old-state broadcast; watchtowers assume connectivity	a node partitioned for days is the specific case Lightning penalises, and mesh partitions are routine

THE FIRST TWO ROWS ARE ECONOMIC AND THE THIRD IS THE ONE MOST OFTEN MISSED. THE FIFTH DISPOSES OF THE OBVIOUS COUNTERARGUMENT.

THE OFFLINE-SIGNING PROBLEM, STATED PRECISELY

A UTXO transaction is only valid if its inputs are unspent. A sender who has been partitioned for hours cannot verify that, and cannot know whether a wallet on another device has spent the same output. In an account model the sequence number or nonce resolves this deterministically: the transaction either applies at that sequence or it does not, and the failure is visible and safe. In a UTXO model the failure mode is a double spend the sender did not intend and cannot detect, and the loss falls on the recipient who accepted the offline transaction. **The exposure sits with the wrong party for a corridor where the recipient is the party with less power.**

LIGHTNING DOES NOT RESCUE IT

Lightning is the natural rebuttal: 120 bytes, one packet, the smallest instrument in Table 20. But payment channels require liveness. A channel partner who is offline cannot contest an old-state broadcast, and watchtowers exist precisely because that assumption fails. A mesh node partitioned for days is the normal case in this architecture, it is the normal operating condition. **Lightning penalises exactly the failure mode the mesh is built to survive.**

WHERE THIS LEAVES BITCHAT

Bitchat is cited throughout this report as the clearest demand signal in the category, and §2 treats its 430,000 daily active users in India as evidence that people want permissionless local messaging. That reading stands. As a supply-side architecture we are bearish, and the reasons compound.

It runs on Bluetooth Low Energy mesh at 30 to 300 metres, so it works in a crowd and nowhere else. Its predecessors FireChat and Bridgefy both spiked during protests and both degraded under exactly the density that made them useful: RF congestion, battery drain and message loss at the moment of peak need. FireChat shut down in 2019; Bridgefy failed security audit on encryption and user tracking. BLE mesh has no scheduling discipline and no hop accounting, making it a worse case of the flooding problem quantified in §19 rather than a better one. It relays Bitcoin, which fails on all five rows above, and Cashu, which is the correct shape but still requires a mint the partitioned user must trust. **Bitchat is a demand signal, not a supply solution, and it will not scale past the crowd.** The 430,000 figure is a shutdown spike with no published retention curve, and the

38 Mesh Throughput Will Scale Like L1s Did

THE STRONGEST REASON TO BE EARLY

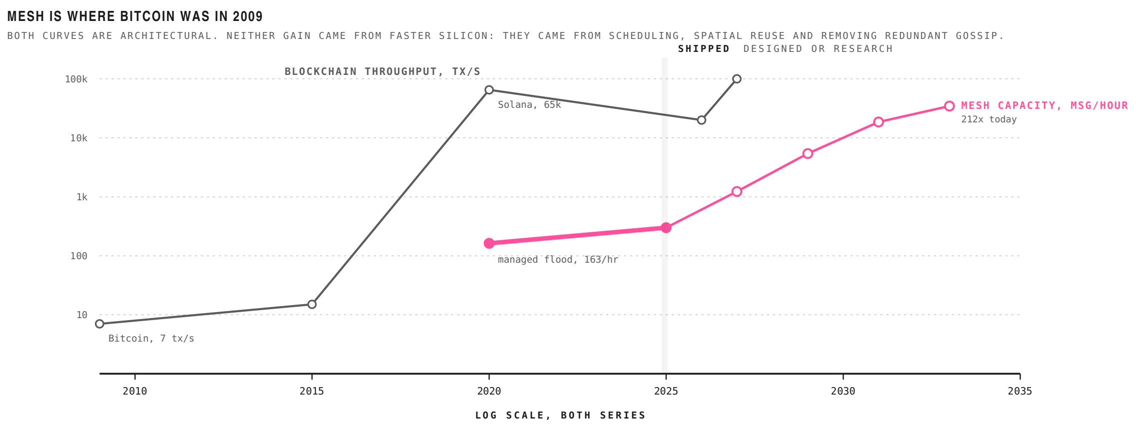
Every capacity figure in this report is computed at the architecture that exists in 2026: managed flood, contention-based access, one channel, omnidirectional antennas. That is not a physical limit. It is the same kind of limit Bitcoin had at 7 transactions per second, and it yielded to the same kind of fix.

THE PRECEDENT

Bitcoin in 2009 flood-gossiped every transaction to every node and had every node validate everything. Ethereum kept the model. Solana in 2020 reached roughly 65,000 transactions per second by removing mempool gossip, scheduling a leader in advance, and executing in parallel. Tempo reached 20,000 on testnet in 2026 against a stated architectural target above 100,000. **None of that gain came from faster silicon.** It came from scheduling, from eliminating redundant broadcast, and from letting non-conflicting work proceed concurrently.

Those are the three things a flooded LoRa mesh does not do.

FIGURE 16: TWO ARCHITECTURAL SCALING CURVES



BLOCKCHAIN THROUGHPUT AND MESH CAPACITY, BOTH LOG SCALE. THE HEAVY RED SEGMENT IS SHIPPED; THE REST IS DESIGNED OR RESEARCH. THE SHAPE IS THE ARGUMENT.

TABLE 22: THE ROADMAP, COSTED IN CAPACITY

ARCHITECTURE	YEAR	REUSE	R	UTIL.	MSG/HOUR	VS TODAY	STATUS
Managed flood	2020	1×	7.36	20%	**163**	1×	shipped
Source routing	2025	1×	4.00	20%	**300**	2×	shipped, MeshCore
Scheduled access	2027	1×	2.20	45%	**1,227**	8×	designed elsewhere
Multi-channel reuse	2029	4×	2.00	45%	**5,399**	33×	no protocol does this
Directional and spatial reuse	2031	9×	1.60	55%	**18,561**	114×	requires new hardware
Network coding	2033	12×	1.25	60%	**34,556**	212×	research

EACH STEP IS A PROTOCOL CHANGE ON THE SAME RADIOS EXCEPT THE FIFTH, WHICH NEEDS SECTORISED REPEATERS. CAPACITY IS DELIVERED MESSAGES PER HOUR FOR ONE COLLISION DOMAIN, COMPUTED FROM THE SAME MODEL AS §19.

Scheduled access alone, which is well understood and deployed in other LPWAN systems, removes most contention and cuts R by nearly half: an eightfold gain with no new hardware. Coordinated multi-channel operation across the 104 US frequency slots is a further four times. Sectorised repeaters and network coding are further out and require hardware and research respectively. The endpoint is roughly **212 times today's capacity**, which is the same order of magnitude as the blockchain gain over a comparable period.

WHAT SCALING DOES AND DOES NOT FIX

It does **not** rescue the transit business. §27 shows demand saturates at 6,721 nodes at today's architecture; at the 2033 architecture the same demand is served by roughly 32 nodes, because capacity per domain rises while addressable transit demand does not. Scaling makes the oversupply problem worse, not better.

39 Addressable Market

BUILT FROM TRANSACTION COUNTS, NOT FROM PERCENTAGES

The temptation in this category is to multiply a large population by a small penetration rate and report the product. We build from transactions instead, because §19 gives a defensible price per transaction and the physics gives a defensible ceiling on how many of them a network can carry.

TABLE 13: TRANSIT REVENUE, BOTTOM-UP

SEGMENT	ASSUMPTION	USERS	TX/YR	TX TOTAL	TRANSIT REVENUE
Shutdown-corridor settlement	300M people affected by internet shutdowns in a recent year; assume 0.5% adopt an off-grid rail and settle 12 times a year	1,500,000	12	18.0M	**$6.3M**
Remittance in low-connectivity corridors	assume 250,000 recurring users settling twice monthly where the terrestrial rail is unreliable rather than absent	250,000	24	6.0M	**$2.1M**
Merchant and agent settlement	assume 40,000 agents reconciling 200 times a year in markets where connectivity is the binding constraint on cash-out	40,000	200	8.0M	**$2.8M**
Total addressable transit	at $0.35 per settled transaction	–	–	**32M**	**$11.2M**

EACH ROW IS AN ASSUMPTION MULTIPLIED BY A STATED FREQUENCY. THE ASSUMPTIONS ARE DELIBERATELY MODEST AND ARE THE FIRST THING TO ARGUE WITH.

The total is **$11.2 million a year** of addressable transit revenue across every segment where an off-grid rail is plausibly the difference between a transaction happening and not happening. That is a small number and it should be read as one. It is roughly the current annualised revenue of GEODNET, a single-payload positioning network, and it is the *entire* transit opportunity rather than one company's share of it.

TABLE 14: LINES THAT DO NOT PRICE PER MESSAGE

NON-TRANSIT LINE	ANNUAL	BASIS
GNSS interference monitoring	**$40–120M**	aviation, maritime, insurance and critical-timing buyers; GEODNET reaches ~$10M on positioning alone and interference is a newer, more urgent budget
Sensor backhaul displacement	**$25–60M**	every sensor network pays separately for connectivity; Hivemapper alone is 100,000 devices at $19/month; assume 5% of comparable fleets migrate
Public safety and emergency	**$15–40M**	US local emergency management, CERT and wildland fire; a $50 supplement to $5,000–15,000 tactical radios, not a replacement

THESE ARE SIZED SEPARATELY BECAUSE THEY SELL SUBSCRIPTIONS OR DATA RATHER THAN DELIVERED PACKETS. THE FIRST ROW IS THE RECOMMENDATION IN §2.

WHAT THE SIZING ACTUALLY SAYS

Transit is the smallest line in this analysis and observation is the largest. The GNSS interference line alone is three to ten times the entire mesh transit market, it requires no incentive layer, it escapes every regulatory ceiling in §10, and it has a live comparable trading at $85M. An investor reading only these two tables should conclude that the delivery tier is a wedge and the observation tier is the business. That is what we concluded.

40 The Revenue Model

WHERE SUPPLY MEETS DEMAND, AND WHAT HAPPENS AFTER

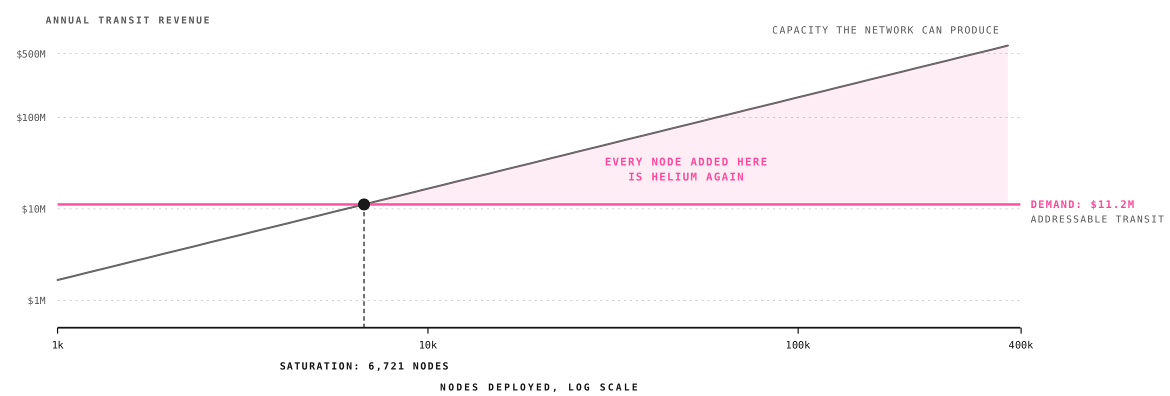
Two independent quantities have now been derived: what a network of a given size can physically produce (§8), and what the market will buy (§21). Plotting them against each other produces the most consequential result in this report.

TABLE 15: CAPACITY AGAINST DEMAND, BY NETWORK SIZE

NODES	DOMAINS	CAPACITY	DEMAND	BINDING	PROTOCOL 10%	STATE
1,000	33	$1.7M	$11.2M	**$1.7M**	$0.17M	demand-limited
5,000	167	$8.3M	$11.2M	**$8.3M**	$0.83M	demand-limited
10,000	333	$16.7M	$11.2M	**$11.2M**	$1.12M	**1× oversupplied**
25,000	833	$41.7M	$11.2M	**$11.2M**	$1.12M	**4× oversupplied**
100,000	3,333	$166.7M	$11.2M	**$11.2M**	$1.12M	**15× oversupplied**
250,000	8,333	$416.8M	$11.2M	**$11.2M**	$1.12M	**37× oversupplied**

CAPACITY ASSUMES 30 NODES PER COLLISION DOMAIN, 20 PERCENT OCCUPANCY, $0.10 PER MESSAGE AND A 35 PERCENT SELL-THROUGH. DEMAND IS THE TOTAL FROM TABLE 13. THE BINDING COLUMN IS THE MINIMUM OF THE TWO.

FIGURE 12: THE POINT WHERE ADDING NODES STOPS ADDING REVENUE



CAPACITY SCALES LINEARLY WITH DEPLOYED NODES. DEMAND DOES NOT SCALE AT ALL. THE SHADED WEDGE IS SUPPLY NOBODY BUYS.

THE SATURATION RESULT

Addressable transit demand is fully served by approximately **6,721 nodes**. Beyond that, every additional deployment adds capacity against fixed demand, which is precisely the failure Helium executed at 500,000 gateways and §4 diagnosed as non-scarcity. **A mesh transit network that succeeds at recruiting operators destroys its own unit economics.** The incentive design must therefore cap or geographically target emissions deployment, which is the opposite of every DeWi token model shipped to date.

TABLE 16: PROTOCOL REVENUE SENSITIVITY AT SATURATION

ASSUMPTION	$0.01	$0.05	$0.10	$0.50
15% sell-through	$0.05M	$0.24M	$0.48M	$2.40M
35% sell-through	**$0.11M**	**$0.56M**	**$1.12M**	**$5.60M**
60% sell-through	$0.11M	$0.56M	$1.12M	$5.60M

BOLD ROW IS THE BASE CASE. PROTOCOL REVENUE IS 10 PERCENT OF GROSS TRANSIT AND IS DEMAND-CAPPED IN EVERY CELL.

Protocol revenue tops out near **$1.12 million a year** in the base case. Against the 8 to 16 times revenue multiples in §12 that implies a terminal network value in the tens of millions: a respectable outcome for a seed cheque and an unacceptable one for a fund underwriting a category-defining position. This is the arithmetic behind the recommendation in §1 to treat the delivery tier as a wedge into the observation tier.

41 The Node as Substrate

COST PER UNIT OF PARTICIPATION

Sections 5 through 10 price the delivery tier as a standalone business and find the ceiling low. That analysis treats a mesh node as a device that sells transit. The second half of the delivery-tier case is that it is not, and that the transit market is the wedge rather than the business.

TABLE 7: WHAT A UNIT OF DEPIN PARTICIPATION COSTS

NETWORK	DEVICE	COST	ALSO REQUIRES	PAYLOADS	SELLS
Hivemapper / Bee Maps	Bee LTE dashcam	$19/mo × 24mo	car + driver	1	street imagery
GEODNET	RTK base station	~$500	rooftop, sky view	1	GNSS corrections
Helium Mobile	5G radio	$500–2,500	mains power, backhaul	1	cellular offload
WeatherXM	weather station	~$500	mounting, backhaul	1	meteorological data
LoRa mesh node	Heltec / RAK / T-Deck	**$15–50**	none: solar, no backhaul	**n**	transit + any sensor payload

COST IS THE OPERATOR'S DEPLOYED CAPITAL PER PARTICIPATING DEVICE. "ALSO REQUIRES" IS THE NON-OBVIOUS PRECONDITION THAT NARROWS THE ADDRESSABLE OPERATOR POPULATION. PAYLOADS IS HOW MANY DISTINCT DATA PRODUCTS THE SAME DEPLOYED DEVICE CAN SELL.

Every other network in that table requires an operator to hold something scarce beyond the device itself. Hivemapper needs a car and a driver. GEODNET needs a rooftop with sky view. Helium Mobile needs mains power and backhaul. Each device then sells exactly one product, and the operator population is bounded by the precondition hardware price.

A LoRa mesh node is the cheapest general-purpose sensing-and-relaying primitive that exists: $15–50, solar-viable, and with no recurring connectivity cost *because it is the connectivity*. Once deployed for one reason, someone wanted off-grid messaging: the same box can carry environmental telemetry, position beacons, RF spectrum survey data, asset tracking and agricultural sensing at near-zero marginal hardware cost.

> **THE REFRAME THAT MATTERS**
>
> Mesh is the transport substrate those verticals currently pay to replicate one at a time. Hivemapper pays $19 a month per device for LTE backhaul. GEODNET requires internet at every base station. Every sensor network independently solves connectivity and independently pays for it. A mesh with real transit economics is the horizontal layer underneath all of them, and horizontal layers are where durable value in this sector accrues.

WHAT THIS DOES NOT CLAIM

It is tempting to say the hardware "backs" the token. It does not, and the language should be avoided: the protocol does not own the devices, operators do, and no claim on that hardware accrues to a token holder. The substrate argument is about *customer acquisition cost across payloads*, not collateral. A node deployed for messaging is a paid-for distribution point for the next sensor network, which is a real economic advantage and a different one.

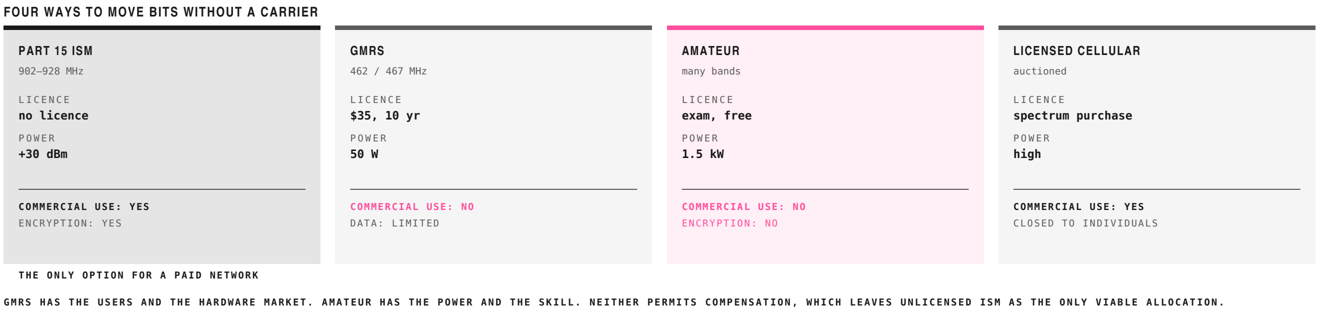
The argument is also unproven. It requires a second payload to ride the substrate within a reasonable window, which is condition 5 in §31 and the weakest of the five. We state it as the bull case §15 is done without it.

42 Ham, GMRS and the Licensed Adjacency

WHERE THE USERS ALREADY ARE

LoRa mesh is not the only way to move bits without a carrier, and it is not the one with the most users. Two licensed services sit adjacent to it, both with larger installed bases and established hardware markets, and neither is available to a paid network. The comparison is worth making because it explains where the operators come from and why the incentive layer cannot follow them.

FIGURE 14: FOUR WAYS TO MOVE BITS WITHOUT A CARRIER



POWER, LICENCE COST AND COMMERCIAL PERMISSION FOR EACH ALLOCATION. THE RIGHTMOST TWO COLUMNS ARE WHERE THE PAID-NETWORK QUESTION IS DECIDED.

TABLE 17: THE FOUR ALLOCATIONS COMPARED

SERVICE	BAND	LICENCE	POWER	PAID USE	MARKET	NOTE
Part 15 ISM	902–928 MHz	none	+30 dBm	yes	unlimited	LoRa mesh lives here
GMRS	462/467 MHz	$35, 10 yr, no exam	50 W	no	~$1.2B hardware	voice, family, off-road
Amateur	many bands	exam, free	1.5 kW	no	733,531 US licences	self-training, no pecuniary interest
Licensed cellular	auctioned	spectrum purchase	high	yes	hundreds of billions	closed to individuals

PAID-USE COLUMN IS THE DECISIVE ONE. GMRS AND AMATEUR BOTH PROHIBIT COMPENSATION, WHICH REMOVES THEM FROM CONSIDERATION REGARDLESS OF THEIR TECHNICAL MERITS.

GMRS: THE LARGER CONSUMER MARKET

The General Mobile Radio Service is a licensed allocation at 462 and 467 MHz permitting 50 watts on base stations, repeater operation, and since the 2017 rule update, limited data including text and GPS position. A licence costs $35, covers an entire household for ten years, and requires no examination. When the FCC halved the fee in April 2022, adoption surged.

The hardware market was roughly $1.2 billion in 2024 and is forecast near $2.1 billion by 2033 at about 6.5 percent compound growth. That is an order of magnitude larger than anything LoRa mesh will address this decade, and it is the clearest evidence that demand for off-grid communication is real and paid for. **People already buy radios. What they do not buy is a network.**

AMATEUR RADIO: THE SKILL BASE, NOT THE MARKET

There are 733,531 active US amateur licences. The average operator is in their mid-60s and fewer than 8 percent are under 40. The service supplies exactly what a mesh deployment needs and cannot buy: people who own towers, understand antenna systems, and can diagnose a feedline problem. It supplies none of what a paid network needs, because 47 CFR §97.113 forbids compensation, pecuniary interest and encryption, as §20 sets out.

The practical consequence is that the amateur community is a *recruiting pool and a distribution channel*, not an operating substrate. A mesh protocol that treats hams as users will work. One that treats amateur spectrum as an asset will not.

WHAT THE ADJACENCY IMPLIES

The addressable operator population is not the Meshtastic subreddit. It is the intersection of the GMRS buyer, who has demonstrated willingness to pay for off-grid hardware, and the amateur licensee, who has the siting and the skill. Both populations are larger than the mesh community and neither can be paid on their own allocation. Unlicensed ISM is where they have to meet.

43 Bitchat Over LoRa

THE BRIDGE LAYER, AND WHAT IT COSTS

§29 treats Bitchat as a Bluetooth network limited to 30 to 300 metres. That was true when this analysis began and it is no longer true. The community solved the range problem without waiting for the project to, and the way they solved it puts Bitchat traffic squarely inside the capacity ceiling measured in §27.

TABLE 36: BRIDGES IN THE WILD

PROJECT	AUTHOR	SINCE	STACK	WHAT IT DOES	NOTE
MeshCore-BitChat	jooray	Jan 2026	MeshCore	bridge node appears as a Bitchat BLE peer; relays to a #mesh hashtag channel	$15–80 boards; tested on Heltec Wireless Stick Lite V3, LilyGo 2.1
meshtastic-bitchat-bridge	GigaProjects	2026	Meshtastic	bidirectional; Meshtastic device over USB plus a Bluetooth adapter	public message exchange only
Meshtastic BitChat peer firmware	evansmj	Jan 2026	Meshtastic	firmware makes a Meshtastic node act as an ordinary Bitchat peer	endorsed publicly by the Cashu author
Bitle	Bitle project	Jul 2026	native Bitchat	ESP32-S3 and C3 solar relay nodes adding LoRa directly to the Bitchat protocol	MIT licensed; 2–10 km hops depending on terrain
Issue #973	permissionlesstech/bitchat	Jan 2026	—	open request for official Meshtastic bridge support as a transport extension	not yet merged

FIVE INDEPENDENT EFFORTS, FOUR OF THEM SHIPPING CODE. THE COMMON PATTERN IS A LORA NODE THAT ADVERTISES ITSELF AS AN ORDINARY BITCHAT BLE PEER AND FORWARDS TRAFFIC ONTO THE MESH.

A MeshCore or Meshtastic node running bridge firmware appears to a nearby phone as a Bitchat peer. Messages sent in Bitchat relay onto the LoRa mesh and back out at another bridge. Neither side needs to know the other exists. Bitle goes further and adds LoRa to the Bitchat protocol directly, on MIT-licensed ESP32 solar relay nodes, claiming 2 to 10 kilometre hops depending on terrain.

> **WHAT THE BRIDGE ACTUALLY BUYS AND COSTS**
>
> It buys range: a Bluetooth bubble becomes a city. It costs the flood tax. Every bridged message is now a LoRa message, subject to R = 7.36, the 600 ms channel occupancy, and the reach collapse in §29. **Bitchat did not escape the delivery-tier ceiling by adding LoRa. It moved inside it.** The 430,000 daily users in India were on Bluetooth, where airtime is free. Route that volume through bridge nodes and it meets the same 163 messages an hour per collision domain as everything else in §28.

This sharpens the position in §29 instead of softening it. Bitchat is the clearest demand signal in the category, the bridges are evidence the demand is real enough that people build hardware for it unpaid, and the economics of carrying that traffic are the economics computed here. Somebody is going to have to pay for the relays, and nobody in Table 36 is being paid.

43 Bitchat Over LoRa

TABLE 37: WHAT THE SHIPPING FIRMWARE CANNOT DO

CAPABILITY	STATUS	WHAT IS MISSING	CONSEQUENCE FOR THIS THESIS
Per-message signing	absent in both	channel and room traffic is encrypted, but no device can prove a given message came from it or was unaltered	the delivery receipt in §19 does not exist in shipping firmware
Store and forward	absent in both	delivery is immediate or it fails; nothing holds a message for a node that is offline	a signed transaction cannot wait in the mesh (§32)
Per-device revocation	absent in Meshtastic	access is a shared channel key; a leak compromises every device at once and no single operator can be removed	a paid network cannot eject a defecting relay
Frequency agility	absent in both	fixed regional channels, so a single-frequency jammer is sufficient	denial of service is cheap against a paid network

FROM A JUNE 2026 COMPARISON WRITTEN BY THE DEVELOPER OF A COMPETING CLOSED-SOURCE SYSTEM, WHO DISCLOSES THE CONFLICT. THE CLAIMS CONCERN MESHTASTIC AND MESHCORE AND ARE CHECKABLE IN SOURCE.

TWO OF THESE DAMAGE OUR OWN ARGUMENT

§19 argues the delivery tier verifies because a relay cannot fabricate a delivery to a counterparty that received nothing, since the receipt is a signature over a path. **Neither Meshtastic nor MeshCore signs individual messages.** Channel traffic is encrypted, and within an encrypted channel no device can prove a message came from it. The receipt this thesis depends on has to be built; it does not ship.

§32 argues a signed transaction can be carried across a mesh for hours before broadcast. **Neither firmware implements store and forward.** Delivery is immediate or it fails. The deferred-broadcast case therefore depends on a courier device holding the transaction, not on the mesh holding it, which is a weaker claim than the one we made.

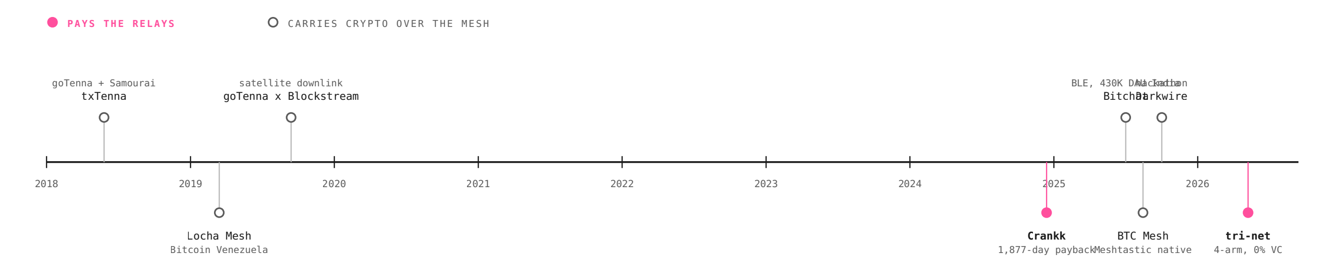
Both gaps are buildable and neither is exotic. Per-message signing is an ECDSA signature over the payload; store and forward is a queue with an expiry. But they are work that a protocol team would have to do before any of the settlement economics in §32 apply, and no shipping firmware has done it. A team that has not noticed this is a team that has read about mesh networking without running it.

44 Prior Attempts

EIGHT YEARS, AND WHAT NOBODY BUILT

At least eight projects have connected mesh radio to a blockchain since 2018. The pattern among them is uniform and it is the strongest evidence that the delivery tier is unbuilt.

FIGURE 10: MESH AND BLOCKCHAIN, 2018–2026



FILLED MARKERS PAY THE RELAYS; HOLLOW MARKERS USE THE MESH AS TRANSPORT FOR CRYPTOCURRENCY WITHOUT COMPENSATING THE NODES THAT CARRY IT.

txTenna (2018, goTenna with Samourai Wallet) relayed signed Bitcoin transactions across a mesh to an internet-connected exit node; one documented field test covered 12.6 km with no cellular service. **Locha Mesh** (2019, Bitcoin Venezuela) built Turpial and Harpia hardware forming an IPv6 mesh over licence-free bands, prompted by the March 2019 national blackout, and ran a 22-hour continuous test bridging Blockstream Satellite downlink into the mesh. **Bitchat** (2025) carries Bitcoin and Cashu over Bluetooth mesh and reached 430,000 daily active users in India during the July 2026 internet shutdowns, prompting an unsuccessful government takedown order against its GitHub repositories. **BTC Mesh**, the **MeshtasticBitcoinCore Bridge** and **Darkwire** do variations of the same thing natively on Meshtastic.

Every one of these uses the mesh to *carry* cryptocurrency. None pays the mesh. The value flows one way: the mesh is the cost and the blockchain is the product.

THE TWO EXCEPTIONS

Crankk integrated with Meshtastic firmware on Kadena in December 2024 and did attempt to reward participation. Its published economics show an average device cost near $150 against estimated daily earnings of eight cents: a break-even horizon of 1,877 days. It is the only prior attempt at the thing this report describes, and its failure is instructive: it built on LoRaWAN star topology and a Meshtastic fork, which means it inherited exactly the flood-routing tax quantified in §18 and §25.

tri-net is the closest live implementation, an Apache-licensed Rust mesh daemon on Zynq-7020 boards with AD9361 software-defined radios, structured around four supply-side arms, transport, compute, coverage and sensor, settling through one contract. Its transport proof payload is source, destination, byte count, and start and end timestamps: proof of delivery implemented. Its sensor arm is described as an RF spectrum atlas with GPS-jamming detection, which is §11 as a product specification. The engineering is serious, expected-transmission-count routing citing the sensor-network literature, hop-by-hop authenticated encryption, and a maturity marker on every claim separating hardware-verified results from simulation.

It is nonetheless uninvestable. The compute arm depends on silicon its own documentation describes as having no route to fabrication; everything past the first milestone is unrun; the bus factor is one; and the allocation to venture investors is zero. That last fact is the signal. **The most complete implementation of this thesis is being built by one person who has structurally refused to make it ownable.** The category is real enough that someone builds it seriously, and the version a fund can own does not yet exist.

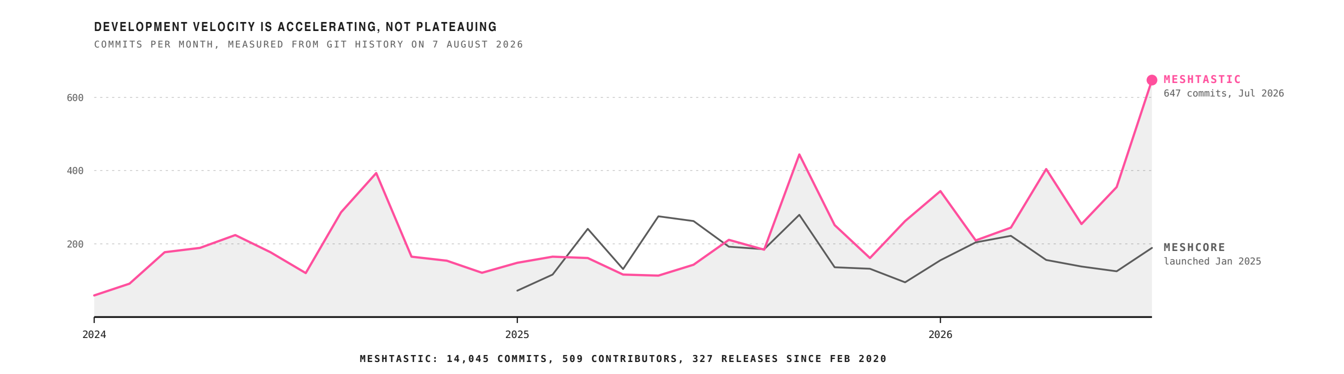
One caveat on comparison: tri-net runs wideband software-defined radio at 5.8 GHz rather than LoRa, so the airtime ceiling derived in §6 does not bind it. It escapes that ceiling and pays instead in capital expenditure, power draw and the need for a spectrum licence. Mesh DePIN is two categories with different binding constraints, not one, and this report treats only the LoRa case quantitatively.

45 Where This Goes

2026 TO 2036

Two things about this category are measurable today and both point the same way: the software is accelerating and the deployed base is growing without anyone funding it. What is not measurable is whether an economic layer ever attaches, which is what the gates below are for.

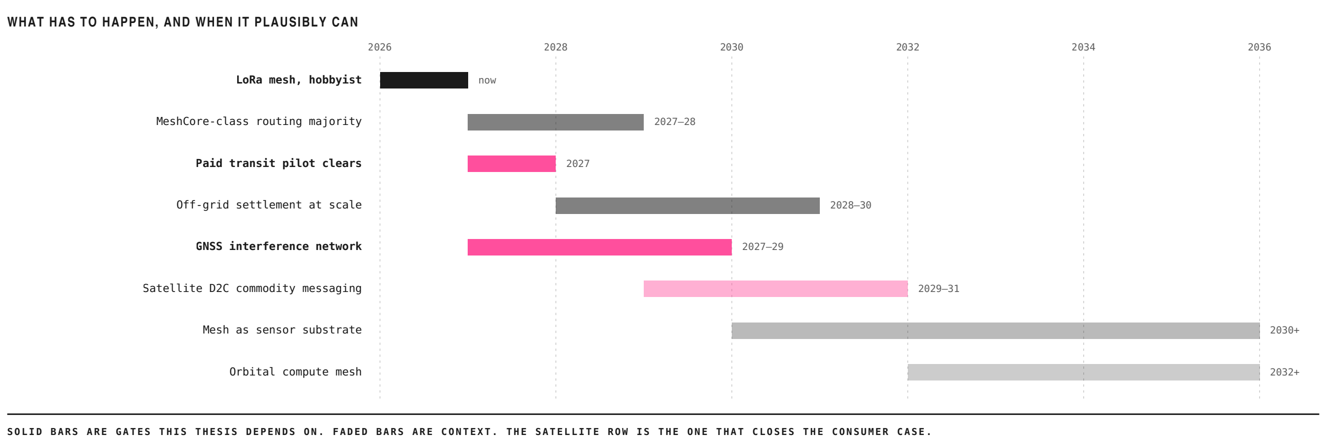
FIGURE 15: DEVELOPMENT VELOCITY, MEASURED FROM GIT HISTORY



COMMITS PER MONTH TO EACH FIRMWARE. MESHTASTIC RECORDED ITS HIGHEST-EVER MONTH IN JULY 2026, SIX YEARS AFTER FIRST COMMIT. MESHCORE HAS SUSTAINED ROUGHLY 150 TO 250 COMMITS A MONTH SINCE LAUNCHING IN JANUARY 2025.

Meshtastic has 14,045 commits from 509 contributors across 327 releases since February 2020, and its busiest month on record was July 2026. MeshCore reached 3,324 commits and 214 contributors in nineteen months. Neither project has a token, a treasury, or a revenue model. This is what an unpaid substrate looks like when it is working.

FIGURE 16: THE GATES, AND WHEN THEY PLAUSIBLY CLEAR



SOLID BARS ARE CONDITIONS THIS THESIS DEPENDS ON; FADED BARS ARE CONTEXT THAT SHAPES IT. THE SATELLITE ROW IS THE ONE THAT CLOSES THE CONSUMER CASE.

WHAT THE FORECASTS ARE WORTH

SOURCE	HORIZON	FORECAST	CAGR
Coherent Market Insights	2033	**$54.6B**	34.9%
Future Market Insights	2035	**$163.3B**	41.1%
Market.us	2033	**$124.4B**	32.6%
Dimension Market Research	2033	**$183.9B**	36.5%
Research and Markets	2030	**$48.4B**	36.8%

PUBLISHED LORAWAN MARKET FORECASTS. THE SPREAD IS ROUGHLY FOURFOLD ACROSS OVERLAPPING HORIZONS, WHICH IS ITSELF THE FINDING.

Five commercial forecasts for the same market over overlapping horizons differ by about four times. We cite none of them in the sizing in §21 and neither should anyone else. The bottom-up transaction build is smaller and defensible; these are larger and are

46 Adjacent Applications

WHAT ELSE RIDES A DEPLOYED NODE

§12 argued that a mesh node is the cheapest general-purpose sensing and relaying primitive available, and that its value is the payloads it can carry rather than the transit it sells. This section names them, because the substrate argument is only as strong as the second application.

TABLE 18: PAYLOADS THAT RIDE THE SAME HARDWARE

APPLICATION	WHAT RIDES THE NODE	CONVICTION	WHERE IT CONNECTS
Drone and UAV command	MAVLink over mesh; Beechat integrated it over Reticulum	high	contested-environment C2 is the tactical buyer in §17
Environmental sensing	soil, water, air quality on the same $15 board	high	the substrate argument in §12
GNSS interference	receive-only monitoring; regulators have stated the gap	highest	the recommendation in §2
Asset and livestock tracking	position beacons at zero marginal hardware cost	medium	competes with cellular IoT on price, not capability
Satellite backhaul	LoRa edge to a D2C or LEO uplink at the gateway	medium	complementary to Starlink, not competitive (§23)
Wildfire and utility monitoring	solar nodes where there is no power or backhaul	medium	public-safety budget, slow procurement

CONVICTION IS OURS. THE FINAL COLUMN POINTS TO THE SECTION THAT SUPPORTS OR CONSTRAINS EACH.

DRONES

Uncrewed systems need command and telemetry in exactly the conditions a mesh is built for: no infrastructure, contested spectrum, and a requirement that the link degrade gracefully. Beechat has integrated MAVLink over Reticulum with encrypted multi-hop transport and peer authentication, and goTenna's acquisition by Forterra was explicitly about autonomous mission systems needing low-cost, low-bandwidth, hard-to-detect links. LoRa is not a video downlink and will never be one. It is a control and telemetry channel that survives when the primary link does not, which is a real requirement with a defence buyer attached.

SENSING AND SATELLITE

Environmental, agricultural and utility sensing are the classic LoRa applications and the ones that motivated LoRaWAN in the first place. The difference under a mesh topology is that no gateway and no backhaul contract are required, which removes the fixed cost that made Helium's IoT network uneconomic for sparse deployments.

Satellite is a complement. A LoRa mesh aggregates edge traffic to a gateway; that gateway can uplink through Starlink, a direct-to-device handset, or an LEO IoT constellation. §14 argues the two mesh layers do not compete, and this is the concrete form of that argument: the edge mesh solves the last three kilometres, the constellation solves the last three thousand.

> **THE HONEST WEAKNESS**
>
> Every row in Table 18 is a plausible payload and none of them is a committed customer. The substrate argument requires that at least one second payload actually rides deployed nodes within a reasonable window, which is condition 5 in §31 and the weakest of the five. The sizing in §21 and the saturation result in §22 are both computed *without* it, so the recommendation does not depend on this section being right. It depends on this section being wrong being survivable.

47 What Would Have to Be True

AND WHAT WOULD MAKE US WRONG

CONDITIONS THE THESIS REQUIRES

1. Routing architecture supports settlement-grade receipts at scale. Testable now on deployed MeshCore networks.
2. Willingness to pay clears operator airtime cost at $0.10 or more per message. A paid-transit pilot in one metropolitan mesh settles it.
3. Rushing and collusion are bounded by protocol design. Simulatable.
4. Observation networks can re-intermediate volunteers without crowding out the intrinsic motivation that built them. The largest unknown.
5. At least one high-value payload rides the substrate within 24 months.

WHEN WE WOULD CLOSE THE POSITION

- If a paid-transit pilot cannot clear above operator airtime cost in a dense metropolitan mesh within twelve months, the delivery tier is dead and no framework rescues it.
- If a rewarded observation network shows measurable quality degradation against its volunteer predecessor within two reward epochs, close that position.
- If satellite direct-to-cell reaches consumer handsets at scale before a mesh protocol reaches revenue, the consumer case evaporates entirely.

THE STRONGEST ARGUMENT AGAINST

Starlink. The FCC approved SpaceX's acquisition of approximately 65 MHz of nationwide exclusive-use contiguous spectrum from EchoStar on 12 May 2026 against a seventeen-billion-dollar transaction, alongside a filing for 15,000 additional very-low-earth-orbit satellites and a stated target of 150 Mbps direct to handset. If every phone carries satellite connectivity by 2029, consumer off-grid messaging becomes a free feature.

The counterargument is narrow. Satellite direct-to-cell requires sky view and performs poorly indoors, underground, in urban canyons and under jamming; and it reintroduces the single point of control the observed demand is running from. Users adopting mesh during shutdowns solve a permission problem, not a coverage problem, and a constellation under one jurisdiction does not solve a permission problem. The niche is defensible, not a mass market.

48 Method and Availability

REPRODUCIBILITY

R was measured with the official Meshtastic discrete-event simulator, which models the production firmware's managed-flood routing, LoRa airtime, channel-activity detection, contention windows and collision behaviour. Three repetitions per configuration across node counts 5–75 and hop limits 3, 5 and 7. Time on air follows the Semtech modem design guide. Every figure and table here is generated from `model.py`; no value was transcribed by hand.

Limitation. These are simulation results, not field measurements. Uniform random placement understates real-world clustering, which likely means R is higher still in dense clusters and lower in sparse ones. A live measurement campaign against the public Meshtastic MQTT infrastructure is in progress. We say so here: a framework built on verifiability should hold its own claims to the standard it recommends.

45 NOMENCLATURE

SYMBOL	DEFINITION	UNIT	SECTION
R	rebroadcast factor, transmissions per delivered message	–	§27
T_{oa}	time on air of one transmission	ms	§25
S	receiver sensitivity, from the thermal noise floor	dBm	§25
u	target channel occupancy, taken as 0.20 throughout	–	§28
C	network message capacity, one collision domain	h^{-1}	§28
Π	gross annual revenue, one collision domain	USD	§28
A_{tx}	airtime consumed by one settled transaction	ms	§32
reach	share of the network receiving a given message	%	§29

SECTION COLUMN POINTS TO WHERE EACH QUANTITY IS DERIVED.

49 Conclusion

WHAT WE BELIEVE, AND WHAT WE WOULD DO

The decentralized physical infrastructure category has been sorted by the resource sold for six years. Sorted instead by what a network can check a contribution against, it explains itself: coverage is self-asserted and produced $6,500 a month; delivery is checkable against a counterparty and we measure its ceiling; observation is checkable against physics and has produced the only revenue worth the name.

We set out to underwrite mesh. We measured it properly, obtained R = 7.36 against the 3 to 5 the literature assumes, found that reach falls as nodes are added, and found that addressable transit demand saturates at 6,721 nodes. Every one of those results moved against the idea we started with, and together they moved the recommendation one tier up the ladder.

WHAT WE BELIEVE

That verifiability, not vertical, predicts revenue in this category, and that the position of a network on that ladder is currently unpriced. That the observation tier is where the next decade of DePIN revenue comes from, because physics is the only referee that cannot be bribed. That mesh capacity has two orders of magnitude of architectural headroom which will not rescue transit economics but will create payloads nobody can currently name. And that the emission designs shipped so far are the primary cause of failure in this category, not the hardware, the market or the timing.

WHAT WE WOULD DO

Fund a receive-only GNSS interference monitoring network on Aptos: $26 million addressable against a $260 bill of materials, buyers who have published the capability gap themselves, a live comparable trading at $85 million on the same hardware and operator profile, and an emission design that combines ORE's fixed competitive issuance with Helium's fiat-denominated burn, gated on a signed contract. Build storage on Shelby and transit on DoubleZero, both of which are chain agnostic and neither of which we would have to finance. Prove the emission invariants with the Move Prover, so that the tokenomics are a checked artefact.

Hold GEOD and 2Z while that is built, because they are the liquid expression of the same thesis and will re-rate before anything private does.

> THE ONE SENTENCE
>
> **Pay for physical work that something external can check, meter the reward by marginal contribution token until somebody is paying for the output.** Every network in this report that violated one of those three is a cautionary case; the two that respected them are the only ones producing revenue.

WHAT WOULD CHANGE OUR MINDS

A paid-transit pilot clearing above operator cost in a dense metropolitan mesh would reopen the delivery tier. A rewarded observation network showing quality degradation against its volunteer predecessor would close the recommendation. And aviation or defence buyers declining to purchase measured interference data at all would end it, which is why §9 begins with a customer conversation.

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